

Introduction to Probability and Statistics

Textbook for STAT/DATA 400 at University of Maryland, College Park ¹
Draft Version

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1 | Probability

Sets, Experiment, and Probability

Definition 1.1 (Experiment)

An **Experiment** is a repeatable task with well defined outcomes.

Definition 1.2 (Sample Space)

The **Sample Space** for an experiment is the set of all possible outcomes of that experiment, usually denoted by $\mathcal{S}$

Definition 1.3 (Event)

Any subset E of the sample space $\mathcal{S}$ attached to an experiment will be called an **Event** associated to the experiment.

To say that **an event E has happened** means “we performed the experiment, and the outcome of this experiment was in the set E ”.

Definition 1.4 (Simple Event)

An event is called a simple event if it has exactly one outcome in it.

Example 1.1.1

Experiments involving coin tosses:

- (a). Experiment: Toss a coin once.
All possible outcomes are contained in the set $\{H, T\}$, with H representing obtaining a head and T representing obtaining a tail. The total number of outcomes is 2.
- (b). Experiment: Toss the coin two times.
All possible outcomes are contained in the set $\{HH, HT, TH, TT\}$. Number of outcomes is 4.
- (c). Experiment: Toss the coin 10 times.
All possible outcomes are contained in the set $\{HHH \cdots H, HTHH \cdots H, \cdots\}$. Number of outcomes is $2^{10} = 1024$.

Example 1.1.2

Experiments involving die rolls:

- (a). Roll a six-sided die once. Outcomes are in the set $\{1, 2, 3, 4, 5, 6\}$.
- (b). Roll the die two times. Outcomes are in the set

$$\left\{ \begin{array}{ccccc} (1, 1), & (1, 2), & (1, 3), & \cdots, & (1, 6), \\ (2, 1), & (2, 2), & (2, 3), & \cdots, & (2, 6), \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ (6, 1), & (6, 2), & (6, 3), & \cdots, & (6, 6) \end{array} \right\}.$$

Example 1.1.3

Consider the following two step experiment: In step 1, we toss a coin; In step 2, we roll a 6-sided die. All possible outcomes are contained in the set

$$\left\{ \begin{array}{cccccc} (H, 1), & (H, 2), & (H, 3), & (H, 4), & (H, 5), & (H, 6), \\ (T, 1), & (T, 2), & (T, 3), & (T, 4), & (T, 5), & (T, 6) \end{array} \right\}.$$

Example 1.1.4

We can perform an experiment where we pick a random student from campus, and ask if they have walked more than 3000 steps today. All possible outcomes of this experiment are contained in the set $\{\text{“Yes”}, \text{“No”}\}$.

Example 1.1.5

We can perform an experiment where we pick a student from campus and measure their height. All possible outcomes are contained in the set of positive real numbers, that is $(0, \infty)$.

Our goal is to assign probabilities or chance to events E corresponding to an experiment. Since events and sample spaces are described as sets and subsets, we will first need to remind ourselves of how to work with these mathematical objects.

Basic Set Theory

We will need to understand some basics of sets and set operations. We define the operations along with the visual of the the operation using Venn Diagrams.

Definition 1.5 (Empty Set)

An empty set is a set with nothing in it, usually denoted as $\emptyset$.

Let A, B be two sets.

Definition 1.6 (Containment/Subset)

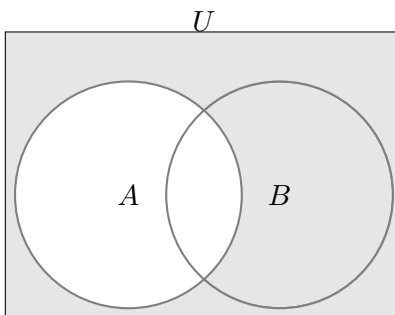
We say A is a **subset** of B if $x \in A$ *implies* $x \in B$, denoted as $A \subset B$.

Definition 1.7 (Equality)

We say A **equals** to B if $A \subset B$ *and* $B \subset A$.

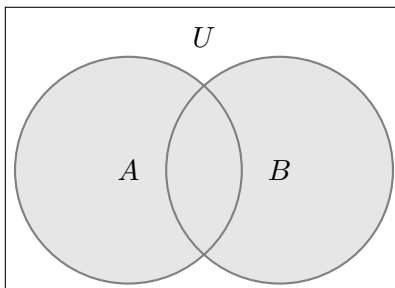
Definition 1.8 (Complement)

The **complement** of A is the set $A^c := \{x : x \notin A\}$



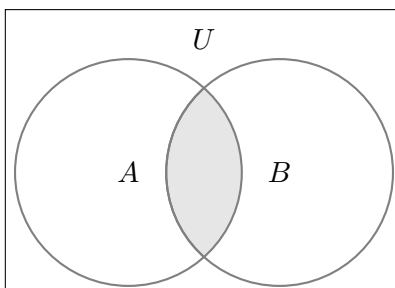
Definition 1.9 (Union)

The **union** of A and B is the set $A \cup B := \{x : x \in A \text{ or } x \in B\}$



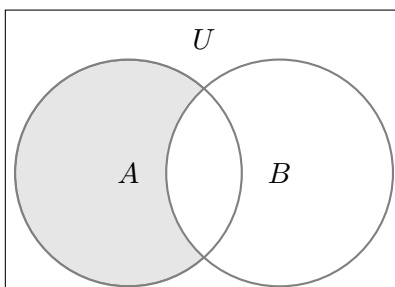
Definition 1.10 (Intersection)

The **intersection** of A and B is the set $A \cap B := \{x : x \in A \text{ and } x \in B\}$



Definition 1.11 (Difference)

The **difference** of set A from set B is defined as $A \setminus B := \{x : x \in A \text{ and } x \notin B\}$. This is the same as $A \setminus B := A \cap B^c$.



Definition 1.12 (Disjoint Sets/Mutually Exclusive)

A and B are said to be **disjoint** if $A \cap B = \emptyset$.

We will often have to work with an infinite collection of sets, to do so, we will first need to label the sets using an “indexing set”. Given a set Γ , a collection of sets indexed by Γ will be denoted by $\{A_\alpha\}_{\alpha \in \Gamma}$. A set is **countable** if there is a one-to-one correspondence with its elements, and the set of natural number $\mathbb{N} = \{1, 2, 3, 4, \dots\}$.

Let $\{A_i\}_{i \in \mathbb{N}}$ be a countable collection of sets.

Definition 1.13 (Pairwise Disjoint Collection)

We say $\{A_i\}_{i \in \mathbb{N}}$ is a collection of pairwise disjoint sets if $A_i \cap A_j = \emptyset$ for all $i \neq j$

Definition 1.14 (Partition)

A collection $\{A_i\}_{i \in \mathbb{N}}$ is said to be a **partition** of $\mathcal{S}$ if $\{A_i\}_{i \in \mathbb{N}}$ are pairwise disjoint **and** $\mathcal{S} = \bigcup_{i=1}^{\infty} A_i$

Definition 1.15 (Countable Union/Intersection)

Let $\{A_i\}_{i \in \mathbb{N}}$ be a countable collection of subsets of $\mathcal{S}$.

The **union** of $\{A_i : i \in \mathbb{N}\}$ is defined as

$$\bigcup_{i \in \mathbb{N}} A_i := \{x : x \in A_i \text{ for some } i \in \mathbb{N}\}$$

The **intersection** of $\{A_i : i \in \mathbb{N}\}$ is defined as

$$\bigcap_{i \in \mathbb{N}} A_i = \{x : x \in A_i \text{ for all } i \in \mathbb{N}\}$$

1.2.1 Axioms of Probability

Let $\mathcal{S}$ be the sample space of an experiment and let $E \subset \mathcal{S}$ be an event. We are interested in assigning a number to E that quantifies the chance/probability that E occurs. We will denote the probability of E as $P(E)$.

The mathematical procedure to assign probabilities to events is to define a function on the “event space”. Assigning probabilities to every possible event can be tedious, and in some case it might not be possible. As a result, we will first need to identify a collection of events that we consider “interesting” and to which we want to assign probabilities. Such a collection of events is called a “sigma algebra”.

Definition 1.16 (Sigma Algebra)

Let $\mathcal{S}$ be a sample space of an experiment. A collection, $\mathcal{B}$, of subset of $\mathcal{S}$ is called a **sigma algebra** if it satisfies the following conditions:

1. $\emptyset \in \mathcal{B}$
2. If $A \in \mathcal{B}$, then $A^c \in \mathcal{B}$
3. If $\{A_i : i \in \mathbb{N}\}$ is countable collection such that $A_i \in \mathcal{B}$ for all i , then $\bigcup_{i \in \mathbb{N}} A_i \in \mathcal{B}$

Note that there are multiple sigma algebras that can be associated to a sample space. The trivial sigma algebra is $\{\emptyset, \mathcal{S}\}$ - the smallest possible sigma algebra. The power set $\mathcal{P}(\mathcal{S})$ of $\mathcal{S}$ also satisfies the axioms to be a sigma algebra, and this is the largest possible sigma algebra associated to $\mathcal{S}$. All other sigma algebras are somewhere in between these two extreme examples. We can use sub-setting as a natural way to define a partial order on all possible sigma algebras.

A probability function assigns probabilities to events in the sigma algebra as follows.

Definition 1.17 (Probability Function)

Consider the pair $(\mathcal{S}, \mathcal{B})$, where $\mathcal{S}$ is the sample space of an experiment and $\mathcal{B}$ is a sigma algebra associated to $\mathcal{S}$.

A probability function for the pair $(\mathcal{S}, \mathcal{B})$, is a function

$$P : \mathcal{B} \longrightarrow \mathbb{R}$$

satisfying the following axioms:

1. (finite measure) $P(\mathcal{S}) = 1$
2. (positivity) $P(A) \geq 0$ for all $A \in \mathcal{B}$
3. (countable additivity) For $A_1, A_2, A_3, \dots$, the collection of pairwise disjoint subsets of $\mathcal{S}$ in $\mathcal{B}$, we must have

$$P\left(\bigcup_{i \in \mathbb{N}} A_i\right) = \sum_{i=1}^{\infty} P(A_i)$$

Theorem 1.18 (Properties of a Probability Function)

Suppose P is a probability function.

- (i) $P(\emptyset) = 0$
- (ii) If $A \subset B$, then $P(A) \leq P(B)$
- (iii) $P(A^c) = 1 - P(A)$
- (iv) $P(A \setminus B) = P(A) - P(A \cap B)$
- (v) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Defining a probability function can often times be a tedious task, as the number of conditions needed to be checked grow exponentially (as a function of the sigma algebra).

When the sample space $\mathcal{S}$ is countable, it is possible to define a probability function on the power set of $\mathcal{S}$ using the following theorem.

Theorem 1.19

Suppose the sample space of an experiment is countable, listed as, $\mathcal{S} = \{s_i\}_{i \in \mathbb{N}} = \{s_1, s_2, s_3, \dots\}$. Let $\mathcal{B} := \mathcal{P}(\mathcal{S})$ be the powerset of $\mathcal{S}$.

Suppose $\{p_i\}_{i \in \mathbb{N}}$ is a sequence of real numbers satisfying:

$$p_i \geq 0 \quad \forall i, \quad \sum_{i=1}^{\infty} p_i = 1$$

define $P : \mathcal{B} \rightarrow [0, 1]$ as follows:

1. For any $s_i \in \mathcal{S}$,

$$P(\{s_i\}) = p_i$$

2. For any $E \in \mathcal{B}$

$$P(E) = \sum_{s \in E} P(\{s\})$$

The function P defines a probability function on $\mathcal{B}$.

Using the theorem above, we can assign probability functions experiments with countable sample space, by assigning probabilities just to the simple events.

Example 1.2.1

Consider the experiment consisting of tossing a coin two times. The sample space is $S = \{HH, HT, TH, TT\}$. Take the largest possible sigma algebra, that is the power set of S , denoted as $\mathcal{P}(S)$, which contains $|\mathcal{P}(S)| = 2^4 = 16$ elements. In this experiment, if the coin is a fair coin, we can assign the following probability function $P : \mathcal{P}(S) \rightarrow \mathbb{R}$:

event E	$P(E)$
$\emptyset$	0
$\{HH\}$	1/4
$\{HT\}$	1/4
$\{TT\}$	1/4
$\{TH\}$	1/4
$\{HH, HT\}$	$1/4 + 1/4 = 1/2$
$\vdots$	$\vdots$
$\{HH, HT, TH\}$	$1/4 + 1/4 + 1/4 = 3/4$
$\vdots$	$\vdots$
$\{HH, HT, TH, TT\}$	1

In the case where the coin is not a fair coin, the probability function can be defined in the following way:

event E	$P(E)$
$\emptyset$	0
$\{HH\}$	1/3
$\{HT\}$	1/3
$\{TT\}$	1/3
$\{TH\}$	0
$\{HH, TH\}$	$1/3 + 0 = 1/3$
$\vdots$	$\vdots$
$\{HH, HT, TT\}$	$1/3 + 1/3 + 1/3 = 1$
$\vdots$	$\vdots$
$\{HH, HT, TH, TT\}$	1

Similarly, the following probability function is also allowed:

event E	$P(E)$
$\emptyset$	0
$\{HH\}$	$1/8$
$\{HT\}$	$1/8$
$\{TT\}$	$3/8$
$\{TH\}$	$3/8$
$\{HH, TH\}$	$1/8+3/8 = 1/2$
$\vdots$	$\vdots$
$\{HH, HT, TT\}$	$1/8+1/8+3/8 = 5/8$
$\vdots$	$\vdots$
$\{HH, HT, TH, TT\}$	1

Notice that the probabilities of events like $\{HH, TH\}$ and $\{HH, HT, TT\}$ are completely determined by the probabilities of the four simple events $\{HH\}$, $\{HT\}$, $\{TH\}$ and $\{TT\}$. While for event E , we note that $P(E)$ must be non-negative. We also observe that $P(S) = 1$, and as $S = S \cup \emptyset$, then $P(S) = P(S \cup \emptyset) = P(S) + P(\emptyset) = 1$, from which we deduce that we must have $P(\emptyset) = 0$. The tables shown above are called the distribution tables, and we see that there can be different distribution tables for the same sample space.

Conditional Probability and Independence

If the probability function attached to an experiments must be useful (say to draw some conclusions about events associated to the experiment), we will need to incorporate the features of the experiment as in our pursuit to calculate probability functions.

Definition 1.20 (Conditional Event)

Suppose E and F are two events, the conditional event $E \mid F$ - verbalized as “E given F” - is the event that “E happens given that F has already happened”

Given two events E and F we can consider the following events:

- 1. $E \cap F$: both events E and F happen.
- 2. $E \mid F$: E happens given that F has happened.
- 3. $F \mid E$: F happens given that E has happened.

There is an interesting relationship between these events given by

Definition 1.21 (Multiplication Principle)

Suppose E and F are two events then we define:

$$P(E \cap F) = \begin{cases} P(E)P(F \mid E) \\ P(F)P(E \mid F) \end{cases}$$

The term $P(F \mid E)$ is called the conditional probability of the event “F given E”, in fact we use the multiplication principle to formally define conditional probability

Definition 1.22 (Conditional Probability)

Suppose E and F are two events then we define:

$$P(E \mid F) = \frac{P(E \cap F)}{P(F)}$$

The multiplication principle informs us that the probability of the simultaneous occurrence of event E and event F , that is $P(E \cap F)$, depends not only on information about the individual occurrence (with no additional information about the other event) of E (that is $P(E)$) and F (that is $P(F)$), **but** also the interaction/influence of how the event E affects the occurrence of event F (given by the conditional probability $P(F \mid E)$) or how the event F affects the occurrence of event E (given by the conditional probability $P(E \mid F)$).

This naturally leads to the notion of independence, which gives the probabilistic description of what it means for the lack of interaction between two events.

Definition 1.23 (Independent Events)
We say two events E and F are independent if

$$P(E \cap F) = P(E)P(F)$$

Equivalently, E and F are independent if

$$P(E \mid F) = P(E) \quad \text{or} \quad P(F \mid E) = P(F)$$

Observe that, $P(E \mid F) = P(E)$, tells us that the fact that F happened did not affect the probability of E happening.

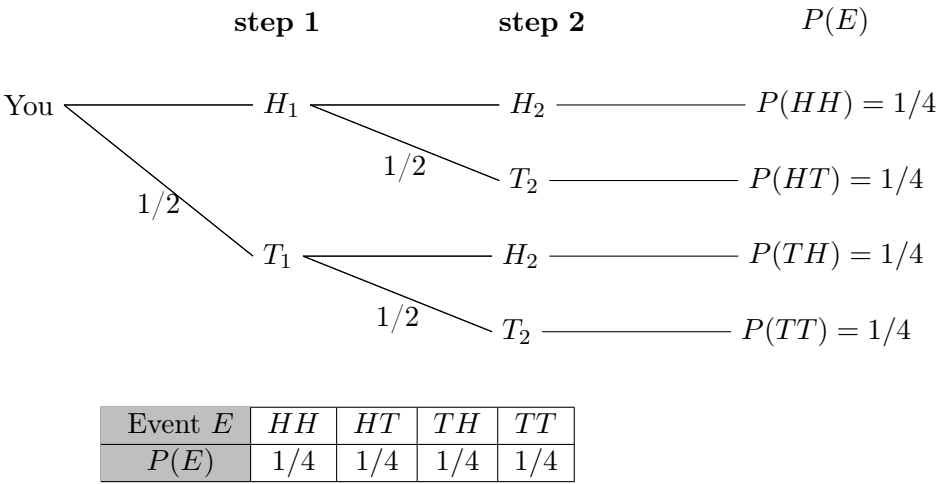
An experiment that can be broken down into a sequence of steps can be represented as tree, where nodes at level i are the outcomes of the $(i - 1)$ th step, and each node at level i will emanate as many nodes as there are possible outcomes for the i th step at that node. Every edge emanating out of a node at level i denotes the conditional event of the outcome at the tail of edge happens given that we are node at the head of the edge.

A unidirectional path from the top of the tree to one of its leaves will describe one outcome for the experiment. The sample space consists of all possible unidirectional paths from the top of the tree to it's leaves.

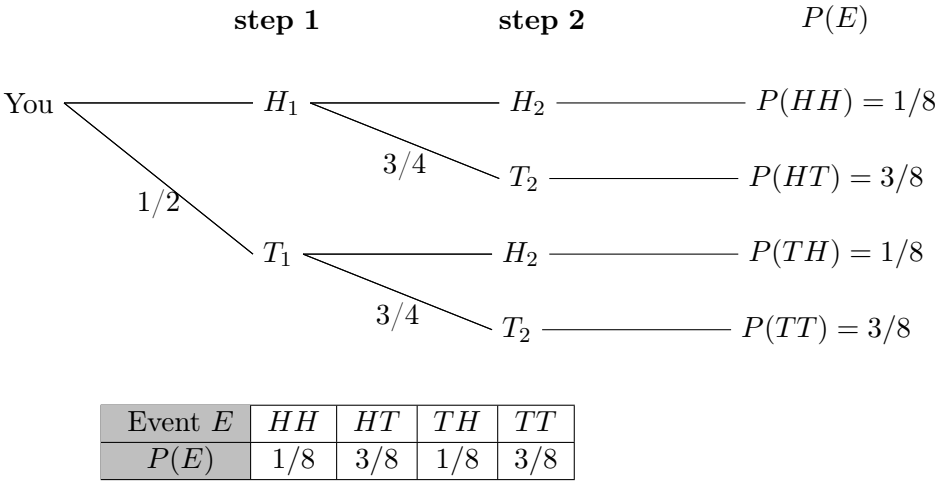
The multiplication principle can be used to assign probabilities to all simple events.

Example 1.3.1
We revisit the experiment where we want to assign probabilities to the simple events associated to tossing a coin two times.

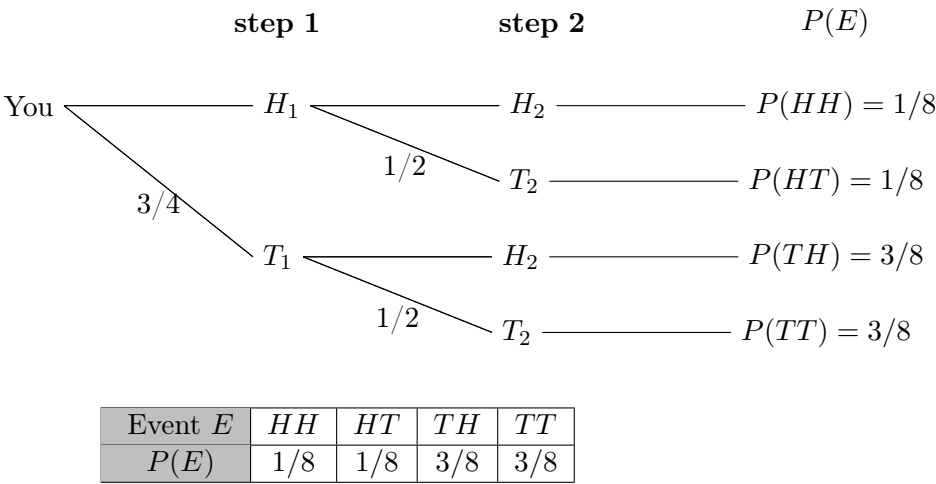
1. Consider the experiment that we toss a fair coin two times.



2. Now consider first we flip a fair coin, then an unfair coin.
The second coin has probability $P(H_2) = 1/4$ and $P(T_2) = 3/4$.



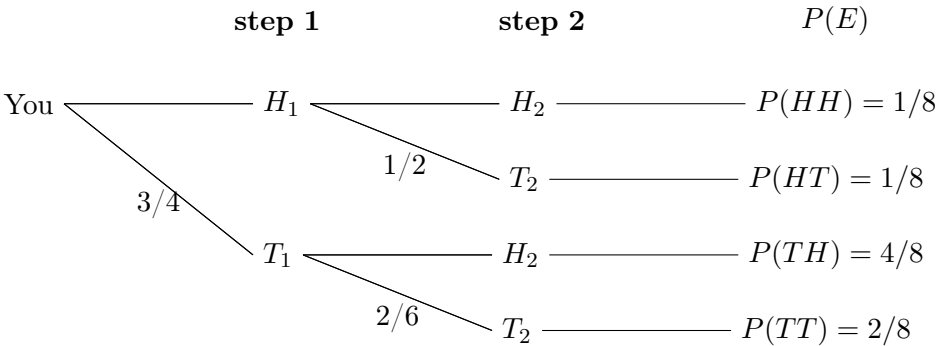
3. Now consider first we flip an unfair coin, then a fair coin.
The first coin has probability $P(H_1) = 1/4$ and $P(T_1) = 3/4$.



4. One might ask if it is possible to construct the following distribution table:

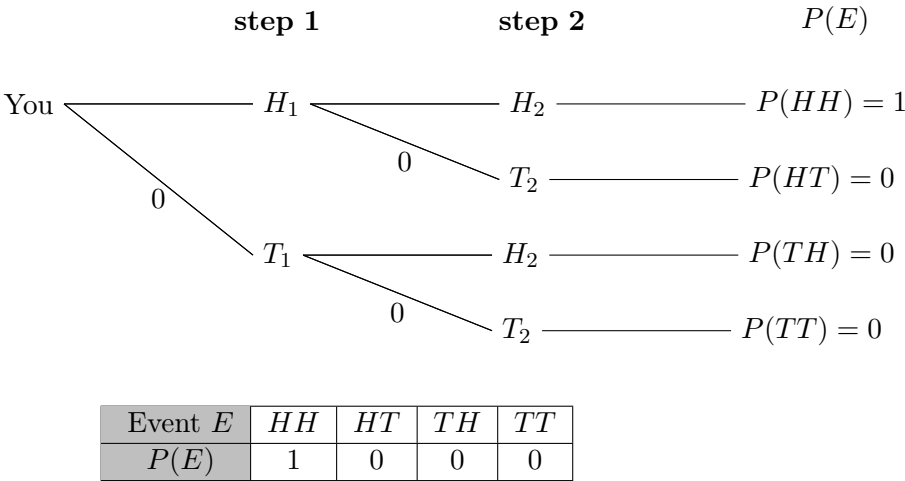
Event E	HH	HT	TH	TT
$P(E)$	$1/8$	$1/8$	$4/8$	$2/8$

To construct such a distribution table, we consider the following experiment: In step 1, we toss a coin with $P(H_1) = 1/4$; In step 2, if we got H_1 in step 1, then we toss a coin with $P(H_2) = 1/2$, if we got T_1 in step 1, then we toss a coin with $P(H_2) = 4/6$.

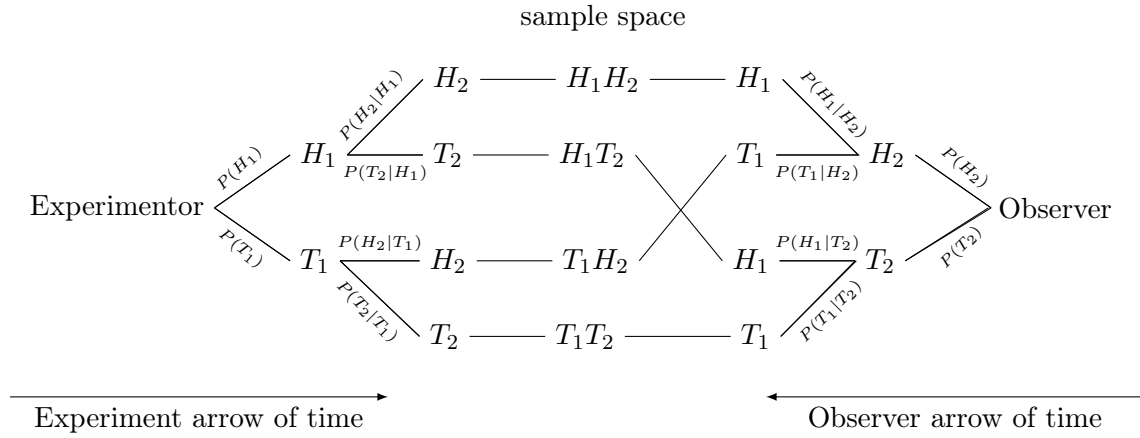


We see that this experiment gives the desired distribution table.

5. Now suppose we toss a two-headed coin two times, that is, $P(H_1) = P(H_2) = 1$.



Observe that in the coin toss examples above, we can consider the the following two tree diagram



The connecting piece between these two trees is the law of total probability and Bayes' theorem.

Bayes' Theorem

We are now ready to state the general theorems

Theorem 1.24 (Law of Total Probability)

Suppose $\{A_1, A_2, \dots, A_k\}$ is a partition for the sample space and B is any event, then

$$\begin{aligned} P(B) &= \sum_{j=1}^k P(B \cap A_j) \\ &= \sum_{j=1}^k P(A_j)P(B | A_j) \end{aligned}$$

The last equation in the theorem is just the application of the multiplication principle.

Theorem 1.25 (Bayes' Theorem)

Suppose $\{A_1, A_2, \dots, A_k\}$ is a partition for the sample space and B is any event, then

$$P(A_i | B) = \frac{P(A_i)P(B | A_i)}{\sum_{j=1}^k P(A_j)P(B | A_j)}$$

Proof. Using the fact that $\{A_1, A_2, \dots, A_k\}$ is a partition and the law of total probability we have:

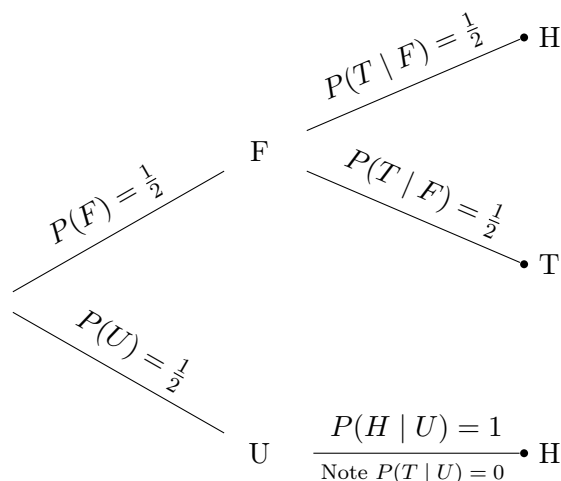
$$\begin{aligned} P(A_i | B) &= \frac{P(A_i \cap B)}{P(B)} \\ &= \frac{P(A_i \cap B)}{\sum_{j=1}^k P(B \cap A_j)} \\ &= \frac{P(A_i)P(B | A_i)}{\sum_{j=1}^k P(A_j)P(B | A_j)} \end{aligned}$$

□

Example 1.4.1

A bag contains two coins, one is a fair coin denoted as F , and the other is a two headed coin denoted as U .

We randomly choose a coin from the bag and toss it once, this can be described by the following tree diagram.



The sample space $\mathcal{S} = \{FH, FT, UH\}$, which is $F \cap H, F \cap T, U \cap H$ respectively.

We can calculate the probability that the tossed coin lands heads as follows

$$\begin{aligned} P(H) &= P(F \cap H) + P(U \cap H) \\ &= P(H | F) \cdot P(F) + P(H | U) \cdot P(U) \\ &= \frac{1}{2} \cdot \frac{1}{2} + 1 \cdot \frac{1}{2} = \frac{1}{4} + \frac{1}{2} = \frac{3}{4} \end{aligned}$$

Observing that $P(H | F) = \frac{1}{2} \neq \frac{3}{4} = P(H)$, helps us conclude that observing a H on the coin toss and choosing the fair coin F are two events that are not **not** independent of each other.

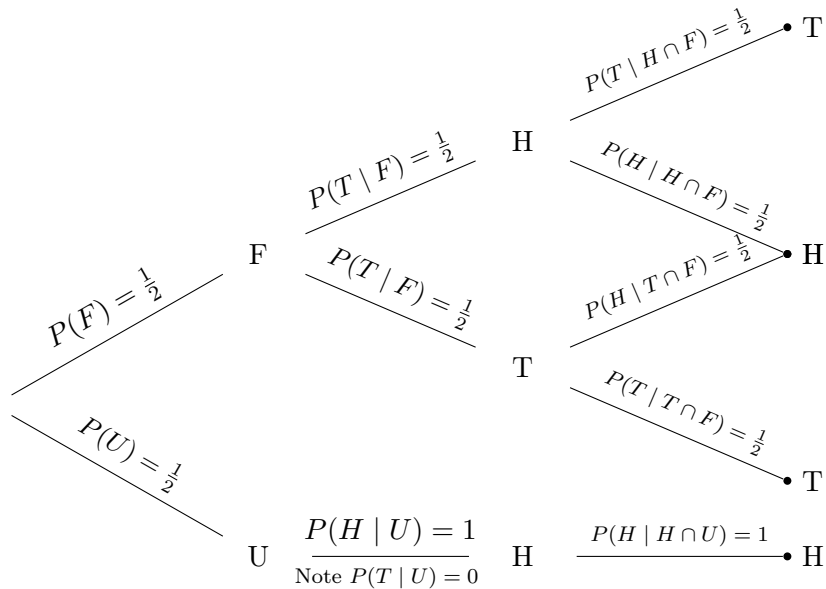
We can also calculate the conditional probability

$$\begin{aligned} P(F|H) &= \frac{P(F \cap H)}{P(H)} \\ &= \frac{P(H | F) \cdot P(F)}{P(H)} \\ &= \frac{\frac{1}{2} \cdot \frac{1}{2}}{\frac{3}{4}} = \frac{1}{4} \cdot \frac{4}{3} = \frac{1}{3} \end{aligned}$$

Therefore, $P(F) = \frac{1}{2} > \frac{1}{3} = P(F|H)$. This makes intuitive sense, as $P(F) = P(\text{tossed coin is a fair coin})$ will be lowered after we observe the H outcome.

Example 1.4.2

Suppose we now randomly choose a coin from the bag containing a fair coin and a two headed coin, and toss it two times independently. This experiment can be visualized in the following tree diagram:



The sample space is $\mathcal{S} = \{FHT, FHH, FTH, FTT, UHH\}$.

We can calculate the probabilities of each of the simple events by taking products along the path to the given outcome, for example:

$$\begin{aligned} P(HHF) &= P(H \cap (H \cap F)) \\ &= P(H | H \cap F) \cdot P(H \cap F) \\ &= P(H | H \cap F) \cdot P(H | F) \cdot P(F) \\ &= \text{Product along path to } HHF \end{aligned}$$

Suppose we want to calculate $P(HH) = P(\text{the tossed coin lands heads both times})$, we do it as follows:

$$\begin{aligned} P(HH) &= P(FHH) + P(UHH) \\ &= P(H | HF) \cdot P(HF) + P(H | HU) \cdot P(HU) \\ &= \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} + \frac{1}{2} \cdot 1 \cdot 1 \\ &= \frac{1}{8} + \frac{1}{2} = \frac{5}{8} \end{aligned}$$

We can also calculate the conditional probability

$$\begin{aligned}
 P(HH \mid F) &= P(\text{observed two consecutive heads given that a fair coin was tossed}) \\
 &= \frac{P(HHF)}{P(F)} \\
 &= \frac{\frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2}}{\frac{1}{2}} \\
 &= \frac{1}{4}
 \end{aligned}$$

Therefore, $P(HH) = \frac{5}{8} > \frac{1}{4} = P(HH \mid F)$, the information that we tossed a fair coin affected the probability of getting two consecutive heads - this information results in a lower probability.

We could also consider the probability

$$\begin{aligned}
 P(F \mid HH) &= P(\text{we tossed a fair coin given that } HH \text{ was observed}) \\
 &= \frac{P(FHH)}{P(HH)} = \frac{P(HH \mid F) \cdot P(F)}{P(HH)} \\
 &= \frac{\frac{1}{4} \cdot \frac{1}{2}}{\frac{5}{8}} = \frac{1}{8} \cdot \frac{8}{5} \\
 &= \frac{1}{5}
 \end{aligned}$$

Therefore, $P(F \mid HH) = \frac{1}{5} < \frac{1}{3} = P(F \mid H) < \frac{1}{2} = P(F)$, which aligns with the intuition that as more heads are observed, our confidence that a fair coin was tossed must decrease.

Example 1.4.3 (Disease Testing)

Suppose we consider a disease in a certain population, and

$$D := \{\text{randomly chosen individual has the disease}\}$$

so that

$$D^c := \{\text{randomly chosen individual does not have the disease}\}$$

Observe that

$$D \cup D^c := \{\text{entire population}\} = \mathcal{S}$$

Additionally, we are given a test, which when administered will determine if the person has the disease or not. Suppose

$$\text{Pos} := \{\text{test result is positive, i.e. test claims you have the disease}\}$$

$$\text{Neg} := \{\text{test result is negative, i.e. test claims you do not have the disease}\}$$

If everyone in the population was administered the test, we can observe that

$$\text{Pos} \cup \text{Neg} := \{\text{entire population}\} = \mathcal{S}$$

We now consider the following probabilities:

$$\begin{aligned}
 P(D) &:= P(\text{a randomly chosen individual from the population has the disease}) \\
 &:= p \in (0, 1)
 \end{aligned}$$

this is sometimes called the prevalence of the disease. Additionally, we are given

$$P(\text{test correctly identifies the disease}) := q \in (0, 1)$$

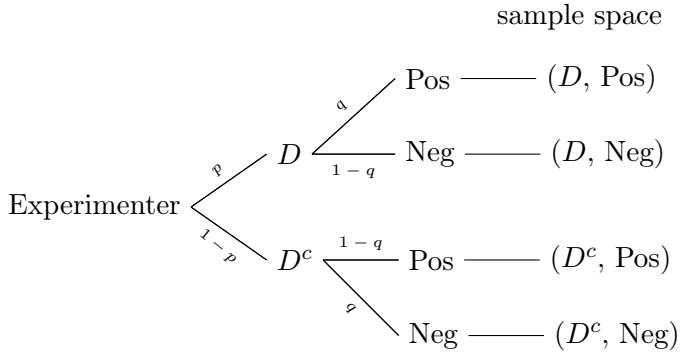
This is sometimes called the accuracy of the test. Using this we can calculate the following conditional probabilities:

$$\begin{aligned}
 P(\text{Pos} \mid D) &= P(\text{Neg} \mid D^c) = q \\
 P(\text{Pos} \mid D^c) &= P(\text{Neg} \mid D) = 1 - q
 \end{aligned}$$

Depending on the information that is available, we can consider two differently ordered experiments.

- Step 1 := Check if you have the disease
- Step 2 := What is the test result?

The tree that we get in this sequence is called the “Experimenter’s Tree”, as the experimenter testing the accuracy of the test needs to know the disease-status of the individual on whom the test is administered. This tree can be visualized as follows:



From the user’s (of the test) point of view, we have the following steps

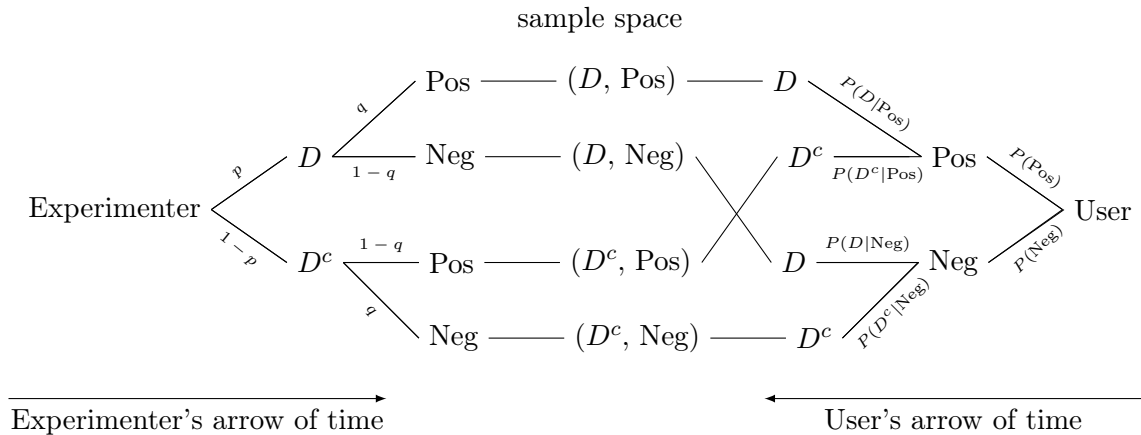
Step 1 := administer the test

Step 2 := Do I have the disease?

We can see that the user’s order of operation is exactly reverse of the experimenter.

The user of the test is aware of the test result, and based on that information wants to know their disease status.

We can put all of this information in the following two tree diagrams.



In this setting, the accuracy of the test relates to $P(\text{Pos}|D^c)$ and $P(\text{Neg}|D)$ being small, as they correspond to the probabilities of incorrectly identifying the disease using the test. On the other hand, $P(\text{Pos}|D)$ and $P(\text{Neg}|D^c)$ should be large as they correspond to the probabilities of correctly identifying the disease using the test.

For the observer, one expects $P(D|\text{Pos})$ and $P(D^c|\text{Neg})$ to be large, and $P(D^c|\text{Pos})$ and $P(D|\text{Neg})$ to be small.

Observe that

1. The trees for the experimenter and the observer result in the same sample space.
2. Using the Law of Total Probability and the Bayes’ theorem, we can calculate all the probabilities in the user’s tree if the all the probabilities in the experimenter’s tree are known.

We calculate as follows:

$$P(\text{Pos}) = P(D, \text{Pos}) + P(D^c, \text{Pos}) = p \cdot q + (1 - p) \cdot (1 - q),$$

Therefore,

$$P(D|\text{Pos}) = \frac{P(D, \text{Pos})}{P(\text{Pos})} = \frac{pq}{pq + (1 - p)(1 - q)},$$

$$P(D^c|\text{Pos}) = \frac{P(D^c, \text{Pos})}{P(\text{Pos})} = \frac{(1 - p)(1 - q)}{pq + (1 - p)(1 - q)}.$$

Similarly, we can calculate

$$P(\text{Neg}) = p \cdot (1 - q) + q \cdot (1 - p),$$

$$P(D^c|\text{Neg}) = \frac{q(1-p)}{q(1-p) + p(1-q)}, \quad P(D|\text{Neg}) = \frac{p(1-q)}{q(1-p) + p(1-q)}.$$

This finishes the calculation of all the probabilities in the user's tree.

As mentioned before, we would like to have

$$P(D|\text{Pos}) = \frac{pq}{pq + (1-p)(1-q)}, \quad \text{and} \quad P(D^c|\text{Neg}) = \frac{(1-p)q}{(1-p)q + p(1-q)}$$

to be large.

We can consider two special cases:

1. Suppose the disease is highly prevalent, for instance, $P(D) = p = 0.9999$. That is, 9999 out of 10000 people on average have the disease. Furthermore, let $P(\text{test correctly identifies the disease}) = q = 0.9999$. That is, the test is on average correct 9999 times out of 10000 times. For the test to be useful, we want to have both $P(D|\text{Pos})$ and $P(D^c|\text{Neg})$ to be large. In this case, we compute their numerical values

$$P(D|\text{Pos}) = \frac{pq}{pq + (1-p)(1-q)} = \frac{0.9999 \cdot 0.9999}{0.9999 \cdot 0.9999 + 0.0001 \cdot 0.0001} \approx 1.$$

However, we see that

$$P(D^c|\text{Neg}) = \frac{(1-p)q}{(1-p)q + p(1-q)} = \frac{0.0001 \cdot 0.9999}{0.0001 \cdot 0.9999 + 0.9999 \cdot 0.0001} = 0.5,$$

from which we see that a negative test result will correctly identify the test with probability 0.5, perhaps the user is better off tossing a fair coin (instead of buying the test). The test is not very useful in this sense.

2. On the other hand, if we have a rare disease, say for instance $P(D) = p = 0.0001$. Suppose that the test accuracy is $P(\text{test correctly identifies disease}) = 0.9999 = q$. For the test to be useful, we again want to have both $P(D|\text{Pos})$ and $P(D^c|\text{Neg})$ to be large, but we again see that

$$P(D|\text{Pos}) = \frac{pq}{pq + (1-p)(1-q)} = \frac{0.0001 \cdot 0.9999}{0.0001 \cdot 0.9999 + 0.9999 \cdot 0.0001} = 0.5,$$

which is again equivalent to tossing a fair coin toss.

Probability and Counting

Suppose we are in the setting where the sample space $\mathcal{S}$ satisfies the following two conditions:

1. $\mathcal{S}$ is a finite set. That is, $\mathcal{S}$ is the sample space of an experiment with finitely many outcomes.
2. Every outcome in $\mathcal{S}$ is equally likely. That is, if $\mathcal{S} = \{s_1, s_2, \dots, s_n\}$ then $P(\{s_i\}) = \frac{1}{n}$ for all $i = 1, 2, 3, \dots, n$

Let $A \subset \mathcal{S}$ be an event. Since every outcome in $\mathcal{S}$ is equally likely

$$\begin{aligned} P(A) &= \text{Probability that } A \text{ occurs} \\ &= \sum_{s \in A} P(\{s\}) \\ &= \sum_{s \in A} \frac{1}{n} \\ &= \frac{n(A)}{n(\mathcal{S})} \end{aligned}$$

where $n(A)$ is the number of objects in A .

Therefore, when the sample space is finite and all outcomes in the sample space are equally likely, calculating the probability of an event A is same as counting outcomes in A . In this scenario, we only need to be able to count!

We state the first important theorem of counting.

Theorem 1.26 (Fundamental Theorem of Counting)

Suppose there are k tasks: $T_1, T_2, \dots, T_k$ that can be performed in $n_1, n_2, \dots, n_k$ ways respectively. Let T be the task of performing $T_1, T_2, \dots, T_k$ sequentially. Then the total number of ways to perform the task T is

$$n_1 \times n_2 \times \dots \times n_k$$

Proof. We prove this theorem using the principle of mathematical induction applied to the number of tasks. We can structure the task T as a tree, with level i in the tree corresponding to the i th task T_i . $\square$

Example 1.5.1

Add examples of counting.

1.5.1 Choosing from n distinct objects

We will often want to count the number of ways of selecting k objects from a set of n distinct objects.

Example 1.5.2

Suppose we want to count the total number of ways one can choose 4 digits from the set of 10 digits $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$.

The answer to this we need to understand how the digits are selected. Do we place the digit back into the set before the subsequent choice (with/without replacement)? Does the order in which the 4 digits are selected matter? This leads to the following 4 possibilities:

	Without Replacement	With Replacement
Ordered	(1,2,4,5) different from (1,5,2,4) (1,1,2,5) is not possible	(1,2,4,5) different from (1,5,2,4) (1,1,2,5) is possible
Unordered	(1,2,3,4) is same as (4,3,2,1) (1,1,2,5) not possible	(1,2,3,4) is same as (4,3,2,1) (1,1,2,4) is possible

We use the Fundamental Theorem of Counting to count the number of ways we can choose k objects from n distinct objects as follows

1. **Order matters and without replacement**

Let T be the task of choosing k objects from n distinct objects, where order matters and without replacement. The task T is the same as filling k ordered spaces, and can be broken into a sequence of sub-tasks

$$T : T_1 \rightarrow T_2 \rightarrow T_3 \rightarrow \cdots \rightarrow T_k$$

where T_i fills the i th spot in the k ordered spaces.

Using the fundamental theorem of counting the total number of ways of doing the task T is

$$n \times (n - 1) \times (n - 2) \times (n - 3) \times \cdots \times (n - k + 1)$$

This number is called “ n permute k ” and is denoted by

$${}_nP_k = \frac{n!}{(n - k)!}$$

(2) **Unordered Without Replacement**

The first step is to calculate the ordered arrangements, which is

$${}_nP_k = \frac{n!}{(n - k)!}$$

Note that each ordered arrangement can be rearranged $k!$ times since we select k objects in total. Therefore, we need to get rid of repeats by dividing by $k!$, which is

$$\frac{{}_nP_k}{k!} = \frac{n!}{(n - k)!k!}$$

So we got

$${}_nC_k = \binom{n}{k} := \frac{n!}{(n - k)!k!}$$

(3) **With Replacement**

The number of ways if choose k objects where order does matter and with replacement in n different things is simply

$$\begin{aligned} & n \times n \times n \times \cdots \times n \\ &= n^k \end{aligned}$$

(4) **With replacement and order does not matter**

We want number of unordered arrangements of size k from n objects with replacement. This can be reformulated as the number of ways to choose k “walls” from $(n + k - 1)$ choices, i.e

$${}_{n+k-1}C_k = \binom{n + k - 1}{k} \text{ ways}$$

Example 1.5.3

We set $n = 10, k = 3$, where $S = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$.

Then we will have $n + k - 1 = 12$ walls

--	--	--	--	--	--	--	--	--	--	--	--

Then, for the event $(1, 1, 2)$, we will have (note that X means the wall that we choose)

1	X	X	2	X	3	4	5	6	7	8	9
---	---	---	---	---	---	---	---	---	---	---	---

For the event $(0, 0, 7)$, we will have

X	X	1	2	3	4	5	6	7	X	8	9
---	---	---	---	---	---	---	---	---	---	---	---

for the event $(5, 9, 7)$, we will have

1	2	3	4	5	X	6	7	X	8	9	X
---	---	---	---	---	---	---	---	---	---	---	---

The **number of possible arrangement of size k from n objects 1. Without replacement and order matters** for each of the four possibilities is listed below

	Without Replacement	With Replacement
Ordered	${}^nP_k = \frac{n!}{n-k}$ is also called “n permute k”	n^k
Unordered	${}^nC_k = \frac{n!}{(n-k)!k!}$ is also called “n choose k”	${}^{n+k-1}C_k$

Example 1.5.4

Consider choosing 3 digits from the set $\{1, 2, 3, 4, 5\}$.

	order matters	order does not matter
with replacement	$(1,2,2)$ is allowed but is different from $(2,1,2)$	$(1,2,2)$ is allowed and is the same as $(2,1,2)$
without replacement	$(1,2,2)$ is not allowed, $(1,2,3)$ is different from $(1,3,2)$	$(1,2,2)$ is not allowed, $(1,2,3)$ is the same as $(1,3,2)$

In this case, $n = 5$ and $k = 3$, and we can compute as follows

$${}^nP_k = {}^5P_3 = \frac{5!}{(5-3)!} = 5 \cdot 4 \cdot 3 = 60, \qquad {}^nC_k = {}^5C_3 = \frac{5!}{(5-3)!3!} = \frac{60}{3!} = 10,$$

$$n^k = 5^3 = 125, \qquad {}^{n+k-1}C_{n-1} = {}^7C_4 = \frac{7!}{4!3!} = 35.$$

	order matters	order does not matter
with replacement	125	35
without replacement	60	10

2 | Random Variables

Random Variables: Examples and Definitions

In practical applications we are typically interested in numerical features associated to an outcome of an experiment.

Example 2.1.1

Suppose we toss a two sided coin three times. The sample space of this experiment is given by:

$$\mathcal{S} := \left\{ \begin{array}{cccc} HHH, & HHT, & HTT, & TTT \\ & HTH, & THT, & \\ & THH, & TTH & \end{array} \right\}$$

We can consider numerical features of an outcome in the sample space, for example:

1. How many heads in the outcome?
2. How many tails in the outcome?
3. Are there more heads than tails?
4. Is the first toss in the outcome a head?
5. Is the last toss a tails?
6. How many tails till the first head?

Example 2.1.2

Suppose we randomly sample a student from the campus. The sample space for this experiment is given by:

$$\mathcal{S} := \{\text{All students on campus}\}$$

We can consider numerical features of an outcome (a student from the campus) in the sample space, for example:

1. What is the height of the chosen student?
2. What is the average daily commute time for the student?
3. The number of credits the student is enrolled in?
4. The number of electronic gadgets the student owns?
5. Average number of people the student interacts with on a given day?
6. Is the student an athlete?

These questions provide answers to very specific numeric/quantitative features of the student.

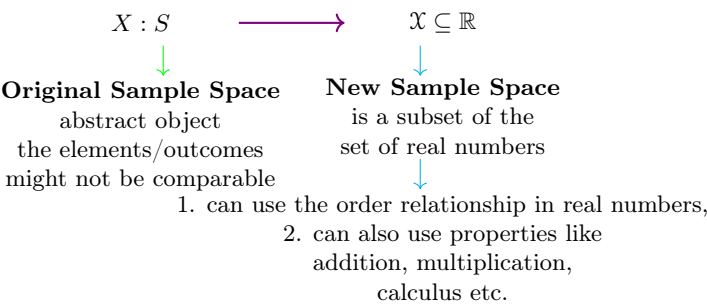
Definition 2.1 (Random Variable)

A random variable X is a real valued function on the sample space $\mathcal{S}$. That is,

$$\begin{aligned} X : \mathcal{S} &\longrightarrow \mathbb{R} \\ \omega &\mapsto X(\omega) \end{aligned}$$

The range of X , that is the set of all possible values that X can take will be denoted as $\mathcal{X}$

Intuitively, a random variable measures a specific quantitative feature of the sample space outcome. Here is a schematic to consider:



Definition 2.2 (Events Associated to a Random Variable)

Given a random variable

$$X : \mathcal{S} \longrightarrow \mathbb{R}$$

we can use the order relationship of $\mathbb{R}$ to consider the following “interesting events” for X :

$$\begin{aligned} \{a \leq X \leq b\} &:= X \text{ takes values in the interval } [a, b] \\ \{a \leq X\} &:= X \text{ takes values in the interval } [a, \infty) \\ \{X \leq b\} &:= X \text{ takes values in the interval } [-\infty, b] \end{aligned}$$

We can also allow for strict inequalities, countable unions, complements, and countable intersections.

We will assume that we are able to calculate the probabilities associated to these events, and these will be denoted by $P(a \leq X \leq b)$, $P(a \leq X)$, $P(X \leq b)$, etc

Example 2.1.3

Suppose the sample space $\mathcal{S}$ is all students on campus.

For $\omega \in \mathcal{S}$, we define:

$$X(\omega) := \text{height of } \omega \text{ (in inches)}$$

What are the values of X ?

$$\mathcal{X} := (0, \infty)$$

Some “interesting events” corresponding to X can be:

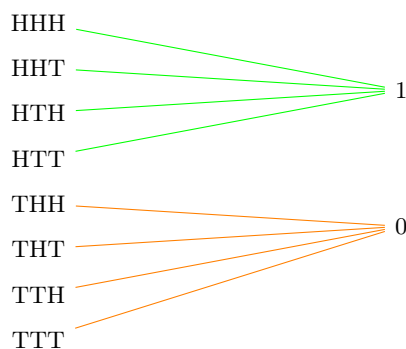
$$\begin{aligned} \{60 \leq X \leq 65\} &:= \{\omega \in \mathcal{S} \mid X(\omega) \in [60, 65]\} \\ &= \{\text{set of all students whose height is between 60 and 65 inches.}\} \end{aligned}$$

$$\begin{aligned} \{X \geq 72\} &:= \{\omega \in \mathcal{S} \mid X(\omega) \in [72, \infty)\} \\ &= \{\text{all students whose height is greater than or equal to 72 inches}\} \end{aligned}$$

Example 2.1.4

Suppose we toss a coin three times, and we want to know if the first toss was a head?

$$X(\omega) = \begin{cases} 0 & \text{if first toss in } \omega \text{ is not H} \\ 1 & \text{otherwise} \end{cases}$$



Another way of looking at it:

$$\begin{aligned} \{X = 0\} &:= X^{-1}(0) = \{\text{all sample space outcomes on which } X \text{ takes value } 0\} \\ &= \{\text{THH, THT, TTH, TTT}\} \end{aligned}$$

Similarly,

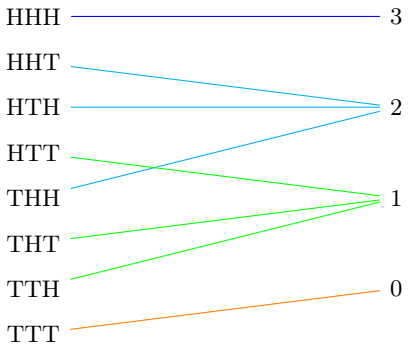
$$\{X = 1\} := X^{-1}(1) = \{\text{HHH, HTH, HHT, HTT}\}$$

$$S := \{\{\text{HHH, HTH, HHT, HTT}\} \mapsto \{1\}, \{\text{THH, THT, TTH, TTT}\} \mapsto \{0\}\}$$

Example 2.1.5

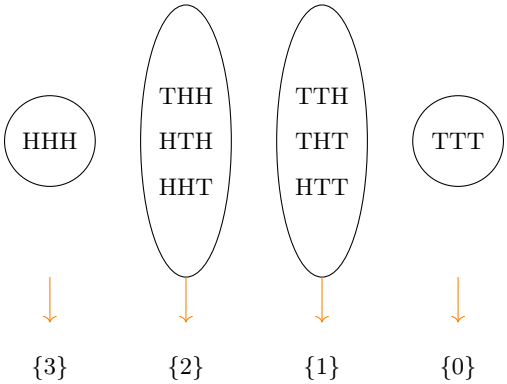
Suppose we tossed a coin three times, and we were interested in counting the number of heads in three tosses. We can work through this as follows:

$$X(\omega) := \# \text{ of heads in three tosses.}$$



$$\begin{aligned} \{X = 0\} &= \{TTT\} \\ \{X = 1\} &= \{HTT, THT, TTH\} \\ \{X = 2\} &= \{HHT, HTH, THH\} \\ \{X = 3\} &= \{HHH\} \end{aligned}$$

$$\mathcal{S} = \left\{ \begin{array}{ll} \{HHH\} & \mapsto 3 \\ \{HHT, HTH, THH\} & \mapsto 2 \\ \{HTT, THT, TTH\} & \mapsto 1 \\ \{TTT\} & \mapsto 0 \end{array} \right\}$$



Definition 2.3 (Cumulative Distribution Function)
The cumulative distribution function associated to a random variable

$$X : \mathcal{S} \longrightarrow \mathbb{R}$$

is defined as

$$\begin{aligned} F_X : \mathbb{R} &\longrightarrow \mathbb{R} \\ x &\mapsto P(X \leq x) \end{aligned}$$

that is

$$F_X(x) := P(X \text{ takes values less than or equal to } x)$$

Example 2.1.6
Suppose we toss a coin with $P(H) = p \in (0, 1)$ once. Let X be the random variable that tracks the outcome of the coin toss, that is

$$X(\omega) = \begin{cases} 1 & \text{if } \omega = H \\ 0 & \text{if } \omega = T \end{cases}$$

We now calculate the cumulative distribution function for X .

1. If $x < 0$:

$$\begin{aligned} \{X \leq x\} &= \{\text{event that is impossible}\} \\ &= \emptyset \end{aligned}$$

Therefore we must have:

$$\begin{aligned} F_X(x) &= P(X \leq x) \\ &= P(\emptyset) \\ &= 0 \end{aligned}$$

for all $x < 0$.

2. If $x = 0$,

$$\begin{aligned} F_X(0) &= P(X \leq 0) \\ &= \underbrace{P(X < 0)}_0 + \underbrace{P(X = 0)}_{P(T)=(1-p)} \end{aligned}$$

Therefore,

$$F_X(0) = 1 - p$$

3. If $x \in (0, 1)$,

$$\begin{aligned} F_X(x) &= P(X \leq x) \\ &= \underbrace{P(X \leq 0)}_0 + \underbrace{P(X = 0)}_{1-p} + \underbrace{P(0 < X < x < 1)}_0 \end{aligned}$$

Therefore,

$$F_X(x) = (1 - p) \quad \text{for all } x \in (0, 1)$$

4. If $x = 1$,

$$\begin{aligned} F_X(1) &= P(X \leq 1) \\ &= \underbrace{P(X < 1)}_{1-p} + \underbrace{P(X = 1)}_{P(H)=p} \end{aligned}$$

Therefore,

$$F_X(1) = 1$$

5. If $x > 1$,

$$\begin{aligned} F_X(x) &= P(X \leq x) \\ &= \underbrace{P(X \leq 1)}_1 + \underbrace{P(X > 1)}_0 \end{aligned}$$

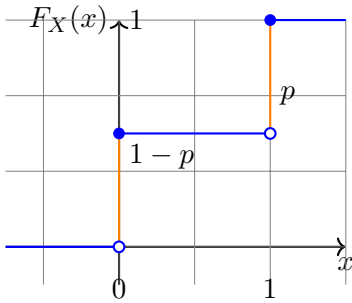
Therefore,

$$F_X(x) = 1 \quad \text{for all } x > 1$$

The cdf F_X can be written as follows:

$$F_X(x) = \begin{cases} 0 & \text{if } x < 0 \\ (1 - p) & \text{if } x \in [0, 1) \\ 1 & \text{if } x \geq 1 \end{cases}$$

The plot of $F_X(x)$:



Example 2.1.7

Suppose we toss a fair coin ($P(H) = \frac{1}{2}$) three times, and let

$$X(\omega) := \text{number of heads in } \omega$$

Values that X can take are

$$\mathcal{X} := \{0, 1, 2, 3\}$$

We now calculate the cdf of X .

1. If $x < 0$:

$$\begin{aligned} F_X(x) &= P(X \leq x) \\ &= 0 \end{aligned}$$

Therefore

$$F_X(x) = 0 \quad \text{for } x < 0$$

2. If $x = 0$:

$$\begin{aligned} F_X(x) &= P(X \leq 0) \\ &= \underbrace{P(X < 0)}_0 + \underbrace{P(X = 0)}_{\substack{P(\text{zero heads in three tosses}) \\ P(TTT) = (\frac{1}{2})^3}} \end{aligned}$$

Therefore

$$F_X(0) = \frac{1}{8}$$

3. If $x \in (0, 1)$:

$$\begin{aligned} F_X(x) &= P(X \leq x) \\ &= \underbrace{P(X \leq 0)}_{\frac{1}{8}} + \underbrace{P(0 < X \leq x)}_0 \end{aligned}$$

Therefore

$$F_X(x) = \frac{1}{8} \quad \text{for } x \in (0, 1)$$

4. If $x = 1$:

$$\begin{aligned} F_X(1) &= P(X \leq 1) \\ &= \underbrace{P(X \leq 0)}_{\frac{1}{8}} + \underbrace{P(0 < X < 1)}_0 + \underbrace{P(X = 1)}_{\substack{P(\text{exactly two heads in three tosses}) \\ P(\{HTT, THT, TTH\}) = \frac{3}{8}}} \\ &= \frac{1}{8} + P(HTT, THT, TTH) = \frac{1}{8} + 3 \left(\frac{1}{2} \right)^3 = \frac{1}{8} + \frac{3}{8} \end{aligned}$$

Therefore

$$F_X(1) = \frac{4}{8} = \frac{1}{2}$$

5. If $x \in (1, 2)$:

$$\begin{aligned} F_X(x) &= P(X \leq x) \\ &= \underbrace{P(X \leq 1)}_{\frac{1}{2}} + \underbrace{P(1 < X \leq x)}_0 \end{aligned}$$

Therefore

$$F_X(x) = \frac{1}{2} \quad \text{for } x \in (1, 2)$$

6. If $x = 2$:

$$\begin{aligned} F_X(2) &= P(X \leq 2) \\ &= \underbrace{P(X \leq 1)}_{\frac{1}{2}} + \underbrace{P(1 < X < 2)}_0 + \underbrace{P(X = 2)}_{\substack{P(\text{exactly two heads in three tosses}) \\ P(\{HHT, HTH, THH\}) = \frac{3}{8}}} \\ &= \frac{1}{2} + P(HHT, HTH, THH) = \frac{1}{2} + 3 \left(\frac{1}{2} \right)^3 = \frac{1}{2} + \frac{3}{8} \end{aligned}$$

Therefore

$$F_X(2) = \frac{7}{8}$$

7. If $x \in (2, 3)$:

$$\begin{aligned} F_X(x) &= P(X \leq x) \\ &= \underbrace{P(X \leq 2)}_{\frac{7}{8}} + \underbrace{P(2 < X < 3)}_0 \end{aligned}$$

Therefore

$$F_X(x) = \frac{7}{8} \quad \text{for } x \in (2, 3)$$

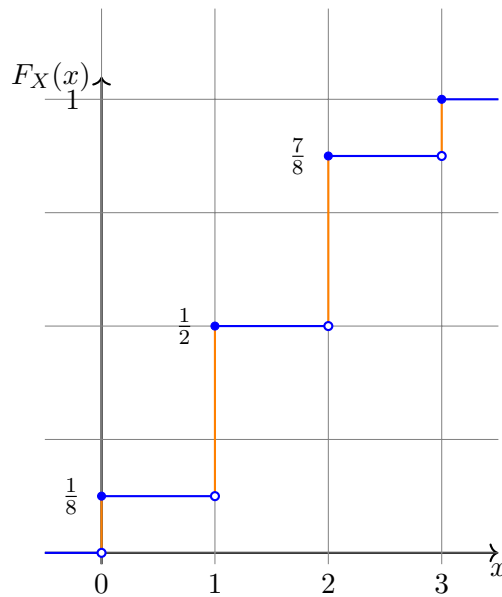
8. If $x \geq 3$:

$$\begin{aligned} F_X(x) &= P(X \leq x) \\ &= \underbrace{P(X \leq 2)}_{\frac{7}{8}} + \underbrace{P(2 < X \leq 3)}_{P(X=3)} \\ &= \frac{7}{8} + P(HHH) = \frac{7}{8} + \left(\frac{1}{2}\right)^3 = \frac{7}{8} + \frac{1}{8} = 1 \end{aligned}$$

We can consolidate all the information about F_X as follows:

$$F_X(x) = \begin{cases} 0 & \text{if } x < 0 \\ \frac{1}{8} & \text{if } x \in [0, 1) \\ \frac{1}{2} & \text{if } x \in [1, 2) \\ \frac{7}{8} & \text{if } x \in [2, 3) \\ 1 & \text{if } x \geq 3 \end{cases}$$

The plot of F_X is given below



It is possible to classify all functions that can arise as cumulative distribution functions of random variable as follows:

Theorem 2.4 (Classification of CDFs)

A function $F(x)$ is a cumulative density function if and only if the following three conditions are satisfied:

1. $\lim_{x \rightarrow -\infty} F(x) = 0$, $\lim_{x \rightarrow \infty} F(x) = 1$
2. $F(x)$ is a non decreasing function
3. $F(x)$ is right continuous. i.e $\lim_{x \rightarrow x_o^+} F(x) = F(x_o)$, $\forall x_o \in \mathbb{R}$

Discrete Random Variables

Definition 2.5 (Discrete Random Variable)

We say a random variable $X : \mathcal{S} \rightarrow \mathbb{R}$ is **discrete** if the cumulative distribution function of X , F_X is a step function.

Definition 2.6 (Probability Mass Function)

If X is a discrete random variable, the probability mass function associated to X is defined as

$$\begin{aligned} p_X : \mathbb{R} &\longrightarrow \mathbb{R} \\ x &\mapsto P(X = x) \end{aligned}$$

that is,

$$p_X(x) := P(X \text{ takes the value } x)$$

It is easy to observe that the only points where p_X can be non-zero are the values, $\mathcal{X}$, that X can take.

Definition 2.7 (Parameters for Discrete Variables)

Suppose X is a discrete random variable with probability mass function p_X and taking values in the set $\mathcal{X}$. We can consider the following parameters associated to X

1. **Expected Value:**

$$E(X) := \mu_X = \sum_{x \in \mathcal{X}} x p_X(x)$$

2. **Statisticians Unconscious Law:** Given a function $g : \mathbb{R} \longrightarrow \mathbb{R}$, then

$$E(g(X)) := \sum_{x \in \mathcal{X}} g(x) p_X(x)$$

3. **Variance:**

$$\begin{aligned} V(X) := \sigma_X^2 &= E((X - \mu_X)^2) \\ &= \sum_{x \in \mathcal{X}} (x - \mu_X)^2 p_X(x) \end{aligned}$$

Continuous Random Variables

Definition 2.8 (Continuous Random Variable)

We say a random variable $X : \mathcal{S} \longrightarrow \mathbb{R}$ is **continuous** if the cumulative distribution function of X , F_X is a continuous function.

Definition 2.9 (Probability Density Function)

We say that the function $f_X : \mathbb{R} \longrightarrow \mathbb{R}$ is the probability density function of the continuous random variable X (with cdf F_X) if f_X satisfies the following:

$$F_X(x) = \int_{-\infty}^x f_X(t) dt$$

Definition 2.10 (Parameters for Continuous Variables)

Suppose X is a continuous random variable with probability density function f_X . We can consider the following parameters associated to X

1. **Expected Value:**

$$E(X) := \mu_X = \int_{-\infty}^{\infty} x f_X(x) dx$$

2. **Statisticians Unconscious Law:** Given a function $g : \mathbb{R} \longrightarrow \mathbb{R}$, then

$$E(g(X)) := \int_{-\infty}^{\infty} g(x) f_X(x) dx$$

3. **Variance:**

$$\begin{aligned} V(X) := \sigma_X^2 &= E((X - \mu_X)^2) \\ &= \int_{-\infty}^{\infty} (x - \mu_X)^2 f_X(x) dx \end{aligned}$$

Definition 2.11 (Identically Distributed Random Variables)

We say two random variables X and Y are identically distributed if the corresponding cdfs F_X and F_Y are pointwise equal (almost) everywhere.

Example 2.3.1

Suppose we toss a fair coin five times, and X tracks the number of heads in five tosses, and Y tracks the number of tails in five tosses. Then, it is possible to show that even though X and Y are not equal as functions on the sample space, the cdfs F_X and F_Y are exactly the same. As a result X and Y will be identically distributed.

Would we be able to conclude the same if we were to toss an unfair coin?

3 | Discrete Random Variable

In this chapter we will go over some standard examples of discrete random variables, some of these distributions are so important they have their own specific names.

Recall that X is said to be a discrete random variables if the cumulative distribution function F_X is a step function. This implies that the set of values of X , $\mathcal{X}$, will be a discrete subset of $\mathbb{R}$ (a set with no limit points).

Uniform Discrete Random Variable

Definition 3.1

The random variable X has **uniform discrete distribution** with parameter $N \in \mathbb{N}$ if it is supported on

$$\mathcal{X} := \{1, 2, 3, \dots, N\}$$

and probability mass function of X is given by:

$$p_X(x) = P(X = x) = \frac{1}{N} \quad \text{for all } x \in \mathcal{X}$$

Theorem 3.2 (Parameters associated to the Uniform Distribution)

Suppose X has the Uniform distribution on $\{1, 2, 3, \dots, N\}$ then,

1. The expected value of X is given by:

$$E(X) = \frac{N + 1}{2}$$

2. The variance of X is given by:

$$V(X) = \frac{(N + 1)(N - 1)}{12}$$

Bernoulli Distribution

The Bernoulli distribution simulates a “trial” whose outcome is either a success or a failure.

Definition 3.3 (Bernoulli Trial)

An experiment is called a Bernoulli trial if it has exactly two outcomes, a success or a failure.

Definition 3.4 (Bernoulli Distribution)

We say that X has the Bernoulli distribution with parameter $p \in (0, 1)$, typically denoted by $X \sim \text{Bernoulli}(p)$ if X takes values $\mathcal{X} := \{0, 1\}$ and the probability mass function of X is given as

$$p_X(x) = \begin{cases} 1 - p & \text{if } x = 0 \\ p & \text{if } x = 1 \end{cases}$$

The pmf can also be written as

$$p_X(x) = p^x(1 - p)^{(1-x)} \quad x \in \{0, 1\}$$

Events associated to the Bernoulli distribution are of the form:

1. $\{X = 0\}$: the trial was a failure.
2. $\{X = 1\}$: the trial was a success.

Theorem 3.5 (Parameters associated to the Bernoulli Distribution)

Suppose $X \sim \text{Bernoulli}(p)$ then,

1. The expected value of X is given by:

$$E(X) = p$$

2. The variance of X is given by:

$$V(X) = p(1 - p)$$

Examples of Bernoulli distribution in applications:

1. Suppose we sample a random student from campus and check if the height of the student is less than or equal to 60 inches.

This experiment can be modelled by $X \sim \text{Bernoulli}(p)$ distribution, where the event

$$\{X = 1\} := \{\text{a randomly chosen student has height} \leq 60 \text{ inches}\}$$

In this setting, $p_X(1) = P(X = 1) = p$, is the proportion of students on campus whose height is less than or equal to 60 inches.

2. Suppose we sample a random student from campus and check if they have taken a statistics course. This experiment can be modelled by $X \sim \text{Bernoulli}(p)$ distribution, where the event

$$\{X = 1\} := \{\text{a randomly chosen student has taken a statistics course}\}$$

In this setting, $p_X(0) = P(X = 0) = (1 - p)$, is the proportion of students on campus who have never taken a statistics class.

3. Suppose we choose a potato at a supermarket, and we put it into our cart if it has no blemishes. This experiment can be modelled by $X \sim \text{Bernoulli}(p)$ distribution, where the event

$$\{X = 1\} := \{\text{a randomly chosen potato has no blemishes}\}$$

In this setting, $p_X(1) = P(X = 1) = p$, is the proportion of potatoes without blemishes in the supermarket aisle.

Binomial Distribution

Suppose an experiment involves performing n independent Bernoulli(p) trials. An outcome for this experiment is a sequence of length n , with the i th entry in the sequence tracking if the i th trial was a success or a failure. In this setting, we might be interested in knowing the number of success in the outcome, and the Binomial random variable does exactly that.

Definition 3.6 (Binomial Distribution)

We say that X has the Binomial distribution with parameters

- $n \in \mathbb{N}$: the number of trials
- $p \in (0, 1)$: $P(\text{Success})$, the probability of a success.

typically denoted by $X \sim \text{Binomial}(n, p)$ if X takes values $\mathcal{X} := \{0, 1, 2, 3, \dots, n\}$ and the probability mass function of X is given as

$$p_X(x) = \binom{n}{x} p^x (1 - p)^{(n-x)} \quad x \in \{0, 1, 2, 3, \dots, n\}$$

Theorem 3.7 (Binomial Experiment)

Suppose $n \in \mathbb{N}$. The experiment involves performing n independent Bernoulli(p) trials. Let ω be an outcome for this experiment, and

$$X(\omega) := \text{number of successes in } \omega$$

Then,

$$X \sim \text{Binomial}(n, p)$$

Proof. The proof involves explicitly calculating the pmf of X and showing that it is the same as the pmf of the Binomial distribution with the given parameters. We leave this as an exercise for the reader. $\square$

Events associated to the Binomial distribution are of the form:

1. $\{X = k\}$: there are exactly k successes in n trials.
2. $\{a < X < b\}$: the number of successes in n trials are greater than a but less than b .
3. $\{a \leq X\}$: there are more than a successes in n trials.

Theorem 3.8 (Parameters associated to the Binomial Distribution)

Suppose $X \sim \text{Binomial}(n, p)$ then,

1. The expected value of X is given by:

$$E(X) = np$$

2. The variance of X is given by:

$$V(X) = np(1 - p)$$

Examples of Binomial distribution in applications:

1. Suppose we sample n students (with replacement) from campus and check if their height of the chosen student is less than or equal to 60 inches.

Since the sampling is with replacement, the result of a given trial is independent of the other trials. Therefore this experiment can be modelled by $X \sim \text{Binom}(n, p)$ distribution, where the event

$$\{X = k\} := \{\text{there are exactly } k \text{ students with height } \leq 60 \text{ inches among the } n \text{ chosen students}\}$$

In this setting, $p_X(k) = P(X = k)$, is probability of finding exactly k students with height ≤ 60 among n randomly chosen students.

Also, $P(X \geq k)$, is probability of finding at least k students with height ≤ 60 among n randomly chosen students.

2. Suppose we sample n students (with replacement) from campus and check if they have taken a statistics course.

This experiment can be modelled by $X \sim \text{Binom}(n, p)$ distribution, where the event

$$\{X = k\} := \{\text{there are exactly } k \text{ students who have taken a statistics class among the } n \text{ chosen students}\}$$

In this setting, $p_X(k) = P(X = k)$, is the probability that of finding exactly k students who have taken a statistics class among the n chosen students.

3. Suppose we choose n potatoes (with replacement, even though that does not really make sense), a potato is a success if it does not have any blemishes.

This experiment can be modelled by $X \sim \text{Binom}(n, p)$ distribution, where the event

$$\{X = k\} := \{\text{there are exactly } k \text{ good potatoes among } n \text{ randomly chosen potatoes}\}$$

In this setting, $p_X(k) = P(X = k)$, is the probability of finding exactly k potatoes without blemishes among n randomly chosen potatoes.

Note that we have be emphasizing that the sampling be with replacement, this ensures that trials are independent of each other.

In many applications, where the population is large and n is small (in comparison to the population), sampling without replacement can still result in “approximately” independent trials.

When we are sampling from a population without replacement, the random variable that tracks the number of successes actually has the “Hypergeometric Distribution”

Hypergeometric Distribution

Fix $N, M, n \in \mathbb{N}$. Suppose we have population of size N with exactly M successes in it. Suppose an experiment involves choosing n objects from the population (without replacement). Given ω an outcome for this experiment, let

$$X(\omega) := \text{number of successes in } \omega$$

, in this setting X is said to have the H/ypergeometric distribution.

Definition 3.9 (Hypergeometric Distribution)

We say that X has the Hypergeometric distribution with parameters

- $N \in \mathbb{N}$: population size
- $M \in \mathbb{N}$: number of success in the population.
- $n \in \mathbb{N}$: number individuals chosen from the population.

typically denoted by $X \sim \text{Hyper}(N, M, n)$ if X takes values $\mathcal{X} := \{0, 1, 2, 3, \dots, \min(n, M)\}$ and the probability mass function of X is given as

$$p_X(x) = \frac{\binom{N}{x} \binom{N-M}{n-x}}{\binom{N}{n}} \quad x \in \{0, 1, 2, 3, \dots, \min(n, M)\}$$

Theorem 3.10 (Parameters associated to the Hypergeometric Distribution)

Suppose $X \sim \text{Hyper}(N, M, n)$ then,

1. The expected value of X is given by:

$$E(X) = n \left(\frac{M}{N} \right)$$

2. The variance of X is given by:

$$V(X) = \left(\frac{N-n}{N-1} \right) \cdot n \cdot \left(\frac{M}{N} \right) \cdot \left(1 - \frac{M}{N} \right)$$

Geometric Distribution

The geometric random variable models phenomena that involve “waiting”.

Definition 3.11 (Geometric Distribution)

We say that X has the Geometric distribution with parameters

- $p \in (0, 1)$: $P(\text{Success})$, the probability of a success.

typically denoted by $X \sim \text{Geom}(p)$ if X takes values $\mathcal{X} := \{1, 2, 3, \dots\} = \mathbb{N}$ and the probability mass function of X is given as

$$p_X(x) = p^x (1-p)^{(x-1)} \quad x \in \{0, 1, 2, 3, \dots\}$$

Theorem 3.12 (Geometric Experiment)

The experiment involves performing independent Bernoulli(p) trials until the first success is observed.

Let ω be an outcome for this experiment, and

$$X(\omega) := \text{number of trials in } \omega$$

Then,

$$X \sim \text{Geom}(p)$$

Proof. We first observe that the sample space for this experiment looks like:

$$\mathcal{S} = \{S, FS, FFS, FFFS, FFFFFS, \dots\}$$

From here, the proof involves explicitly calculating the pmf of X and showing that it is the same as the pmf of the Geometric distribution with parameter p . We leave this as an exercise for the reader. $\square$

The above theorem shows that the Geometric distribution models phenomena that involve waiting for one success.

Theorem 3.13 (Parameters associated to the Geometric Distribution)

Suppose $X \sim \text{Geom}(p)$ then,

1. The expected value of X is given by:

$$E(X) = \frac{1}{p}$$

2. The variance of X is given by:

$$V(X) = \frac{1-p}{p^2}$$

Examples of Geometric distribution in applications:

1. Suppose we sample students (with replacement) from campus until a student with height is less than or equal to 60 inches is found.

Since the sampling is with replacement, the result of a given trial is independent of the other trials. Therefore this experiment can be modelled by $X \sim \text{Geom}(p)$ distribution, where

$$P(\text{Success}) = P(\text{finding a student with height less than or equal to 60 inches})$$

and event

$$\{X = k\} := \{\text{we find } k \text{ students before the first student with height less than or equal to 60 inches}\}$$

In this setting, $p_X(k) = P(X = k)$, is probability of picking exactly k students before finding a student with height less than or equal to 60 inches.

Also, $P(X \geq k)$, picking more than or equal to k students before finding a student with height less than or equal to 60 inches.

2. Suppose we sample students (with replacement) from campus until we find one that has taken a statistics course.

This experiment can be modelled by $X \sim \text{Geom}(p)$ distribution, where

$$P(\text{Success}) = P(\text{finding a student who has taken a statistics course})$$

and the event

$$\{X = k\} := \{\text{we pick } k \text{ students before finding the first student who has taken at least one statistics class}\}$$

In this setting, $p_X(k) = P(X = k)$, is the probability of picking exactly k students before finding a student who has taken at least one statistics class.

3. Suppose we choose potatoes (with replacement, even though that does not really make sense) until we find a potato that does not have a blemish.

In this setting, a potato is a success if it does not have any blemishes.

This experiment can be modelled by $X \sim \text{Geom}(p)$ distribution,

$$P(\text{Success}) = P(\text{finding a potato without any blemishes})$$

and the event

$$\{X = k\} := \{\text{we pick } k \text{ potatoes before finding the first potato without any blemishes}\}$$

In this setting, $p_X(k) = P(X = k)$, is the probability of picking exactly k potatoes before finding the first potato without any blemishes.

Negative Binomial Distribution

The Negative Binomial random variable also models phenomena that involve “waiting”.

Definition 3.14 (Negative Binomial Distribution)

We say that X has the Negative Binomial distribution with parameters

- $r \in \mathbb{N}$: number of successes we are waiting for.
- $p \in (0, 1)$: $P(\text{Success})$, the probability of a success.

typically denoted by $X \sim \text{NegBinom}(r, p)$ if X takes values $\mathcal{X} := \{0, 1, 2, 3, \dots\} = \mathbb{N}$ and the probability mass function of X is given as

$$p_X(x) = \binom{x+r-1}{x} p^r (1-p)^x \quad x \in \{0, 1, 2, 3, \dots\}$$

Theorem 3.15 (Negative Binomial Experiment)

The experiment involves performing independent Bernoulli(p) trials until r successes are observed.

Let ω be an outcome for this experiment, and

$$X(\omega) := \text{number of failures in } \omega$$

Then,

$$X \sim \text{NegBinom}(r, p)$$

Proof. We first observe that the sample space for this experiment is infinite, and looks like

$$\mathcal{S} = \{SS \dots S, FSS \dots S, SFS \dots S, \dots\}$$

The proof involves explicitly counting the outcomes in the sets

$$\{X = k\} = \{\text{exactly } k \text{ failures observed before the } r\text{th success}\}$$

and using this to calculate the pmf of X , and showing that it is the same as the pmf of the Negative Binomial distribution with parameters r and p . We leave this as an exercise for the reader. $\square$

The above theorem shows that the Negative Binomial distribution models phenomena that involve waiting for r successes.

Theorem 3.16 (Parameters associated to the Negative Binomial Distribution)

Suppose $X \sim \text{NegBinom}(r, p)$ then,

1. The expected value of X is given by:

$$E(X) = \frac{r(1-p)}{p}$$

2. The variance of X is given by:

$$V(X) = \frac{r(1-p)}{p^2}$$

Examples of Negative Binomial distribution in applications:

1. Suppose we sample students (with replacement) from campus until we find r students with height is less than or equal to 60inches is found.

Since the sampling is with replacement, the result of a given trial is independent of the other trials. Therefore this experiment can be modelled by $X \sim \text{NegBinom}(r, p)$ distribution, where

$$P(\text{Success}) = P(\text{finding a student with height less than or equal to 60 inchex})$$

and event

$$\{X = k\} := \{\text{we find } k \text{ students of height greater than 60inches before the finding the } r\text{th student with height}$$

In this setting, $p_X(k) = P(X = k)$, is probability of finding exactly k students with height greater than 60inches before finding the r th student with height less than or equal to 60inches.

Also, $P(X \geq k)$, finding more than or equal to k students with height greater than 60inches before finding r th student with height less than or equal to 60inches.

2. Suppose we sample students (with replacement) from campus until we find r students that have taken a statistics course.

This experiment can be modelled by $X \sim \text{NegBinom}(r, p)$ distribution, where

$$P(\text{Success}) = P(\text{finding a student who has taken a statistics course})$$

and the event

$$\{X = k\} := \{\text{we find } k \text{ students who have never taken a statistics class before finding } r\text{th student who has tak}$$

In this setting, $p_X(k) = P(X = k)$, is the probability of finding exactly k students who have never taken a statistics class before finding r th student who has taken at least one statistics class.

3. Suppose we choose potatoes (with replacement, even though that does not really make sense) until we find r potatoes that do not have a blemish.

In this setting, a potato is a success if it does not have any blemishes.

This experiment can be modelled by $X \sim \text{NegBinom}(r, p)$ distribution,

$$P(\text{Success}) = P(\text{finding a potato without any blemishes})$$

and the event

$$\{X = k\} := \{\text{find exactly } k \text{ potatoes with blemishes before finding the } r\text{th potato without any blemishes}\}$$

In this setting, $p_X(k) = P(X = k)$, is the probability of finding exactly k potatoes with blemishes before finding the r th potato without any blemishes.

Poisson Distribution

The Poisson distribution is a very useful distribution that shows up in many applications, especially in the context of Poisson Processes.

Definition 3.17 (Poisson Distribution)

We say that X has the Poisson distribution with parameters

- $\lambda \in \mathbb{R}_+$: rate parameter

typically denoted by $X \sim \text{Pois}(\lambda)$ if X takes values $\mathcal{X} := \{0, 1, 2, 3, \dots\} = \mathbb{N}$ and the probability mass function of X is given as

$$p_X(x) = e^{-\lambda} \frac{\lambda^x}{x!}$$

To show that p_X is indeed a pmf, we observe the series expansion

$$e^\lambda = \sum_{k=0}^{\infty} \frac{\lambda^k}{k!}$$

Theorem 3.18 (Parameters associated to the Poisson Distribution)

Suppose $X \sim \text{Pois}(\lambda)$ then,

1. The expected value of X is given by:

$$E(X) = \lambda$$

2. The variance of X is given by:

$$V(X) = \lambda$$

Relationships Between Discrete Distributions

3.8.1 Binomial and Hypergeometric

Let $N, M, n \in \mathbb{N}$ be given and set $p = \frac{M}{N}$. Suppose,

- (a) $\text{bin}(x; n, p)$ is the probability mass function for binomial distribution with parameters:

$n =$ sample size

$p =$ probability of success

- (b) $\text{hyper}(x; N, M, n)$ is the probability mass function for hypergeometric distribution with parameters

$n =$ sample size

$M =$ number of successes

$N =$ population size

Then, if $N, M \rightarrow \infty$, and $\frac{M}{N} \rightarrow p$, then $\text{hyper}(x; N, M, n) \rightarrow \text{bin}(x; n, p)$.

An alternate way to state this is to say that if sample size N is small compared to population size N , we can assume that samples are approximately independent.

3.8.2 Binomial and Poisson

Suppose

- (a) $\text{bin}(x; n, p)$ is the probability mass function for binomial Distribution with parameters:

$n =$ sample size

$p =$ probability of success

- (b) $\text{pois}(x; \lambda)$ is the probability mass function for Poisson Distribution with parameters $\lambda > 0$

Then, if $n \rightarrow \infty$ and $p \rightarrow 0$ such that $n \cdot p \rightarrow \lambda$, then $\text{bin}(x; n, p) \rightarrow \text{pois}(x; \lambda)$.

If n is large and p is small, $\text{bin}(x; n, p) \approx \text{pois}(x; \lambda)$, this is sometimes stated by saying that the Poisson Distribution is approximately the Binomial Distribution for rare events.

4 | Continuous Random Variable

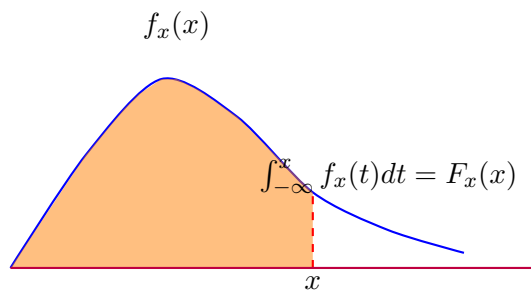
Suppose X is a random variable with cdf $F_X(x)$, that is $F_X(x) = P(X \leq x)$. We say X is a continuous random variable provided that $F_X : \mathbb{R} \rightarrow \mathbb{R}$ is a continuous function.

In practice continuous random variables are useful when studying quantitative features of sample space outcome that can take values in an interval.

A probability density function for a continuous random variable is a function $f_X : \mathbb{R} \rightarrow \mathbb{R}$ satisfying

$$F_X(x) = P(X \leq x) = \int_{-\infty}^x f_X(t) dt$$

The following picture describes the pictorial relationship between the cumulative distribution function and probability density function.



We note that:

1. The density function f_X need not be a continuous function for F_X to be continuous.
2. For events of the form $\{a \leq X \leq b\}$, that is the event

$$\{X \text{ takes values in } [a, b]\}$$

, the probabilities are calculated as follows:

$$P(a \leq X \leq b) = P(X \text{ takes values in } [a, b]) = \int_a^b f_X(x) dx$$

3. The events of the type $\{X = a\}$ have zero probability

$$P(X = a) = P(a \leq x \leq a) = \int_a^a f_X(x) dx = 0$$

That is, the probability that X takes any fixed number equals to zero.

4. The Fundamental Theorem of Calculus suggests that

$$\frac{d}{dx} \left(\int_a^x f(u) du \right) = f(x)$$

from which we obtain the following theorem:

Theorem 4.1

If the F_X is differentiable, we must have

$$F'_X(x) = f_X(x)$$

.

5. One way to calculate the pdf of a random variable X : first calculate the cdf $F_X(x)$ of X , then if $F_X(x)$ is differentiable, $f_X(x) = F'_X(x)$.

6. Also, recall Leibniz Rule for differentiating an integral gives

$$\frac{d}{dx} \left(\int_{a(x)}^{b(x)} f(t) dt \right) = b'(x) \cdot f(b(x)) - a'(x) \cdot f(a(x))$$

Which can be used to calculate the density functions of transformations $g(X)$ of the random variable X .

Theorem 4.2

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function satisfying $f_X(x) \geq 0$ for all x and $\int_{-\infty}^{\infty} f(x) dx = 1$. Then $f(x)$ is a pdf of some random variable. On the other hand, if $f_X(x)$ is a pdf of X , then $f_X(x)$ satisfies $f_X(x) \geq 0$ for all x and $\int_{-\infty}^{\infty} f(x) dx = 1$.

Example 4.0.1

Consider the function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$f(x) = \begin{cases} x^2(1-x) & x \in (0, 1) \\ 0 & \text{otherwise} \end{cases}$$

We would like to find a constant c such that $f_X = c \cdot f$ defines a pdf of a random variable. First we notice that $f(x) = x^2(1-x) \geq 0$ whenever $x \in (0, 1)$. Now we compute

$$\int_{-\infty}^{\infty} f(x) dx = \int_0^1 x^2(1-x) dx = \int_0^1 x^2 - x^3 dx = \left(\frac{x^3}{3} - \frac{x^4}{4} \right) \Big|_0^1 = \frac{1}{12}.$$

Thus if $c = 12$, $f_X = 12 \cdot f$, then f_X is a pdf of a random variable with the correct normalization.

$$f_X(x) = \begin{cases} 12 \cdot x^2(1-x) & x \in (0, 1) \\ 0 & \text{otherwise} \end{cases}.$$



In this case, we can calculate

$$\begin{aligned} F_X(x) &= \int_{-\infty}^x f_X(u) du = \int_{-\infty}^x 12 \cdot u^2(1-u) du \\ &= 12 \cdot \lim_{c \rightarrow -\infty} \int_c^x u^2(1-u) du \\ &= 12 \cdot \lim_{c \rightarrow -\infty} \left(\left(\frac{x^3}{3} - \frac{x^4}{4} \right) - \left(\frac{c^3}{3} - \frac{c^4}{4} \right) \right) \\ &= 12 \left(\frac{x^3}{3} - \frac{x^4}{4} \right). \end{aligned}$$

Here we observe that $F'_X(x) = f_X(x)$.

Example 4.0.2

Consider X being a random variable with pdf F_X and cdf F_X , and let $Y = 4X^5$ be a new random variable. To calculate the pdf of Y , we first note that in general $f_Y(x) \neq 4(f(x))^5$. For $y \in \mathbb{R}$,

$$F_Y(y) = P(Y \leq y) = P(4X^5 \leq y) = P(X^5 \leq y/4) = P(X \leq (y/4)^{1/5}) = F_X((y/4)^{1/5}).$$

Now, using chain rule we can compute

$$f_Y(y) = \frac{d}{dy} (F_Y(y)) = \frac{d}{dy} \left(F_X((y/4)^{1/5}) \right) = F'_X \left((y/4)^{1/5} \right) \cdot (y/4)^{-4/5} / 20.$$

With f_X defined by the last example, we obtain

$$F'_X \left(\left(\frac{y}{4} \right)^{1/5} \right) = f_x \left(\left(\frac{y}{4} \right)^{1/5} \right) = 12 \cdot \left(\frac{y}{4} \right)^{2/5} \cdot \left(1 - \left(\frac{y}{4} \right)^{1/5} \right),$$

and thus

$$f_Y(y) = \begin{cases} \frac{3}{5} \cdot \left(\frac{y}{4} \right)^{-2/5} \cdot \left(1 - \left(\frac{y}{4} \right)^{1/5} \right) & y \in (0, 1) \\ 0 & \text{otherwise} \end{cases}.$$

We remind ourselves of the parameters attached to a continuous random variable: **Parameters Attached to Continuous Variables**

Suppose X is a continuous random variable with pdf $f_X(x)$.

1. The expected value of X is given by

$$E(X) = \mu_X = \int_{-\infty}^{\infty} x f_X(x) dx.$$

2. If $Y = h(X)$, then

$$E(Y) = E(h(X)) = \int_{-\infty}^{\infty} h(x) \cdot f_X(x) dx = \int_{-\infty}^{\infty} y \cdot f_Y(y) dy.$$

3. The variance of X is defined by

$$\sigma_X^2 = V(X) = \int_{-\infty}^{\infty} (x - \mu_X)^2 \cdot f_X(x) dx.$$

Note that $V(X) = E(X^2) - E(X)^2$ still holds.

4. For $\alpha \in (0, 1)$, x_α is the α^{th} critical value for X is x_α satisfies $P(X \geq x_\alpha) = \alpha$, which holds iff $1 - P(X \leq x_\alpha) = \alpha$, iff $P(X \leq x_\alpha) = (1 - \alpha)$, iff $F_X(x_\alpha) = 1 - \alpha$.
5. For $p \in (0, 1)$, n_p is called the p^{th} percentile provided that $P(X \leq n_p) = p$, which is equivalent to $F_X(n_p) = p$. We note that α^{th} critical value is the same as the $(1 - \alpha)^{\text{th}}$ percentile.

Examples of Continuous Random Variables

Uniform Continuous Distribution

Let $[A, B] \subseteq \mathbb{R}$ be a closed interval where $A < B$. Then we can define the probability density function as the following

$$f(x) = \begin{cases} \frac{1}{B-A} & x \in [A, B] \\ 0 & \text{otherwise} \end{cases}$$

Then, the cumulative density function of the uniform continuous distribution is

$$F(x) = \begin{cases} 0 & \text{if } x < A \\ \frac{x-A}{B-A} & \text{if } x \in [A, B] \\ 1 & \text{if } x \geq B \end{cases}$$

Then, the expected value is

$$\begin{aligned} E(X) &= \int_{-\infty}^{\infty} x \cdot f(x) dx = \int_A^B \frac{x}{B-A} dx \\ &= \frac{B+A}{2} \end{aligned}$$

The variance is

$$\begin{aligned} V(X) &= \int_{-\infty}^{\infty} (x - \mu_x)^2 f_X(x) dx = \int_A^B \frac{\left(x - \frac{B+A}{2}\right)^2}{B-A} dx \\ &= \frac{(B+A)^2}{12} \end{aligned}$$

If $a, b \in [A, B]$ and $a < b$, then

$$P(a \leq X \leq b) = \int_a^b \frac{x}{B-A} dx = \frac{b-a}{B-A}$$

Normal Distribution

Definition 4.3

We define **Normal Distribution** using the following two parameters

$$\begin{aligned} \mu &\rightarrow \text{mean} \\ \sigma^2 &\rightarrow \text{variance} \end{aligned}$$

denoted as $N(\mu, \sigma^2)$

Note: $\sigma = \sqrt{\sigma^2}$ is the standard deviation.

Definition 4.4

Suppose X is normally distributed, i.e $X \sim N(\mu, \sigma^2)$. Then, the probability density function of X is defined as

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} \cdot e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \quad x \in (-\infty, \infty)$$

And the cumulative distribution function is

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(t) dt = \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^x e^{-\frac{(t-\mu)^2}{2\sigma^2}} dt$$

Therefore,

$$P(a \leq X \leq b) = \frac{1}{\sqrt{2\pi}\sigma} \int_a^b e^{-\frac{(t-\mu)^2}{2\sigma^2}} dt$$

Note:

1. If $X \sim N(\mu, \sigma^2)$, then

$$E(X) = \mu$$

$$V(X) = \sigma^2$$

2. When $\mu = 0, \sigma^2 = 1$, the distribution is called the **Standard Normal Distribution** $Z \sim N(0, 1)$, and the standard normal distribution probability density function is

$$f(z) = \frac{1}{\sqrt{2\pi}} \cdot e^{-\frac{z^2}{2}}, \quad x \in (-\infty, \infty)$$

and the cumulative distribution function of Z is denoted as $\Phi(z)$,

$$P(Z \leq z) = \Phi(z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^z e^{-\frac{t^2}{2}} dt$$

To be specific about parameters, we denote probability density function of $X \sim N(\mu, \sigma^2)$ as

$$f(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi}\sigma} \cdot e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \quad x \in (-\infty, \infty)$$

3. If $X \sim N(\mu, \sigma^2)$ and $Z \sim N(0, 1)$, then

$$\begin{aligned} P(X \leq x) &= \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^x e^{-\frac{(t-\mu)^2}{2\sigma^2}} dt \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\frac{x-\mu}{\sigma}} e^{-\frac{s^2}{2}} ds && \text{Change of variable with } s = \frac{t-\mu}{\sigma} \\ &= P\left(z \leq \frac{x-\mu}{\sigma}\right) \end{aligned}$$

Therefore, if $X \sim N(\mu, \sigma^2)$, then $\frac{X-\mu}{\sigma}$ has the standard normal distribution.

4. If $X \sim N(\mu, \sigma^2)$, then $Z = \frac{X-\mu}{\sigma} \sim N(0, 1)$. Then

$$\begin{aligned} P(a \leq X \leq b) &= P\left(\frac{a-\mu}{\sigma} \leq z \leq \frac{b-\mu}{\sigma}\right) \\ &= \Phi\left(\frac{b-\mu}{\sigma}\right) - \Phi\left(\frac{a-\mu}{\sigma}\right) \end{aligned}$$

Therefore, we can use the standard normal cumulative distribution function to calculate the cumulative distribution function for the random variable $X \sim N(\mu, \sigma^2)$, i.e

$$X \sim N(\mu, \sigma^2) \longrightarrow Z \sim N(0, 1)$$

$$x \mapsto Z = \frac{x-\mu}{\sigma} \quad \text{It is also Z-score of } x$$

$$z \mapsto x = Z \cdot \sigma + \mu$$

4.3.1 Percentile and Critical Values

Definition 4.5

Suppose $Z \sim N(0, 1)$, the standard normal distribution. Then, for $\alpha \in (0, 1)$, z_α is the α th **critical value** that satisfies the following

$$\begin{aligned} P(Z \geq z_\alpha) &= \alpha \\ \text{i.e.} \quad \int_{z_\alpha}^{\infty} f(z; 0, 1) dz &= \alpha \\ \text{i.e.} \quad 1 - P(Z \leq z_\alpha) = \alpha &\implies 1 - \Phi(z_\alpha) = \alpha \end{aligned}$$

Definition 4.6

Let $p \in (0, 1)$. Then $(100p)$ th **percentile**, $\eta(p)$ satisfies

$$\begin{aligned} P(Z \leq \eta(p)) &= p \\ \text{i.e.} \quad \Phi(\eta(p)) &= p \end{aligned}$$

Suppose $X \sim N(M, \sigma^2)$ and $\alpha \in (0, 1)$, then

$$\alpha^{th} \text{ critical value for } X := x_\alpha = \sigma z_\alpha + M$$

where z_α is the α^{th} critical value for $Z \in N(0, 1)$

4.3.2 Approximating Bin(n,p) using Normal Distribution

Let $X \sim \text{Binom}(n, p)$, so that $\mu_x = np$ and $\sigma_x^2 = np(1-p)$. Then if the $\text{Bin}(n, p)$ is not too skewed, then

$$P(X \leq x) \approx \Phi\left(\frac{x + 0.5 - \mu_x}{\sigma_x}\right)$$

The right hand side does not involve terms of the form $\binom{n}{k}$, the latter can be difficult to compute.

Gamma Distribution

Gamma distribution is useful when modeling component lifetimes and waiting times.

Definition 4.7

Gamma function is defined as

$$\Gamma(\alpha) = \int_0^\infty t^{\alpha-1} \cdot e^{-t} dt, \text{ for } \alpha > 0$$

Theorem 4.8

Properties of Gamma Function

1. $\Gamma(\alpha + 1) = \alpha\Gamma(\alpha)$ for $\alpha > 0$. Since $\Gamma(1) = 1$, we have $\Gamma(n) = (n-1)!$ for $n \in \mathbb{N}$ using induction
2. $\Gamma(\frac{1}{2}) = \sqrt{\pi}$

Definition 4.9

We say X has the **Gamma Distribution** with shape parameter α and scale parameter β if the probability density function of X is

$$f(x; \alpha, \beta) = \begin{cases} \frac{1}{\beta^\alpha \Gamma(\alpha)} x^{\alpha-1} \cdot e^{-\frac{x}{\beta}} & x > 0 \\ 0 & x \leq 0 \end{cases}$$

Since $f(x; \alpha, \beta) \geq 0 \forall x \in \mathbb{R}$ and we can check $\int_{-\infty}^\infty f(x; \alpha, \beta) dx = 1$ by doing a change of variable $y = \frac{x}{\beta}$, therefore f satisfies the property of probability density function.

Note:

1. When $\beta = 1$, we call it a **standard gamma distribution** with shape parameters α , with probability density function

$$f(x; \alpha) = \begin{cases} \frac{x^{\alpha-1} e^{-x}}{\Gamma(\alpha)} & x > 0 \\ 0 & x \leq 0 \end{cases}$$

If T has standard Gamma distribution with shape parameters: α , then $X = \beta T$ has Gamma distribution with shape α , and scale β

2. If $X \sim \text{Gamma}(\alpha, \beta)$, then

$$(a) \quad E(X) = \mu_x = \alpha\beta$$

$$(b) \quad V(X) = \sigma_X^2 = \alpha\beta^2$$

$$(c) \quad \sigma = \sqrt{\alpha}\beta$$

3. The cumulative density function of $X \sim \text{Gamma}(\alpha, \beta)$ is the following

$$F_X(x; \alpha, \beta) = \begin{cases} \int_0^x f(t; \alpha, \beta) dt & \text{if } x > 0 \\ 0 & \text{otherwise} \end{cases}$$

Then $P(X \leq x) = F_X(x; \alpha, \beta) = F\left(\frac{x}{\beta}; \alpha\right)$, the cumulative density function of standard Gamma with parameters α , i.e

$$F_X(x; \alpha) = \begin{cases} \int_0^x \frac{t^{\alpha-1} e^{-t}}{\Gamma(\alpha)} & \text{if } x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

4.4.1 Special Cases of Gamma Distribution

Definition 4.10

When X has a Gamma Distribution with $\alpha = 1, \Gamma - \frac{1}{\lambda}$ for some $\lambda > 0$. Then, we say X has a **Exponential Distribution** with parameter λ . The probability density function of X is the

$$f(x; \lambda) = \begin{cases} \lambda \cdot e^{-\lambda x} & \text{if } x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

And the cumulative density function of X

$$F_X(x; \lambda) = \begin{cases} 1 - e^{-\lambda x} & \text{if } x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

Note:

1. If X has exponential distribution with parameters $\lambda > 0$, then

(a) $E(X) = \mu_x = \frac{1}{\lambda}$

(b) $V(X) = \sigma_X^2 = \frac{1}{\lambda^2}$

(c) $\sigma = \frac{1}{\lambda}$

2. If we have a Poisson process with rate α , then the exponential distribution with $\lambda = \alpha$ models the distribution of elapsed time between the occurrence of two successive events.

3. If $X \sim \text{Exp}(\lambda)$, then

$$\begin{aligned} P(X \geq t + t_0 \mid X \geq t_0) &= \frac{[\{X \geq t + t_0\} \cap \{X \geq t_0\}]}{P(X \geq t_0)} \\ &= \frac{P(X \geq t + t_0)}{P(X \geq t_0)} = \frac{1 - F(t + t_0; \lambda)}{1 - F(t_0; \lambda)} \\ &= e^{-\lambda t} = P(X \geq t) \end{aligned}$$

This means that if X was modeling lifetime of a component, the distribution of additional lifetime is exactly the same as the original distribution of lifetime, i.e the exponential distribution has **meaningless property**

Definition 4.11

X is said to have **chi-squared** distribution with parameter v (degree of freedom) if $X \sim \text{Gamma}(\alpha, \beta)$ with $\alpha = \frac{v}{2}, \beta = 2$. The probability density function of X is defined as the following

$$f(x; v) = \begin{cases} \frac{1}{2^{\frac{v}{2}} \cdot \Gamma(\frac{v}{2})} \cdot x^{\frac{v}{2}-1} \cdot e^{-\frac{x}{2}} & \text{if } x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

Note:

1. If X is a chi-squared distribution with parameters v , then

(a) $E(X) = \mu_x = \alpha \cdot \beta = v$

(b) $V(X) = \sigma_X^2 = \alpha \beta^2 = 2v$

(c) $\sigma = \sqrt{2v}$

2. Chi-square distribution plays important role in statistical inference.

3. If $X \sim N(M, \sigma^2)$, then $\left(\frac{X-M}{\sigma}\right)^2$ has chi-square distribution with $v = 1$

Lognormal Distribution

Definition 4.12

We say X has a **Lognormal Distribution** if $\ln(X)$ has normal distribution with parameters M and σ^2 . i.e

$$\ln(X) \sim N(M, \sigma^2)$$

If X has lognormal distribution, then the probability density function is

$$f(x; M, \sigma) = \begin{cases} \frac{1}{\sqrt{2\pi}\sigma x} \cdot e^{-\frac{(\ln(x)-M)^2}{2\sigma^2}} & \text{if } x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

Note:

1. If $\sim N(M, \sigma^2)$ then

(a) $E(\ln(X)) = M$

(b) $V(\ln(X)) = \sigma^2$

2. If X has lognormal distribution, and $\ln(x) \sim N(M, \sigma^2)$, then

(a) $E(X) = \mu_x = e^{M + \frac{\sigma^2}{2}}$

(b) $V(X) = \sigma_X^2 = e^{(2M + \sigma^2)} \cdot (e^{\sigma^2} - 1)$

3. We can use standard normal table to calculate the probabilities

$$\begin{aligned} F_X(x; M, \sigma) &= P(X \leq x) = P(\ln(x) \leq \ln(x)) \\ &= P(Z \leq \frac{\ln(x) - M}{\sigma}) \quad \because \ln(x) \sim N(M, \sigma^2) \\ &= \Phi\left(\frac{\ln(x) - M}{\sigma}\right) \quad \text{for } x > 0 \end{aligned}$$

Beta Distribution

Beta distribution takes values in a finite interval, which is good to model proportions that naturally lie in $(0, 1)$

Definition 4.13

X is said to have **Beta Distribution** with parameters: $\alpha, \beta > 0$ if the probability density function is

$$f(x; \alpha, \beta) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} \cdot x^{\alpha-1}(1-x)^{\beta-1} \quad x \in (0, 1)$$

Note:

1. We called Beta distribution because the Beta function is

$$B(\alpha, \beta) = \frac{\Gamma(\alpha) \cdot \Gamma(\beta)}{\Gamma(\alpha + \beta)} \quad \text{for } \alpha, \beta > 0$$

So that

$$f(x; \alpha, \beta) = \frac{1}{B(\alpha, \beta)} \cdot x^{\alpha-1}(1-x)^{\beta-1} \quad x \in (0, 1)$$

2. If $X \sim \text{Beta}(\alpha, \beta)$, then

(a) $E(X) = \mu_x = \frac{\alpha}{\alpha + \beta}$

(b) $V(X) = \sigma_X^2 = \frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)}$

Note that both μ_x and σ_X^2 are in $(0, 1)$

3. Depending on values of α, β , the probability density function $f(x; \alpha, \beta)$ has different shapes.

(a) If $\alpha > 1, \beta = 1$, then it is strictly increasing

(b) If $\alpha = 1, \beta > 1$, then it is strictly decreasing

(c) If $\alpha < 1, \beta < 1$, then it is U-shaped

(d) If $\alpha = \beta$, then it is symmetric about $\frac{1}{2}$ with $\mu_x = \frac{1}{2}$ and $\sigma_X^2 = \frac{1}{4(2\alpha + 1)}$

(e) If $\alpha = \beta = 1$, then it is a uniform distribution on $(0, 1)$

Cauchy Distribution

Definition 4.14

X is said to have **Cauchy Distribution** with parameter θ if the probability density function of X is

$$\begin{aligned} f(x; \theta) &= \frac{1}{\pi} \cdot \frac{1}{1 + (x - \theta)^2} \quad x \in (-\infty, \infty) \\ &\quad \theta \in (-\infty, \infty) \end{aligned}$$

Note:

1. $E(X)$ and $V(X)$ do not exist if X has Cauchy distribution since

$$\begin{aligned} E(X) &= \int_{-\infty}^{\infty} x f(x) dx = \int_{-\infty}^{\infty} \frac{1}{\pi} \cdot \frac{x}{1 + (x - \theta)^2} dx \\ &= 0 \text{ since } \frac{x}{1 + (x - \theta)^2} \text{ is an odd function} \end{aligned}$$

This is not the case.

2. The graph of $f(x; \theta)$ is bell shaped like the Normal density but has heavier tails than Normal density.

5 | Joint Distribution

Many applications require that we use more than one characteristic or variable related to the experimental unit. For example:

1. when studying Body Mass Index (BMI) one needs the height and the weight
2. Predicating house prices needs many factors: location, house area, number of bedroom and bath-rooms, etc.
3. Predicting the price of a stock depends on many performance metrics.

In all these cases, we need to calculate the probability of two or more events simultaneously, therefore, we want the **Joint Distribution** of variable of interest

Two Discrete Random Variable

Definition 5.1

Suppose X, Y are two discrete random variables. Then $\mathcal{X} :=$ values of X and $\mathcal{Y} :=$ values of Y . Then, the cartesian product of $\mathcal{X}, \mathcal{Y}$

$$\mathcal{X} \times \mathcal{Y} = \{(x, y) : x \in \mathcal{X}, y \in \mathcal{Y}\}$$

is the **Joint Sample Space**.

The **Joint Probability Mass Function**, i.e Joint pmf of X and Y is

$$P(x, y) := P(X = x \text{ and } Y = y)$$

Which is $P(X \text{ takes value } x \text{ and } Y \text{ takes value } y)$

Note:

1. $p(x, y)$ satisfies
 - (a) $p(x, y) \geq 0 \forall (x, y) \in \mathcal{X} \times \mathcal{Y}$
 - (b) $\sum_{(x, y) \in \mathcal{X} \times \mathcal{Y}} p(x, y) = 1$
2. If $A \subseteq \mathcal{X} \times \mathcal{Y}$, i.e A is a set of events in the joint distribution, then

$$P(A) = \sum_{(x, y) \in A} p(x, y)$$

Definition 5.2

Given a joint sample space of X, Y , we define the **Marginal Probability Mass Function of X** as

$$p_X(x) := \sum_{y \in \mathcal{Y}} p(x, y) \text{ for fixed } x \in \mathcal{X}$$

Marginal Probability Mass Function of Y as

$$p_Y(y) := \sum_{x \in \mathcal{X}} p(x, y) \text{ for fixed } y \in \mathcal{Y}$$

Definition 5.3

Given a joint sample space of X, Y , we define the **conditional distribution function of X given that $Y = y$** as

$$\begin{aligned} p_{X|Y} &:= P(X = x \text{ given that } Y = y) \\ &:= \frac{p(x, y)}{p_Y(y)} \end{aligned}$$

conditional distribution function of Y given that $X = x$

$$\begin{aligned} p_{Y|X} &:= P(Y = y \text{ given that } X = x) \\ &:= \frac{p(x, y)}{p_X(x)} \end{aligned}$$

Important Identity:

$$p(x, y) = p_{X|Y}(x | y) \cdot p_Y(y) = p_{Y|X}(y | x) \cdot p_X(x)$$

Two Continuous Random Variables

Definition 5.4

Let X, Y be two continuous random variables. Then, the **Joint Sample Space** is given by

$$\mathbb{R} \times \mathbb{R} = \mathbb{R}^2$$

Then, the **joint probability density function**, i.e joint pdf for the random variables X and Y is

$$f : \mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{R}$$

that satisfies the following

1. $f(x, y) \geq 0 \quad \forall (x, y) \in \mathbb{R}^2$
2. $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1$

Note:

1. If $A \subseteq \mathbb{R}^2$, i.e A is a set of events in the joint sample space, then

$$P((x, y) \in A) = \int_A \int f(x, y) dx dy$$

If $A = \{(x, y) : x \in [a, b], y \in [c, d]\}$ = a rectangle $[a, b] \times [c, d]$ Then

$$\begin{aligned} P((x, y) \in A) &= \int_a^b \int_c^d f(x, y) dx dy \\ &= P(a \leq X \leq b \text{ and } c \leq Y \leq d) \end{aligned}$$

Definition 5.5

Given a joint sample space of X, Y , where X, Y are continuous, we define the **Marginal Probability Mass Function for X** as

$$f_X(x) := \int_{-\infty}^{\infty} f(x, y) dy \text{ for } x \in (-\infty, \infty)$$

Marginal Probability Mass Function for Y as

$$f_Y(y) := \int_{-\infty}^{\infty} f(x, y) dx \text{ for } y \in (-\infty, \infty)$$

Definition 5.6

Given a joint sample space of X, Y , where X, Y are continuous, we define the **conditional density function** of Y given that $X = x$ as

$$f_{Y|X}(y|x) = \frac{f(x, y)}{f_X(x)} \quad y \in (-\infty, \infty)$$

conditional density function of X given that $Y = y$

$$f_{X|Y}(x|y) = \frac{f(x, y)}{f_Y(y)} \quad x \in (-\infty, \infty)$$

Important Identity:

$$f(x, y) = f_{X|Y}(x|y) \cdot f_Y(y) = f_{Y|X}(y|x) \cdot f_X(x)$$

Independent Random Variables

Definition 5.7

We say two discrete variables X and Y are **independent** if

$$p(x, y) = p_X(x) \cdot p_Y(y) \quad \forall (x, y) \in \mathcal{X} \times \mathcal{Y}$$

otherwise, we say X and Y are **dependent**

Definition 5.8

We say two continuous variables X and Y are **independent** if

$$f(x, y) = f_X(x) \cdot f_Y(y) \quad \forall (x, y) \in \mathbb{R} \times \mathbb{R}$$

otherwise, we say X and Y are **dependent**

Note: When X and Y are independent, we can calculate the joint probability density function by multiplying the probability density functions of X and Y

Expected Values and Variances of Joint Distribution

Suppose X, Y are two random variables that are either both discrete or both continuous. Then, define

$$\begin{aligned} p(x, y) &:= \text{joint probability mass function of } X, Y \text{ if both discrete} \\ f(x, y) &:= \text{joint probability density function of } X, Y \text{ if both continuous} \end{aligned}$$

Definition 5.9

Then, if $h(x, y)$ is a function of X and Y , then we can define the **expected value** of $h(x, y)$ as

$$E(h(x, y)) = \begin{cases} \sum_{(x, y) \in \mathcal{X} \times \mathcal{Y}} h(x, y) \cdot p(x, y) & \text{if } X, Y \text{ are discrete} \\ \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} h(x, y) \cdot p(x, y) & \text{if } X, Y \text{ are continuous} \end{cases}$$

Let

$$\begin{aligned} \mu_x &:= E(X) && \text{Expected Value of } X \\ \mu_y &:= E(Y) && \text{Expected Value of } Y \end{aligned}$$

Then, we can define

$$\begin{aligned} h(X, Y) &= (X - \mu_x)(Y - \mu_y) \\ &= \text{Product of the deviation of } X \text{ and } Y \\ &\quad \text{from their respective expected values} \end{aligned}$$

Note:

1. If large values of X are associated with large values of Y , i.e $(X - \mu_x) \geq 0 \implies (Y - \mu_y) \geq 0$ and small values of X are associated with small values of Y , i.e $(X - \mu_x) \leq 0 \implies (Y - \mu_y) \leq 0$ Then, we have

$$h(X, Y) = (X - \mu_x)(Y - \mu_y) \geq 0$$

as $(X - \mu_x)$ and $(Y - \mu_y)$ are both positive or both negative.

2. If large X values are associated with small values, i.e $(X - \mu_x) \geq 0 \implies (Y - \mu_y) \leq 0$ and large values of Y are associated with small values of X , i.e $(Y - \mu_y) \geq 0 \implies (X - \mu_x) \leq 0$. Then, we have

$$h(X, Y) = (X - \mu_x)(Y - \mu_y) \leq 0$$

as $(X - \mu_x)$ and $(Y - \mu_y)$ has opposite sign

Then the expected value of $h(x, y) = (X - \mu_x)(Y - \mu_y)$, is a good candidate to measure relationship between X and Y .

Definition 5.10

We define the **covariance** of X and Y as

$$\begin{aligned} Cov(X, Y) &= E[(X - \mu_x)(Y - \mu_y)] = E(XY) - E(X) \cdot E(Y) \\ &= \text{Expected value of the product of deviation} \\ &\quad \text{of } X \text{ from } \mu_x \text{ and } Y \text{ from } \mu_y \end{aligned}$$

If X, Y are both discrete, then

$$Cov(X, Y) = \sum_x \sum_y (x - \mu_x)(y - \mu_y)p(x, y)$$

If X, Y are both continuous, then

$$Cov(X, Y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x - \mu_x)(y - \mu_y)f(x, y)dx dy$$

Note:

1. If $X = Y$, then $Cov(X, Y) = Cov(X, X) = V(X)$
2. If X and Y have strong positive relationship, i.e large values of X associated with large values of Y and small values of X associated with small values of Y , then $Cov(X, Y)$ will be positive
3. If $Cov(X, Y)$ is positive, then we can say X and Y are positively associated.
Similarly, if $Cov(X, Y)$ is negative, then we can say X and Y are negatively associated.

However, covariance is not a good measure of association. Depends on the unit in which X and Y are measured, it can be made arbitrarily large or small by changing units.

Correlation Coefficient

Let $\sigma_X = \sqrt{V(X)}$ and $\sigma_Y = \sqrt{V(y)}$, we can make the $Cov(X, Y)$ unit-less by dividing $\sigma_X \cdot \sigma_Y$.

Definition 5.11

The **correlation** of X and Y is

$$\rho_{X,Y} = Corr(X, Y) = \frac{Cov(X, Y)}{\sigma_X \cdot \sigma_Y}$$

Note:

1. $\rho_{X,Y}$ does not depend on the units of X and Y
2. $\rho_{X,Y}$ satisfies $-1 \leq \rho_{X,Y} \leq 1$
3. If a, c are both positive or both negative, then

$$Corr(aX + b, cY + d) = Corr(X, Y)$$

In particular, if $Y = aX + b$, then

$$\begin{aligned} Corr(X, Y) &= Corr(X, aX + b) = sign(a) \cdot Corr(X, X) \\ &= sign(a) = \pm 1 \end{aligned}$$

Therefore, $Corr(X, Y)$ measures the degree of linear relationship between X and Y .

If $\rho_{X,Y}$ close to 1, then there is a strong positive linear association between X and Y

If $\rho_{X,Y}$ close to -1 , then there is a strong negative linear association between X and Y

If $\rho_{X,Y} = 0$, then X, Y are uncorrelated, i.e no linear relationship. Note that it does not mean that there is no relationship between X, Y

4. If X, Y are independent, then

$$\begin{aligned} Cov(X, Y) &= 0 \\ Corr(X, Y) &= 0 \end{aligned}$$

However, $Corr(X, Y) = 0$ does not necessarily imply that X and Y are independent.

5. Correlation does not imply causation. It only implies association between variables. It is often the case that there is an underlying causal variable (unrelated to those already considered) that is causes the positive association.
6. Note that

$$\rho_{X,Y} = \frac{Cov(X, Y)}{\sigma_X \cdot \sigma_Y} \implies Cov(X, Y) = \rho_{X,Y} \cdot \sigma_X \cdot \sigma_Y$$

Linear Combination of Random Variables

Definition 5.12

Suppose $X_1, X_2, X_3, \dots, X_n$ are n -random variables such that

- (i) $E(X_i) = M_i, i = 1, 2, 3, \dots, n$
- (ii) $V(X_i) = \sigma_i^2, i = 1, 2, 3, \dots, n$

Let $a_1, a_2, \dots, a_n \in \mathbb{R}$, we can define the **Linear Combination** of $X_1, \dots, X_n$ as

$$Y = a_1X_1 + a_2X_2 + \dots + a_nX_n$$

Theorem 5.13

Given $Y = a_1X_1 + a_2X_2 + \dots + a_nX_n$ for $X_1, X_2, X_3, \dots, X_n$ are n -random variables and $a_1, a_2, \dots, a_n \in \mathbb{R}$, then

- 1.

$$\begin{aligned} E(Y) &= E(a_1X_1 + a_2X_2 + \dots + a_nX_n) \\ &= a_1M_1 + a_2M_2 + \dots + a_nM_n \end{aligned}$$

- 2.

$$\begin{aligned} V(Y) &= V(a_1X_1 + a_2X_2 + \dots + a_nX_n) \\ &= \sum_{i=1}^n \sum_{j=1}^n a_1a_j \cdot Cov(X_i, X_j) \end{aligned}$$

In particular, if $X_1, X_2, X_3, \dots, X_n$ are independent, then

$$\begin{aligned} V(Y) &= a_1^2 V(X_1) + a_2^2 V(X_2) + \dots + a_n^2 V(X_n) \\ &= a_1^2 \sigma_1^2 + a_2^2 \sigma_2^2 + \dots + a_n^2 \sigma_n^2 \end{aligned}$$

Example: If $Z = aX + bY$, then

$$\begin{aligned} E(Z) &= a\mu_x + b\mu_y \\ V(Z) &= a^2 \sigma_x^2 + b^2 \sigma_y^2 + 2ab \text{Cov}(X, Y) \end{aligned}$$

If $a = 1, b = -1$, i.e $Z = X - Y$. Then

$$\begin{aligned} E(X - Y) &= \mu_x - \mu_y \\ V(X - Y) &= \sigma_x^2 + \sigma_y^2 + 2\rho_{x,y} \cdot \sigma_x \cdot \sigma_y \end{aligned}$$

6 | Random Sample and Statistics

Collecting data it is important aspect of doing statistic. We want to use a subset of the population to draw conclusion/inference about the entire population, and this subset is called a **sample**. The way a sample is obtained will affect our confidence in the conclusion we drive from using it.

Example 6.0.1

suppose we have a vaccine for Covid 19 and We would be interested in knowing its efficacy. Consider the following experiment:

1. First, get two samples of people, group **A** and group **B**
2. Then, give vaccine to group A and give placebo to group B.
3. Expose both groups to the virus, and measure the appropriate response variable
4. Apply statistical tests to check if there is statistically significant difference between the group responses

To get correct conclusions about the vaccine, group A and B need to be carefully selected. Therefore, we need to consider the following questions

- (i) Is it possible that certain people can never make it to the samples?
- (ii) Does the sample contain only young individuals?
- (iii) Are the samples representative of the population?
- (iv) Are the sample size large enough?

And the statistical tools are usually based on principles of probability which need a precise notion of randomness of a sample.

Random Sample

Definition 6.1

We say $\{X_1, X_2, \dots, X_n\}$, a set of random variable, is a **random sample** if

- (i) X_i has the same probability distribution for $i = 1, 2, \dots, n$
- (ii) the set $\{X_1, X_2, \dots, X_n\}$ is an independent set of random variables

Note:

1. Sometimes a random sample is also referred as a set of independent and identically distributed random variable
2. The common distribution of all the X_i 's is called **population distribution**
- 3.

$$\{X_1, X_2, X_3, X_n\} \xrightarrow{\text{Sampling Procedure}} \{x_1, x_2, \dots, x_n\}$$

note that $\{X_1, X_2, X_3, X_n\}$ is random sample, i.e a collection of random variables and $\{x_1, x_2, \dots, x_n\}$ is the sample data that changes every time we run the sampling procedure.

Example 6.1.1

Suppose we are getting a random sample of size 3 from the population $\{1, 2, 3, 4, 5\}$. To get a random sample, we need to sample with replacement.

Let $\{Num1, Num2, Num3\}$ be a random sample, i.e collection of random variables. Since we are sampling with replacement, there are $5 \times 5 \times 5$ ways of getting sample data. so each run of sampling procedure gets one of the element in

$$\left\{ \begin{array}{ccccc} (1, 1, 1) & (1, 1, 2) & (1, 2, 1) & (2, 1, 1) & \dots \\ \dots & \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & (5, 5, 4) & (5, 5, 5) \end{array} \right\}$$

Statistics

We are often interested in some features or characteristics of the random sample.

Definition 6.2

A **statistic** is a quality calculated using a random sample, say $\{x_1, x_2, \dots, x_n\}$. Since the calculation of the statistic involves random variables, it will also be a random variable

Note:

1. The value of the statistic changes as the sample data changes
2. A statistic is a random variable defined on sample data corresponding to a random sample.
For example, if $\mathcal{X} :=$ values of the random variables in the random sample, and sample data is $\{(x_1, x_2, \dots, x_n) \mid x_i \in \mathcal{X}, \forall i = 1, 2, \dots, n\}$. Since $\{X_1, X_2, \dots, X_n\}$ is an independent set, the **joint distribution** for $X_1, x_2, \dots, X_n$ is given as

$$p(x_1, x_2, x_3, \dots, x_n) = p_X(x_1) \cdot p_X(x_2) \cdots p_X(x_n)$$

where $p_X(x_i)$ is the probability mass function of the common distribution X , If X is continuous, then we can use probability density function $f_X(x)$ to define the joint probability density function of $X_1, X_2, \dots, X_n$.

Then, we can use the joint probability mass function or probability density function of $X_1, X_2, \dots, X_n$ to calculate the probability distribution of the statistic of interest, i.e the **sampling distribution** of the statistic

Example 6.2.1

Recall the example that Population = $\{1, 2, 3, 4, 5\}$ and random sample = $\{Num1, Num2, Num3\}$. Then we can have the following different statistic

Name of the statistic	Definition of the statistic	Examples of evaluation on sample data
Sample Mean	$\overline{X} = \frac{Num_1, Num_2, Num_3}{3}$	$\overline{X}(1, 1, 1) = \frac{1+1+1}{3} = 1$ $\overline{X}(2, 1, 2) = \frac{2+1+2}{3} = \frac{5}{3}$
Sample Median	$\widetilde{X} = median\{Num_1, Num_2, Num_3\}$	$\widetilde{X}(1, 1, 1) = 1$ $\widetilde{X}(2, 1, 4) = 2$
Sample Maximum	$Max = \max\{Num_1, Num_2, Num_3\}$	$Max(1, 1, 1) = 1$ $Max(1, 2, 5) = 5$
Sample Total	$T_o = Num_1 + Num_2 + Num_3$	$T_o(1, 1, 1) = 1 + 1 + 1 = 3$ $T_0(1, 3, 5) = 1 + 3 + 5 = 9$
Sample variance	$S^2 = \frac{(Num_1-\overline{X})^2+(Num_2-\overline{X})^2+(Num_3-\overline{X})^2}{2}$	$S^2(1, 1, 1) = \frac{(1-\overline{X}(1,1,1))^2+(1-\overline{X}(1,1,1))^2+(1-\overline{X}(1,1,1))^2}{2}$ $S^2(1, 1, 1) = 0$ $S^2(1, 2, 3) = \frac{(1-\overline{X}(1,2,3))^2+(2-\overline{X}(1,2,3))^2+(3-\overline{X}(1,2,3))^2}{2}$ $S^2(1, 2, 3) = \frac{(1-2)^2+(2-2)^2+(3-2)^2}{2} = 1$

Sampling Distribution and Central Limit Theorem

6.3.1 Sampling Distribution

Definition 6.3

Let $\{X_1, X_2, \dots, X_n\}$ be a random sample size n where

- (i) X_i 's have the same distribution for $i = 1, 2, 3, \dots, n$
- (ii) The set $\{X_1, X_2, \dots, X_n\}$ is an independent set of random variables

Let $\mathcal{X} :=$ values of the random variables in the random sample, call it

$$\mathcal{X} = \{x_1, x_2, x_3, \dots, x_n\}$$

Then, the **joint sample space** for the random variables $X_1, X_2, \dots, X_n$ is the n-Cartesian product of $\mathcal{X}$, i.e

$$\mathcal{X}^n = \{(x_1, x_2, \dots, x_n); x_i \in \mathcal{X}\}$$

Since $X_1, X_2, \dots, X_n$ are independent random variables, the **joint probability distribution** of $X_1, X_2, X_3, \dots, X_n$ is

1. $p(x_1, x_2, \dots, x_n) = p_X(x_1) \cdot p_X(x_2) \cdots p_X(x_n)$ if X_i 's are discrete and $p_X(x)$ is the common probability mass function of all X_i 's
2. $f(x_1, x_2, \dots, x_n) = f_X(x_1) \cdot f_X(x_2) \cdots f_X(x_n)$ if X_i 's are continuous and $f_X(x)$ is the common probability density function of all X_i 's

For brevity, we will work with discrete random variables. But continuous variables case is similar

Definition 6.4

Let T be a statistic for the random sample $\{X_1, X_2, \dots, X_n\}$, i.e a random variable calculated using random variables $X_1, X_2, \dots, X_n$. Then, T is a function defined as

$$T : \mathcal{X}^n \longrightarrow \mathbb{R}$$

where $\mathcal{X}^n =$ joint sample space of $X_1, X_2, \dots, X_n$

To calculate the sampling distribution of T , we need to use the joint distribution of the random variables $X_1, X_2, \dots, X_n$. Let $p(x_1, x_2, \dots, x_n)$ be the joint probability mass of $X_1, X_2, \dots, X_n$ assuming the X_i 's are discrete.

Then, we want the probability mass function of T , i.e for $t \in \mathcal{T}$, we want $P(T = t)$ where $\mathcal{T} :=$ values of T

Definition 6.5

Let $E(t) = \{(x_1, x_2, \dots, x_n) \in \mathcal{X}^n; T(x_1, x_2, x_3, \dots, x_n) = t\}$. Then

$$\begin{aligned} P(T = t) &= \sum_{(x_1, x_2, \dots, x_n) \in E(t)} p(x_1, x_2, \dots, x_n) \\ &= \sum_{(x_1, x_2, \dots, x_n) \in E(t)} p_X(x_1) p_X(x_2) \cdots p_X(x_n) \end{aligned}$$

where $p_X(x)$ is the common probability mass function of $X_1, X_2, \dots, X_n$

Example 6.3.1

The Sample Mean and Sample Total:

Let $X = \{X_1, X_2, \dots, X_n\}$ be random sample coming from a population with mean $= \mu$ and variance $= \sigma^2$, i.e

$$\begin{aligned} E(X_i) &= \mu & \text{for } i = 1, 2, \dots, n \\ V(X_i) &= \sigma^2 & \text{for } i = 1, 2, \dots, n \end{aligned}$$

The sample total for $\{X_1, X_2, \dots, X_n\}$ is

$$T_o := X_1 + X_2 + \cdots + X_n$$

And the sample mean for $\{X_1, X_2, \dots, X_n\}$ is

$$\begin{aligned} \bar{X} &:= \frac{X_1 + X_2 + \cdots + X_n}{n} \\ &= \frac{T_o}{n} \end{aligned}$$

Then, we can calculate

$$\begin{aligned} E(T_o) &= E(X_1 + X_2 + \cdots + X_n) \\ &= E(X_1) + E(X_2) + \cdots + E(X_n) \\ &= n\mu \end{aligned}$$

and

$$\begin{aligned} V(T_o) &= V(X_1 + \cdots + X_n) \\ &= V(X_1) + \cdots + V(X_n) \\ &= \sigma^2 + \cdots + \sigma^2 \\ &= n\sigma^2 \end{aligned}$$

Similarly, $E(\bar{X}) = \mu$ and $V(\bar{X}) = \frac{\sigma^2}{n}$

If we want to calculate the sampling distribution of $\bar{X}$ and T_o , we have to use Central Limit Theorem!

6.3.2 Central Limit Theorem

Theorem 6.6

Central Limit Theorem (Normal Case):

Suppose $\{X_1, \dots, X_n\}$ is a random sample coming from a population with Normal distribution $N(\mu, \sigma^2)$. Then, for any $n \in \mathbb{N}$, the distribution of the sample mean $\bar{X}$ and the sample total T_o are

$$\begin{aligned} \bar{X} &\sim N\left(\mu, \frac{\sigma^2}{n}\right) \\ T_o &\sim N(n\mu, n\sigma^2) \end{aligned}$$

Similarly, the random variables

$$\frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} \text{ and } \frac{T_o - n\mu}{\sqrt{n}\sigma}$$

have the standard normal distribution

Note: For Central Limit Theorem to hold for all $n \in \mathbb{N}$ (especially when n is small), we need the population to be normally distributed. However, we can drop normality assumption if n is large

Theorem 6.7

Central Limit Theorem (General Case):

If $\{X_1, X_2, \dots, X_n\}$ is a random sample from a population with

$$\begin{aligned} \text{mean} &= \mu \\ \text{variance} &= \sigma^2 \end{aligned}$$

Then, if n is large enough, the distribution of $\bar{X}$ and T_o is approximately normal, i.e

$$\begin{aligned} \bar{X} &\sim N\left(\mu, \frac{\sigma^2}{n}\right) \\ T_o &\sim N(n\mu, n\sigma^2) \end{aligned}$$

the approximation improves as $n \rightarrow \infty$, i.e the distribution of the random variables

$$\frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} \text{ and } \frac{T_o - n\mu}{\sqrt{n}\sigma}$$

approaches the standard normal distribution as $n \rightarrow \infty$

Example 6.3.2

Suppose the population = $\{0, 1\}$. The distribution of the population is the Bernoulli distribution, i.e

$$\begin{aligned} P(X = 1) &= P(\text{success}) = p = P_X(1) \\ P(X = 0) &= P(\text{failure}) = 1 - p = P_X(0) \end{aligned}$$

Let $\{X_1, X_2, \dots, X_n\}$ be a random sample from $\{0, 1\}$. Then, we want to calculate the sampling distribution of T_o .

Suppose $n = 2$, then $\mathcal{X}^2 = \{(0, 0), (0, 1), (1, 0), (1, 1)\}$, and values of $T_o = \{0, 1, 2\}$. Also, $p(x, y) =$

$p_X(x) \cdot p_X(y)$ be the joint probability mass function of X_1 and X_2 .

Then,

$$P(T_o = 0) = p(0, 0) = p_X(0) \cdot p_X(0) = (1 - p)^2$$

$$\begin{aligned} P(T_o = 1) &= p(1, 0) + p(0, 1) = p_X(0) \cdot p_X(1) + p_X(1) \cdot p_X(0) \\ &= (1 - p)p + p(1 - p) = 2p \cdot (1 - p) \end{aligned}$$

$$P(T_o = 2) = p(1, 1) = p_X(1) \cdot p_X(1) = p^2$$

Putting this into a table gives

t	0	1	2
$P(T_o = t)$	$(1 - p)^2$	$2p(1 - p)$	p^2

Suppose $n = 3$, now

$$\mathcal{X}^3 = \left\{ \begin{array}{cccc} (0, 0, 0) & (1, 0, 0) & (1, 1, 0) & (1, 1, 1) \\ & (0, 1, 0) & (1, 0, 1) & \\ & (0, 0, 1) & (0, 1, 1) & \end{array} \right\}$$

values of $T_o = \{0, 1, 2, 3\}$

$$p(x_1, x_2, x_3) = p_X(x_1) \cdot p_X(x_1) \cdot p_X(x_2)$$

Now, we have

$$P(T_o = 0) = p(0, 0, 0) = p_X(0) \cdot p_X(0) \cdot p_X(0) = (1 - p)^3$$

$$\begin{aligned} P(T_o = 1) &= p(1, 0, 0) + p(0, 1, 0) + p(0, 0, 1) \\ &= p_X(1) \cdot p_X(0) \cdot p_X(0) + p_X(0) \cdot p_X(1) \cdot p_X(0) + p_X(0) \cdot p_X(0) \cdot p_X(1) \\ &= p(1 - p)(1 - p) + (1 - p)p(1 - p) + (1 - p)(1 - p)p \\ &= 3(1 - p)^2 p \end{aligned}$$

$$\begin{aligned} P(T_o = 2) &= p(1, 1, 0) + p(0, 1, 1) + p(1, 0, 1) \\ &= p_X(1) \cdot p_X(1) \cdot p_X(0) + p_X(0) \cdot p_X(1) \cdot p_X(1) + p_X(1) \cdot p_X(0) \cdot p_X(1) \\ &= 3p^2(1 - p) \end{aligned}$$

$$P(T_o = 3) = p(1, 1, 1) = p_X(1) \cdot p_X(1) \cdot p_X(1) = p^3$$

Then the probability distribution table of T_o is

t	0	1	2	3
$P(T_o = t)$	$(1 - p)^3$	$3p(1 - p)^2$	$3(1 - p)p^2$	p^3

Suppose $n = 4$, now,

$$\mathcal{X}^3 = \left\{ \begin{array}{ccccc} (0, 0, 0) & (1, 0, 0, 0) & (1, 1, 0, 0) & (1, 1, 1, 0) & (1, 1, 1, 1) \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ \cdots & (0, 0, 0, 1) & (0, 0, 1, 1) & (0, 1, 1, 1) & \cdots \end{array} \right\}$$

And the joint distribution is

$$p(x_1, x_2, x_3, x_4) = p_X(x_1) \cdot p_X(x_2) \cdot p_X(x_3) \cdot p_X(x_4)$$

And the values of T_o is

$$T_o = \{0, 1, 2, 3, 4\}$$

And the distribution of T_o can be calculated as

t	0	1	2	3	4
$P(T_o = t)$	$(1 - p)^4$	$4p(1 - p)^3$	$4(1 - p)^2 p^2$	$4(1 - p)p^3$	p^4

In general for arbitrary n , values of $T_o = \{0, 1, 2, 3, \dots, n\}$ and

$$P(T_o = k) = \binom{n}{k} (1-p)^{n-k} \cdot p^k \quad k = 0, 1, 2, \dots, n$$

Note that this is the same as probability mass function of the Binomial distribution with parameters n and p , Therefore,

$$T_o \sim \text{Bin}(n, p)$$

This make sense because to count the number of successes (1) for given sample of size n , where each sample entry is an outcome of independent Bernoulli trial is the Binomial Experiment. Then

$$\begin{aligned} T_o \sim \text{Bin}(n, p) &\implies E(T_o) = np \\ &V(T_o) = np(1-p) \end{aligned}$$

Also, the central limit theorem says for large enough n , $T_o \sim N(n\mu_X, n\sigma_X^2) = N(np, np \cdot (1-p))$ where $\mu_X = \text{mean}(X) = p$ and $\sigma_X^2 = \text{Var}(x) = p(1-p)$.

$T_o \sim \text{Bin}(n, p) \implies$ for large n , $N(np, np(1-p))$. This is Normal approximation for Binomial

7 | Statistical Inference

Statistical inference is the process of using data to make inferences about the underlying distribution that generated the data. All of the following can be phrased in these terms.

Example 7.0.1

We are given a coin that comes up heads with some probability p . Estimate the value of p .

Example 7.0.2

Estimate the CDF of heights for male students (at a fixed point in time, say this current semester) at UMD.

Example 7.0.3

A company claims that their new line of batteries last 100 hours under regular use. Is their claim accurate?

Example 7.0.4

Four fertilizers are available for use that, on paper, look as good as each other. Is this really the case?

In the following chapters, we try to develop techniques that answer some of these questions. We start with point estimators.

Point estimators

Point estimators are an attempt at estimating a quantity of interest about the population. We have already seen this idea in play: the sample average is how we estimate the population mean (i.e the sample average is a point estimator for the population mean) and the sample variance is a point estimator for the population variance.

Definition 7.1

Suppose that θ is a parameter. An **estimator** for θ , denoted by $\hat{\theta}$ (and $\hat{\theta}_n$ if we want to emphasize the sample size) is a statistic (i.e a function defined on the random sample).

Example 7.1.1

$$\bar{X} = \frac{X_1 + \dots + X_n}{n}$$
$$T = \frac{X_1^2 + X_2^2 + \dots + X_n^2}{n^2 + 1}$$

$\bar{X}$ and T are both statistics, and hence qualify as estimators for $E(X)$.

Once a *random sample has been realized* (i.e once observations are made) we can use the observed values along with the estimator to *create* an estimate. For example, suppose $n = 3$, $x_1 = 1$, $x_2 = 2$, and $x_3 = 3$ in the above. Then

$$\bar{x} = \frac{1 + 2 + 3}{3}$$
$$t = \frac{1^2 + 2^2 + 3^2}{3} = \frac{1 + 4 + 9}{3} = \frac{14}{3}$$

The above example seems a little strange. We know $\bar{X}$ has been used repeatedly to estimate $E(X)$ but we have never used T . The reason is that $\bar{X}$ has mathematical properties that make it *useful* in estimating $E(X)$ while T does not have these properties. This shows that simply being an estimator is likely not enough to be a *useful* estimator. We need something more.

7.1.1 Unbiasedness, Consistency and Mean Squared Error

We now turn our attention towards what makes a good (useful) estimator. There are several notions that are interrelated. We look at a couple below.

Let $X_1, \dots, X_n$ be a sample of size n . Let θ_n be an estimator for θ (i.e. θ_n is a statistic that we want to use to make inferences about θ). Now θ_n is said to be:

1. is said to be *unbiased* if $E(\theta_n) = \theta$ for all possible values of θ .
2. has a *mean square error* (MSE) given by

$$\text{MSE}(\theta_n) = E((\theta_n - \theta)^2)$$

3. is said to be *consistent* if for every $\varepsilon > 0$,

$$\lim_{n \rightarrow \infty} P(|\theta_n - \theta| > \varepsilon) = 0$$

Note:

- Unbiasedness indicates that we get the correct answer on average.
- Consistency means that as the sample size (denoted by n) increases, the probability of making a *significant error* (measured by ε) decreases towards zero.
- The mean squared error measures how much the estimator deviates from the actual value on average. Given two (or more) estimators the one with the least mean squared error can be deemed to be “good”.
- In order to keep the notation somewhat simple, in many situations we will abuse notation slightly and drop the subscript n which is used to emphasize the sample size. For example, we may talk about $\bar{X}$ being a consistent estimator for μ when it is more precise to talk about $\bar{X}_n$ being consistent for μ .
- $\bar{X}$ is both an unbiased and consistent estimator for $E(X) = \mu$. The latter half of the statement relating to $\bar{X}$ being consistent is called the weak law of large numbers.
- S^2 (the sample variance) is an unbiased estimator for σ^2 . In fact the (up to now mysterious) division by $n - 1$ of $\sum_{i=1}^n (X_i - \bar{X})^2$ is related to this. However the more “natural estimator” $T = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2$ for σ^2 is a biased estimator.
- The principle of unbiased estimation states that we prefer unbiased estimators over once that are not.

Let $X_1, \dots, X_n$ be a random sample with $E(X_1) = \mu$ and $V(X_1) = \sigma^2$. Let’s show that $\bar{X}$ is an unbiased estimator for μ .

$$\begin{aligned} E(\bar{X}) &= E\left(\frac{1}{n} \sum_{i=1}^n X_i\right) \\ &= \frac{1}{n} E\left(\sum_{i=1}^n X_i\right) \\ &= \frac{1}{n} \sum_{i=1}^n E(X_i) \\ &= \frac{1}{n} n\mu = \mu \end{aligned}$$

However, this is not the only unbiased estimator.

Exercise: Show that X_1 and $\frac{3X_1+4X_2}{7}$ are also all unbiased estimators of μ .

To determine which of these estimators is better, we can turn towards comparing their mean squared errors. But before we look at the details, it helps to develop a little bit more terminology and results that relate the concepts of unbiasedness and mean squared error together.

Definition 7.2

The quantity $E(\hat{\theta}) - \theta$ is called the *bias* of the estimator θ . We will denote it by $\text{bias}(\theta)$.

We can now rewrite the mean squared error as follows:

$$\text{MSE}(\hat{\theta}) = (\text{bias}(\hat{\theta}))^2 + V(\hat{\theta})$$

For unbiased estimators which have a bias of 0, the equation becomes

$$\text{MSE}(\hat{\theta}) = V(\hat{\theta})$$

So computing the mean squared errors we obtain:

$$\begin{aligned}
 V(\bar{X}) &= V\left(\frac{1}{n} \sum_{i=1}^n X_i\right) \\
 &= \frac{1}{n^2} V\left(\sum_{i=1}^n X_i\right) \\
 &= \frac{1}{n^2} \sum_{i=1}^n V(X_i) \text{ (as } X_i \text{ are independent)} \\
 &= \frac{1}{n^2} n \sigma^2 = \frac{\sigma^2}{n}
 \end{aligned}$$

$$V(X_1) = \sigma^2$$

$$\begin{aligned}
 V\left(\frac{3X_1 + 4X_2}{7}\right) &= \frac{1}{49} (9V(X_1) + 16V(X_2)) \\
 &= \frac{25}{49} \sigma^2 \approx \frac{\sigma^2}{2}
 \end{aligned}$$

So, for a larger sample (in fact, for $n \geq 3$ in this example),

$$V(\bar{X}) < V\left(\frac{3X_1 + 4X_2}{7}\right) < V(X_1)$$

Thus $V(\bar{X})$ seems like the better choice.

Definition 7.3

The *MVUE* (*minimum variance unbiased estimator*) tells us that when we have multiple unbiased estimators, we should pick the one with the lowest variance.

Constructing Estimators

So far we have discussed what makes a point estimator good. What we have not done is discuss how to construct a point estimator. In this section, we talk about how to construct point estimators that under certain technical conditions have good properties.

The setup: We have a random sample $X_1, \dots, X_n$ with the common probability mass function/probability density function $f(x; \theta)$. We use θ to emphasize the fact that there may be more than one unknown parameter (for example, a normal distribution has two parameters, mean and variance, that are required to fully determine it). The notions we have discussed above such as consistency, can be extended to vectors of parameters as well. We do not attempt to do so here.

7.2.1 The method of moments

We start our discussion with the method of moments (abbreviated as **MM**). The method of moments relies on:

1. $\frac{1}{n} \sum_{i=1}^n X_i^k$ being a good point estimator for the k th moment $E(X^k)$, where X is an R.V with the population distribution. This is due to the law of large numbers.
2. Exploiting various relationships between the parameters and the moments.

Example 7.2.1

Suppose $X_1, \dots, X_n$ is a random sample with $X_i \sim \text{Ber}(p)$. p is a fixed but unknown number. Construct a point estimator $\hat{p}_{MM}$ using the method of moments.

Let X be a random variable with the same population distribution as the X_i s, i.e $X \sim \text{Ber}(p)$. Now $E(X) = p$. Also, $\frac{1}{n} \sum_{i=1}^n X_i$ is a point estimator for $E(X)$, thus the first moment $E(X)$ can be estimated by $\frac{1}{n} \sum_{i=1}^n X_i$. But since $p = E(X)$, then p can be estimated by

$$\hat{p}_{MM} = \frac{1}{n} \sum_{i=1}^n X_i$$

Example 7.2.2

Let $X_1, \dots, X_n$ be a random sample with $X_i \sim \mathcal{N}(\mu, \sigma^2)$, where μ and σ^2 are fixed but unknown numbers. Construct point estimators $\hat{\mu}_{MM}$ and $\hat{\sigma}_{MM}^2$ for σ^2 .

Let $X \sim \mathcal{N}(\mu, \sigma^2)$. Now $E(X) = \mu$ and $V(X) = \sigma^2$. But $V(X) = E(X^2) - [E(X)]^2$. Note that $\frac{1}{n} \sum X_i$ is a point estimator for $E(X)$, and $\frac{1}{n} \sum X_i^2$ is a point estimator for $E(X^2)$. Thus we can construct the method of moments estimators

$$\hat{\mu}_{MM} = \frac{1}{n} \sum_{i=1}^n X_i$$

$$\hat{\sigma}_{MM}^2 = \frac{1}{n} \sum_{i=1}^n X_i^2 - \left(\frac{1}{n} \sum_{i=1}^n X_i \right)^2$$

In algorithmic form, the method of moments can be written down as:

1. Step 1: Given parameters $\theta_1, \dots, \theta_k$, write as many moments $E(X), \dots, E(X^m)$ as needed as functions of the parameters θ_i .
2. Step 2: Solve for θ_i in terms of the moments.
3. Step 3: Form the estimators $\hat{\theta}_{iMM}$.

Based on our examples so far, it is tempting to conclude that $k = m$. However, this is not the case: If $X_1, \dots, X_n$ be a random sample with $X_i \sim U[-\theta, \theta]$, then $E(X_i) = 0$ so we actually need to consider $E(X_i^2)$ in order to construct an estimator for θ .

Note:

1. Our example with a random sample $X_1, \dots, X_n$ with $X_1 \sim \text{Ber}(p)$ is related to our example in the previous section of attempting to determine the probability with which a coin comes up heads (Example 7.0.1). A natural approach to estimating such a probability would be tossing the coin multiple times and then computing the estimate as

$$\frac{\# \text{ of heads}}{\# \text{ of tosses}}$$

This is exactly what the $\hat{p}_{MM}$ suggests that you do!

2. The method of moments can yield biased estimators (for example, $\hat{\sigma}_{MM}^2$ in Example 7.2.2 is biased).
3. Under certain technical conditions, the method of moments gives a consistent estimator. These topics are beyond the scope of this class.

7.2.2 Maximum likelihood estimation

We now turn our attention to maximum likelihood estimation (abbreviated as **MLE**). The idea behind maximum likelihood estimators is to find an estimator $\hat{\theta}$ that maximizes the probability of the observed sample.

Maximum likelihood has two steps.

- *Step I:* Given a random sample $X_1, \dots, X_n$ with a common probability mass/probability density function $f(x; \underline{\theta})$, compute the joint probability mass/probability density function. We typically write:

$$\prod_{i=1}^n f(x_i; \underline{\theta})$$

Here, we are using the independence of the X_i 's to write the joint pmf/joint pdf as a product of the individual pmfs/pdfs.

- *Step II:* Define the likelihood function:

$$\mathcal{L}(\underline{\theta}) = \prod_{i=1}^n f(x_i; \underline{\theta})$$

Find the $\hat{\theta}$ which maximizes $\mathcal{L}(\underline{\theta})$.

Notice that we are thinking of the joint pmf/joint pdf as a function of $\underline{\theta}$. The idea is that when we are given observations $x_1, \dots, x_n$ *they are fixed* values. We seek to find the $\underline{\theta}$ that maximizes $\mathcal{L}(\underline{\theta})$, where $\underline{\theta}$ is the parameter (or collection of parameters) with values which are unknown to us. Further, step II requires us to optimize $\mathcal{L}(\underline{\theta})$. This suggests that calculus techniques will come into play.

Example 7.2.3

Let $X_1, \dots, X_n$ be a random sample with $X_1 \sim \text{Ber}(p)$. We find the MLE for p .

Note that given $X \sim \text{Ber}(p)$, the probability mass is given by

$$f(x; p) = p^x(1 - p)^{1-x}$$

for $x = 0, 1$ and 0 otherwise.

Let's take $\theta = p$. Now we have

$$\mathcal{L}(\theta) = \prod_{i=1}^n \theta^{x_i} (1 - \theta)^{1-x_i}$$

We want to find the value of p that maximizes the above. A natural way to do this is by using calculus: Find the critical value for $\mathcal{L}(p)$. However, we will run into some computational difficulties if we try to compute $\mathcal{L}'(p)$ (we would need to use the product rule multiple times). We now pause the example for the moment and introduce the log-likelihood function that allows us to tackle this problem:

Definition 7.4

The log-likelihood function $\ell(\underline{\theta})$ is defined as $\ell(\underline{\theta}) = \ln \mathcal{L}(\underline{\theta})$.

Note that:

$$\ell(\underline{\theta}) = \ln \prod_{i=1}^n f(x_i, \underline{\theta}) = \sum_{i=1}^n \ln f(x_i; \theta)$$

Thus, the *log-likelihood is far easier to differentiate*, since the derivative of a sum is easier to compute than that of a product.

Exercise: Explain why finding a $\underline{\theta}$ that maximizes $\mathcal{L}(\underline{\theta})$ is equivalent to finding a $\underline{\theta}$ that maximizes $\ell(\underline{\theta})$. (Hint: $\ln(x)$ is an increasing function).

Example continued.

Now, we will write out the log-likelihood, take the derivative, and complete the derivation.

$$\begin{aligned}
 \mathcal{L}(\theta) &= \prod_{i=1}^n \theta^{x_i} (1 - \theta)^{1-x_i} \\
 \ell(\theta) &= \ln \prod_{i=1}^n \theta^{x_i} (1 - \theta)^{1-x_i} \\
 &= \sum_{i=1}^n \ln (\theta^{x_i} (1 - \theta)^{1-x_i}) \\
 &= \sum_{i=1}^n (\ln \theta^{x_i} + \ln (1 - \theta)^{1-x_i}) \\
 &= \sum_{i=1}^n (x_i \ln \theta + (1 - x_i) \ln (1 - \theta)) \\
 \ell'(\theta) &= \frac{d}{d\theta} \left[\sum_{i=1}^n (x_i \ln \theta + (1 - x_i) \ln (1 - \theta)) \right] \\
 &= \sum_{i=1}^n \left(\frac{d}{d\theta} (x_i \ln \theta + (1 - x_i) \ln (1 - \theta)) \right) \\
 &= \sum_{i=1}^n \left(\frac{x_i}{\theta} + \frac{1 - x_i}{1 - \theta} (-1) \right) \\
 &= \sum_{i=1}^n \left(\frac{(1 - \theta)x_i - \theta(1 - x_i)}{\theta(1 - \theta)} \right) \\
 &= \sum_{i=1}^n \left(\frac{(1 - \theta)x_i - \theta(1 - x_i)}{\theta(1 - \theta)} \right) \\
 &= \sum_{i=1}^n \left(\frac{x_i - \theta x_i - \theta + \theta x_i}{\theta(1 - \theta)} \right) \\
 &= \sum_{i=1}^n \left(\frac{x_i - \theta}{\theta(1 - \theta)} \right)
 \end{aligned}$$

Setting $\mathcal{L}'(\theta) = 0$ we obtain

$$\begin{aligned}
 \frac{1}{\hat{\theta}(1 - \hat{\theta})} \sum_{i=1}^n (x_i - \hat{\theta}) &= 0 \\
 \left(\sum_{i=1}^n x_i \right) - n\hat{\theta} &= 0 \\
 \hat{\theta} &= \frac{1}{n} \sum_{i=1}^n x_i
 \end{aligned}$$

Thus, the maximum likelihood estimate is $\hat{p}_{MLE} = \frac{1}{n} \sum_{i=1}^n x_i$ and the maximum likelihood estimator is

$$\hat{p}_{MLE} = \frac{1}{n} \sum_{i=1}^n X_i$$

Example 7.2.4

Suppose that $X_1, \dots, X_n$ is an (iid) random sample with $X_1 \sim \mathcal{N}(\mu, \sigma^2)$ where both μ, σ^2 are fixed but unknown. Find the MLEs for μ, σ^2 .

Take $\theta_1 = \mu, \theta_2 = \sigma^2$. Then the pdf of a single random variable X_i is given by:

$$\frac{1}{\sqrt{2\pi\theta_2}} e^{-\frac{(x-\theta_1)^2}{2\theta_2}}$$

Now, we proceed to write the likelihood and log-likelihood.

$$\begin{aligned}
 \mathcal{L}(\theta_1, \theta_2) &= \prod_{i=1}^n \frac{1}{\sqrt{2\pi\theta_2}} e^{-\frac{(x_i - \theta_1)^2}{2\theta_2}} \\
 \ell(\theta_1, \theta_2) &= \sum_{i=1}^n \ln \left(\frac{1}{\sqrt{2\pi\theta_2}} e^{-\frac{(x_i - \theta_1)^2}{2\theta_2}} \right) \\
 &= \sum_{i=1}^n \left[-\frac{1}{2} \ln(2\pi\theta_2) + \ln e^{-\frac{(x_i - \theta_1)^2}{2\theta_2}} \right] \\
 &= \sum_{i=1}^n -\frac{1}{2} \ln(2\pi) - \frac{1}{2} \ln(\theta_2) - \frac{(x_i - \theta_1)^2}{2\theta_2} \\
 &= -\frac{n}{2} \ln(2\pi) - \frac{n}{2} \ln(\theta_2) - \frac{1}{2\theta_2} \sum_{i=1}^n (x_i - \theta_1)^2
 \end{aligned}$$

Next, the critical points $(\hat{\theta}_1, \hat{\theta}_2)$ of $\ell(\theta_1, \theta_2)$ should satisfy the equations

$$\begin{aligned}
 \left. \frac{\partial \ell}{\partial \theta_1} \right|_{(\theta_1, \theta_2) = (\hat{\theta}_1, \hat{\theta}_2)} &= 0 \\
 \left. \frac{\partial \ell}{\partial \theta_2} \right|_{(\theta_1, \theta_2) = (\hat{\theta}_1, \hat{\theta}_2)} &= 0
 \end{aligned}$$

Writing out the derivatives, we get

$$\begin{aligned}
 \frac{\partial \ell}{\partial \theta_1} &= -\frac{1}{2\theta_2} \sum_{i=1}^n 2(x_i - \theta_1)(-1) \\
 &= \frac{1}{\theta_2} \sum_{i=1}^n (x_i - \theta_1) \\
 \frac{\partial \ell}{\partial \theta_2} &= -\frac{n}{2\theta_2} + \frac{1}{2\theta_2^2} \sum_{i=1}^n (x_i - \theta_1)^2
 \end{aligned}$$

Solving for $(\hat{\theta}_1, \hat{\theta}_2)$ that simultaneously make $\frac{\partial \ell_1}{\partial \theta_1}, \frac{\partial \ell_2}{\partial \theta_2}$ equal 0 we obtain:

$$\begin{aligned}
 \hat{\theta}_1 &= \frac{1}{n} \sum_{i=1}^n x_i \\
 \hat{\theta}_2 &= \frac{1}{n} \sum_{i=1}^n (x_i - \hat{\theta}_1)^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2
 \end{aligned}$$

So far, the method of moments and maximum likelihood estimators have both given us the same estimators. The following example illustrates this is not always the case.

Example 7.2.5

Let $X_1, \dots, X_n$ be a random sample with $X_1 \sim U[0, \theta]$ where $\theta > 0$ is fixed but unknown. Now

$$\hat{\theta}_{MM} = \frac{2}{n} \sum_{i=1}^n X_i$$

but on the other hand,

$$\hat{\theta}_{MLE} = \max\{X_1, \dots, X_n\}$$

Under suitable conditions, maximum likelihood estimators are consistent as well. In addition, they are:

1. Invariant: Given a function g , the MLE for $g(\theta)$ is $g(\hat{\theta}_{MLE})$.
2. Asymptotically unbiased: Even if the MLE is biased, as the sample size increases, the bias goes to zero.

There are further properties that we do not discuss here.

A quick look back

Now that we have discussed point estimators, here are some important facts about how we approached the topics.

1. Values of parameters are *fixed* but they are not known to us.
2. When constructing estimators we took a *parametric approach*, i.e we assumed that we knew the distribution of the population but not the value of the parameters. If these assumptions are incorrect, the information given by the estimators will not be useful.
3. Notions of good for estimators depend on having good long run properties: on average, over a large sample (i.e. over a large number of observations from repeated repetitions).

Exercises for Chapter 7

Exercise 7.1. Show that S^2 is an unbiased estimator for σ^2 , but $T = \frac{1}{n} \sum (X_i - \bar{X})^2$ is not. *Hint: $T = \frac{n}{n-1} S^2$, so $E(T)$ is easy to compute. The hard work goes into showing S^2 is unbiased.*

Exercise 7.2. Compute the bias, variance and mean squared error for $T = \frac{n}{n-1} S^2$.

Exercise 7.3. Chebychev's inequality tells us that given any random variable X with $E(X) = \mu$ and $V(X) = \sigma^2$, we have (for any $c > 0$)

$$P(|X - \mu| \geq c) \leq \frac{\sigma^2}{c^2}$$

Let $X_1, \dots, X_n, \dots$ be i.i.d random variables that have the same distribution as X . Use this to prove the *weak law of large numbers* for all such random variables.

Exercise 7.4. Consider Example 7.0.1 where you are attempting to determine the probability p of a coin coming up heads. Suggest an experiment to approximate the value of p .

Exercise 7.5. The following 10,000 numbers (hosted on ELMS) were generated from a Uniform distribution on $[0, \theta]$. Suggest an estimator $\hat{\theta}$ to estimate θ and use R to obtain the estimate based on the data.

Exercise 7.6. The following 10,000 numbers (hosted on ELMS) were generated from an Exponential distribution, i.e random samples of the random variable $X \sim \text{Exp}(\lambda)$. Suggest an estimator $\hat{\lambda}$ to estimate λ and use R to obtain the estimate based on the data.

Exercise 7.7. Consider a random variable X with $E(X) = \mu$ and $V(X) = \sigma^2$. Let $X_1, \dots, X_n$ be i.i.d, with each $X_i \sim X$ (i.e, a random sample of the variable X). Let

$$T = k \sum_{i=1}^n i X_i$$

You are given that T is unbiased. What is the value of k ?

Exercise 7.8. Suppose that $X_1, \dots, X_n$ is a random sample with $X_1 \sim \text{exp}(\lambda)$. Compute the MM estimator and the MLE for λ .

Exercise 7.9. Suppose that $X_1, \dots, X_n$ is a random sample where the pdf of X_1 is given by f (defined below).

$$f(x; \theta) = \begin{cases} \theta x^{\theta-1} & 0 \leq x \leq 1 \\ 0 & \text{o.w} \end{cases}$$

Let $\theta > 1$ be fixed. Compute the MM estimator and MLE for θ .

Exercise 7.10. Let $X_1, \dots, X_n$ be a random sample with $X_1 \sim U[0, \theta]$ where $\theta > 0$ is fixed but unknown.

1. Compute $\hat{\theta}_{MM}$.

2. Compute $\hat{\theta}_{MLE}$.

Hint: $\mathcal{L}(\theta) = \frac{1}{\theta^n}$. To obtain the estimator, explain why $\theta \geq \max\{x_1, \dots, x_n\}$. Also, explain why if we have $\theta_1 < \theta_2$, then we have $\mathcal{L}(\theta_1) > \mathcal{L}(\theta_2)$.

3. Which do you think is better in this case, $\hat{\theta}_{MM}$ or $\hat{\theta}_{MLE}$? Verify your guess empirically by writing a suitable program in *R*.

8 | Confidence Intervals

In the previous chapter, we used a given sample $X_1, \dots, X_n$ to construct estimates of various parameters of interest ($\bar{X}$ for μ , S^2 for σ^2 , etc.) In this chapter, we focus on using a sample to *create intervals* that are likely to contain a parameter of interest.

Note: It is the *interval being constructed* that depends on the sample, **not the value of the parameter** (which while unknown to us is **fixed** and does not depend on the sample).

Definition 8.1

A $1 - \alpha$ confidence interval (abbreviated as **CI**) for a parameter θ is an interval

$$I(X_1, \dots, X_n) = ((a(X_1, \dots, X_n), b(X_1, \dots, X_n)))$$

such that the equality

$$P(\theta \in I(X_1, \dots, X_n)) = 1 - \alpha$$

holds for all possible values of θ .

Note:

- We could require $P(\theta \in I(X_1, \dots, X_n)) \geq 1 - \alpha$.
- α is usually taken to be 0.01, 0.05 or 0.1.

Constructing Confidence Intervals

Let's start with an example to get an idea of what the process might look like.

Example 8.1.1

Suppose that we have a random sample $X_1, \dots, X_n$ with $X_1 \sim \mathcal{N}(\mu, \sigma^2)$. σ^2 is known, but μ is not known. We want to construct an interval $I(X_1, \dots, X_n) = ((a(X_1, \dots, X_n), b(X_1, \dots, X_n)))$ such that

$$P(\mu \in I(X_1, \dots, X_n)) = P(a(X_1, \dots, X_n) < \mu < b(X_1, \dots, X_n)) = 0.99$$

It seems reasonable to try to use our knowledge of $\bar{X}$ in the construction. We start with the following fact:

Fact. If $Y_1, \dots, Y_m$ are independent random variables such that for each i we have $Y_i \sim \mathcal{N}(\mu_i, \sigma_i^2)$, then we have that

$$\sum_{i=1}^m a_i Y_i \sim \mathcal{N}\left(\sum_{i=1}^m a_i \mu_i, \sum_{i=1}^m a_i^2 \sigma_i^2\right)$$

A consequence of this fact is that $\bar{X} \sim \mathcal{N}(\mu, \frac{\sigma^2}{n})$.

Note: This is not an application of the CLT. This result is true even when the sample size is small ($n < 30$) and is exact (i.e, not an approximation) compared to our previous applications of the CLT.

Now note that from our previous knowledge of normal random variables,

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim \mathcal{N}(0, 1)$$

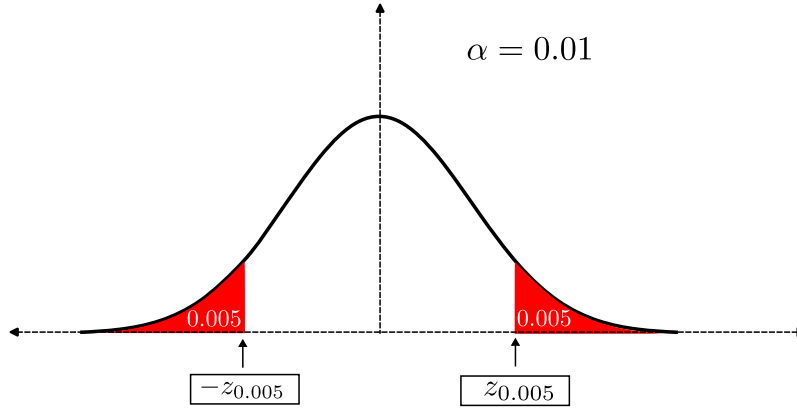


Figure 8.1: Normal distribution, illustrating the meaning of $z_{0.005}$. It is a cutoff point selected so that the total probability mass of the portion shaded in red adds up to α , which is equal to 0.01 in our example.

Thus, we have that (See Figure 8.1)

$$P\left(-z_{0.005} < \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} < z_{0.005}\right) = 0.99$$

$$P\left(-2.5758 < \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} < 2.5758\right) = 0.99$$

$$P\left(\bar{X} - 2.5758 \frac{\sigma}{\sqrt{n}} < \mu < \bar{X} + 2.5758 \frac{\sigma}{\sqrt{n}}\right) = 0.99$$

So, it seems reasonable to have the interval be

$$I(X_1, \dots, X_n) = \left(\bar{X} - 2.5758 \frac{\sigma}{\sqrt{n}}, \bar{X} + 2.5758 \frac{\sigma}{\sqrt{n}}\right)$$

as we have that $P(\mu \in I) = 0.99$.

Let's consider a more concrete example.

Example 8.1.2

Take $n = 10$ and $\sigma^2 = 4$, i.e we have a random sample $X_1, \dots, X_{10}$ where $X_i \sim \mathcal{N}(\mu, 4)$. Recall again that μ is fixed, but unknown to us.

Suppose that we carry out an experiment and obtain a realization of the random variables $x_1, \dots, x_n$. We calculate the sample average $\bar{X} = \frac{1}{10} \sum_i x_i$ and find that $\bar{X} = 10$.

We can use the formula from the previous example (8.1.1) to calculate the CI (with confidence level 99%) for this realization:

$$\left(10 - 2.5758 \frac{4}{\sqrt{10}}, 10 + 2.5758 \frac{4}{\sqrt{10}}\right) = (6.7418, 13.2582)$$

Note:

- The results above heavily depended on the fact that $\bar{X} \sim \mathcal{N}(\mu, \frac{\sigma^2}{n})$. This in turn came from the fact that $X_i \sim \mathcal{N}(\mu, \sigma^2)$, where σ^2 is known.
- The value 0.99 in the above equation gives us the *confidence level*. In general we can specify the confidence level by $1 - \alpha$ (α is called the *significance level*). It is customary to choose α to be 0.01, 0.05 or 1 as was noted earlier.
- Regarding Example 8.1.2, it is **not correct** to say that

$$P(\mu \in (6.7418, 13.2582)) = 0.99$$

In fact, since μ is fixed, we cannot make meaningful statements along the above lines. Confidence intervals should be interpreted using long-run frequencies.

A related (and perhaps more realistic) scenario is one where the observations are from a normal distribution but the population variance is unknown. In such a scenario we can use the following fact.

Fact. Let $X_1, \dots, X_n$ be iid RVs with $X_i \sim \mathcal{N}(\mu, \sigma^2)$, where μ and σ^2 are unknown. Then

$$\frac{\bar{X} - \mu}{s/\sqrt{n}} \sim t_{n-1}$$

where t_{n-1} has a t -distribution with $n - 1$ degrees of freedom. The pdf of the t -distribution with $n - 1$ degrees of freedom (where $n > 1$) is given by:

$$f_T(t) = \frac{\Gamma(\frac{n}{2})}{\sqrt{\pi(n-1)}\Gamma(\frac{n-1}{2})} \left(1 + \frac{t^2}{n-1}\right)^{-\frac{n}{2}}$$

where the function $\Gamma(x)$ is given by

$$\Gamma(x) = \int_0^\infty t^{x-1} e^{-t} dt$$

Much like the normal distribution, it is cumbersome to work directly with the pdf when doing computations. We use tables (or computer software) to work with t -distributions. Further, like the normal distributions, we rely on certain properties of the distribution to carry out computations.

Two important properties that we use when employing t -distributions are:

1. The pdf of t_n is bell-shaped and is symmetric about its mean, which is 0.
2. Let $Z \sim N(0, 1)$. For sufficiently large values of n , $P(Z < t) \approx P(t_n < t)$ for all t .

More precisely,

$$\lim_{n \rightarrow \infty} P(t_n < t) = P(Z < t)$$

Definition 8.2

Suppose that $T \sim t_n$ (with $n > 1$) and $\alpha > 0$. Then the t critical value $t_{\alpha,n}$ is such that

$$P(T > t_{\alpha,n}) = \alpha$$

By manipulating the inequality

$$-t_{\frac{\alpha}{2},n-1} < \frac{\bar{X} - \mu}{s/\sqrt{n}} < t_{\frac{\alpha}{2},n-1}$$

and using the fact that

$$P\left(-t_{\frac{\alpha}{2},n-1} < \frac{\bar{X} - \mu}{s/\sqrt{n}} < t_{\frac{\alpha}{2},n-1}\right) = 1 - \alpha$$

we obtain the following $1 - \alpha$ CI for μ :

$$I = \left(\bar{X} - t_{\frac{\alpha}{2},n-1} \frac{s}{\sqrt{n}}, \bar{X} + t_{\frac{\alpha}{2},n-1} \frac{s}{\sqrt{n}}\right)$$

Example 8.1.3

Suppose that after sampling 15 male students, the average sample height $\bar{X}$ is found to be 178cm, and the *sample* standard deviation s is found to be 7.4cm. Construct a 95% CI from for the population mean. You may assume that the heights are normally distributed.

Note that the true underlying standard deviation σ is not known. Since $1 - \alpha = 0.95$, $\alpha = 0.05$.

The CI is thus given by

$$\left(178 - t_{0.025,14} \frac{7.4}{\sqrt{15}}, 178 + t_{0.025,14} \frac{7.4}{\sqrt{15}}\right)$$

Using a t -table look up or computer software,

$$t_{0.025,14} \approx 2.15$$

Thus, the final confidence interval is

$$(173.90, 182.10)$$

Approximate Confidence Intervals

In the previous section we made a key assumption, *the data is being generated from a normal distribution*.

This assumption places a strong constraint on constructing confidence intervals, even for μ . It turns out that we can address these issues if we are willing to allow for approximate confidence intervals.

Definition 8.3

We say that $I(X_1, \dots, X_n)$ is an $1 - \alpha$ approximate confidence interval for θ if

$$\lim_n P(\theta \in I(X_1, \dots, X_n)) = 1 - \alpha$$

By setting

$$P\left(-z_{\alpha/2} \leq \frac{\bar{X} - \mu}{s/\sqrt{n}} \leq z_{\alpha/2}\right) \approx 1 - \alpha$$

and manipulating the above equation as in the previous example, we obtain an approximate CI (valid when the sample size is large)

$$I(X_1, \dots, X_n) = \left(\bar{X} - \frac{s}{\sqrt{n}}z_{\alpha/2}, \bar{X} + \frac{s}{\sqrt{n}}z_{\alpha/2}\right)$$

Exercise: Carry out the computations necessary to obtain the above formula.

Example 8.2.1

A sample of size 393 from a certain population has a sample average of 12.1 and a sample standard deviation of 2.37. Compute a 0.95 (or 95%) approximate CI for μ . Note that $1 - \alpha = 0.95$, $\alpha = 0.05$.

Plugging the numbers into the formula derived from Definition 8.3, and noting that $z_{0.025} = 1.959964$, we obtain

$$\left(12.1 - \frac{2.37}{\sqrt{393}}z_{0.025}, 12.1 + \frac{2.37}{\sqrt{393}}z_{0.025}\right) = (11.86568, 12.33432)$$

Definition 8.4

The standard error of an estimator $\hat{\theta}$, denoted by $\hat{\theta}$, is given by $\sqrt{V(\hat{\theta})}$. We can estimate $\hat{\theta}$ by introducing a point estimator $\widehat{SE}(\hat{\theta})$ for the standard error.

As an example, consider $\bar{X}$. It has a standard error given by $\bar{X} = \sqrt{\frac{\sigma^2}{n}}$ (since $V(\bar{X}) = \frac{\sigma^2}{n}$ as we have seen). Any many cases the value of the standard error would be unknown but we can use the sample standard deviation S as a point estimator for the standard error, i.e. we can use $S = \widehat{SE}(\bar{X})$.

Fact (based on CLT and other results).

For a large enough sample $X_1, \dots, X_n$ (the rule of thumb being $n > 30$), the distribution of the following statistic is approximately normal.

$$\frac{\bar{X} - \mu}{\widehat{SE}(\bar{X})} \sim \mathcal{N}(0, 1) \text{ approximately}$$

Since $V(\bar{X}) = \frac{\sigma^2}{n}$ and s^2 is a consistent estimator for σ^2 we can rewrite the above as

$$\frac{\bar{X} - \mu}{s/\sqrt{n}} \sim \mathcal{N}(0, 1) \text{ approximately}$$

Example 8.2.2

A sample of 75 households from a particular neighborhood contained 12 households with annual income that exceeded \$200,000. Use this information to construct a 99% confidence interval for the population proportion p of households that have annual incomes that exceed \$200,000 in this particular neighborhood.

Note that we can interpret the natural point estimator for p , i.e

$$\hat{p} = \frac{\# \text{ of households that earn more than \$200,000}}{\# \text{ of households}}$$

as a sample average. Take $X_i \sim \text{Ber}(p)$ with p unknown, i.e X_i is 1 if household i exceeds \$200,000 annual income and 0 otherwise. Then

$$\hat{p} = \frac{\sum_{i=1}^n X_i}{n}$$

is exactly the same estimator as above.

Since $E(X_i) = p$ and $V(X_i) = p(1 - p)$, we obtain

$$\frac{\hat{p} - p}{\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}} \sim \mathcal{N}(0, 1) \text{ approximately}$$

Thus we obtain the confidence interval

$$\left(\hat{p} - \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} z_{\alpha/2}, \hat{p} + \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} z_{\alpha/2} \right)$$

Note that this is similar to the previous formula we derived

$$\left(\bar{X} - \frac{s}{\sqrt{n}} z_{\alpha/2}, \bar{X} + \frac{s}{\sqrt{n}} z_{\alpha/2} \right)$$

where $s = \sqrt{V(X_1)}$.

By substituting in the values, we obtain

$$\left(\frac{12}{75} - \sqrt{\frac{(\frac{12}{75})(\frac{63}{75})}{75}} z_{0.005}, \frac{12}{75} + \sqrt{\frac{(\frac{12}{75})(\frac{63}{75})}{75}} z_{0.005} \right) = (0.05096, 0.26904)$$

Note:

- Under certain technical conditions,

$$\frac{\hat{\theta}_{MLE} - \theta}{\widehat{\text{SE}}(\hat{\theta})} \sim \mathcal{N}(0, 1) \text{ approximately for large samples}$$

We can use this to construct approximate CIs for θ using the above computations as a blueprint.

- If $X_1, \dots, X_n \sim \mathcal{N}(\mu, \sigma^2)$ with σ^2 unknown, we can use the fact that

$$\frac{(n-1)s^2}{\sigma^2} \sim \chi_{n-1}^2$$

to construct CIs for σ^2 .

- If we were to construct 0.95 and 0.99 confidence intervals based on the same sample, we'd see that the length of the 0.99 CI is greater than that of the 0.95 CI. This trade-off, between the accuracy (given by the probability) and the precision (related to the length in these examples) is unavoidable: increasing one decreases the other, and vice versa.
- It is possible to construct one sided confidence intervals as well. For example, you can construct a $1 - \alpha$ interval

$$I(X_1, \dots, X_n) = (-\infty, a(X_1, \dots, X_n))$$

by considering the following approach:

$$\begin{aligned} P(\mu \in (-\infty, a(X_1, \dots, X_n))) \\ &= P(\mu < a(X_1, \dots, X_n)) \\ &= P\left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} < z_\alpha\right) \\ &= P(\mu < \bar{X} - z_\alpha \frac{\sigma}{\sqrt{n}}) \end{aligned}$$

- If the population variance is known, it is possible to determine the sample size required to guarantee that the resulting length is below a certain value. To see this, note that the length ℓ of the interval is given by

$$\ell = \frac{2z_{\alpha/2}\sigma}{\sqrt{n}}$$

which, after solving for n , yields

$$n = \left(\frac{2z_{\alpha/2}\sigma}{\ell} \right)^2$$

In case σ is not known, it may be possible to use a stand in value if one has prior knowledge about the situation.

Hypothesis testing with confidence intervals

Consider the following example:

Example 8.3.1

A company creates a new line of soda. They advertise that the new soda contains only 5g of sugar per bottle. You conduct an experiment to see if there is evidence to the contrary for the company's claim. After sampling 36 bottles you find that the average sugar content was 5.1g, with a sample standard deviation of 1g. Does the data contradict the company's claim?

We approach this by first taking a closer look at the claim. A reasonable way to understand the claim is as stating that on average, each bottle contains 5g of soda (think about why!). So, we can view the company's claim as a claim about a parameter (the mean) of an underlying distribution (the amount of sugar in a bottle).

Intuitively (all else being equal), a sample average of 4.99g would agree with the company's claim more than an average of 20g. Phrased differently, a sample average of 20g would suggest that the actual average falls closer to 20g, while a sample average of 4.99g would suggest that the actual average falls close to 4.99.

Thus, we could reasonably expect to find the sample average in an interval around the sample average. Confidence intervals are such intervals. For example, we could choose to reject the company's claim if the stated amount (5g) is not within a 0.95 CI that we construct from the experimental data.

We construct a 0.95 CI using the given data.

$$\begin{aligned} I &= \left(5.1 - 1.96 \frac{1}{\sqrt{36}}, 5.1 + 1.96 \frac{1}{\sqrt{36}} \right) \\ &= (5.1 - 0.3267, 5.1 + 0.3267) \\ &= (4.7733\text{g}, 5.4267\text{g}) \end{aligned}$$

Since 5 falls within the interval, we would not reject the company's claim: the data does not appear to contradict the company's claim.

On the other hand, suppose that the sample average had been 5.6g and the sample standard deviation had remained the same (1g). We would have obtained the interval

$$(5.2733\text{g}, 5.9267\text{g})$$

Since 5g is not inside this confidence interval, we would reject the company's claim: the data does appear to contradict the company's claim.

Note:

1. This question can be phrased in the form of a *hypothesis testing* problem. We check to see whether the null hypothesis H_0 (in our case $H_0 : \mu = 5$ is contradicted by the sample). If it is, we adopt the alternate hypothesis H_a (in our case $H_a : \mu \neq 5$).
2. Hypothesis testing is an important technique in statistics. It can and often is developed independently of confidence intervals (sometimes even before discussing confidence intervals). We have followed a different approach that relies on the (mathematical) connections between the two in certain situations. Hypothesis testing has much broader applications than the example suggests.
3. The general form of the hypothesis test we have conducted is of the following form:

$$\begin{aligned} H_0 : \mu &= \mu_0 \\ H_a : \mu &\neq \mu_0 \end{aligned}$$

For a large sample (rule of thumb $n > 30$) we could follow the above procedure and reject H_0 at a significance level of α if and only if

$$\mu_0 \notin \left(\bar{X} - z_{\alpha/2} \frac{s}{\sqrt{n}}, \bar{X} + z_{\alpha/2} \frac{s}{\sqrt{n}} \right)$$

Otherwise, we fail to reject the null hypothesis.

4. There are two types of errors possible with hypothesis tests.

Type I - The null hypothesis is true, but we reject it.

Type II - The null hypothesis is false, but we fail to reject it.

The significance level α is the probability of making a type I error.

Exercises for Chapter 8

Exercise 8.1. In Example 8.2.2, we used the fact that

$$\frac{\hat{p} - p}{\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}} \sim \mathcal{N}(0, 1) \text{ approximately}$$

to construct a large-sample $1 - \alpha$ CI for p . However, it is more accurate to say that

$$\frac{\hat{p} - p}{\sqrt{\frac{p(1-p)}{n}}} \sim \mathcal{N}(0, 1) \text{ approximately}$$

Use the above to construct a $1 - \alpha$ CI for p .

Exercise 8.2. An experiment was conducted, and you received the following data (shown below). You are told that the underlying distribution is normal, and the standard deviation σ is equal to 1. Construct a 95% CI for μ .

6.5	7.09	6.56	6.91
7.13	7.1	7.76	8.76
6.92	6.8	7.26	6.86
7.89	7.74	7.77	6.89
7.12	7.12	6.19	6.31
7.32	6.97	6.56	6.78
6.42	6.61	6.28	7.18
7.71	7.51	7.23	7.42
6.17	6.09	5.84	8.07
6.64	9.31	7.25	7.97

Exercise 8.3. A botanist working in a greenhouse wants to estimate the average height of a certain species of plant. They measure 10 plants of the same age and recorded the following values (in feet, shown below). Calculate the mean height and construct a 95% CI based upon the t -distribution.

2.00	3.26	2.84	4.77	3.23
3.64	1.84	4.43	1.35	2.28

Exercise 8.4. A student is curious about how much time their classmates spend studying. They surveyed 12 of their classmates and recorded the following times (in hours, shown below). Calculate the mean time and construct a 90% CI based upon the t -distribution.

4.0	5.3	4.8	6.8
5.2	5.6	3.8	6.4
3.3	4.3	4.2	5.2

Exercise 8.5. An experiment was conducted, and you received the following data. You are told that the underlying distribution is normal, but the standard deviation is unknown. Construct an approximate 99% CI for μ .

4.75	5.04	4.78	4.95
5.07	5.05	5.38	5.88
4.96	4.9	5.13	4.93
5.44	5.37	5.39	4.94
5.06	5.06	4.59	4.65
5.16	4.99	4.78	4.89
4.71	4.81	4.64	5.09
5.36	5.26	5.12	5.21
4.59	4.54	4.42	5.53
4.82	6.16	5.12	5.49

Exercise 8.6. An experiment was conducted, and you received the following data (shown below). The underlying distribution is unknown. Construct an approximate 90% CI for the mean.

9.24	5.25	13.74	2.07
7.24	3.38	17.58	8.37
1.05	20.23	5.41	9.21
30.97	11.23	1.93	10.15
6.25	11.31	9.35	19.76
11.74	3.81	3.91	15.77
0.93	0.72	1.25	7.07
17.48	4.22	0.9	22.26
2.5	0.77	5.55	11.42
1.94	4.98	6.55	3.53

Exercise 8.7. An experiment was conducted, and you received the following data. There are 1000 observations (contained in file `1000obvs.csv` on ELMS). Use R to calculate the sample mean and an approximate 95% CI for the mean of the data.

Exercise 8.8. Suppose that customers on a certain website are able to rate a product by inputting a number between 0 and 5. 1000 ratings were collected, and stored in the file `customer_ratings.csv` on ELMS. Use R to calculate the average rating and provide an approximate 90% CI for the mean.

Exercise 8.9. Suppose that you design an experiment to measure the performance of two machines at a factory, Machine 1 and Machine 2. The measure of performance is the amount of time (in minutes) it takes to produce one widget (the smaller the better). You measured the times for 1000 widgets, for each machine, and recorded the data (found in `machine_times.csv` on ELMS).

1. Use R to calculate the mean times for Machine 1 and Machine 2. Construct approximate 95% CIs for the mean times for both machines.
2. Which machine would you pick based on the constructed CIs? Why?

Supplementary exercises:

Exercise 8.10. An experiment was conducted, and you received the following data (shown below). You are told that the underlying distribution is normal, and the standard deviation σ is equal to 3. Construct a 95% CI for μ .

8.49	10.27	8.69	9.73
10.39	10.29	12.29	15.27
9.76	9.4	10.79	9.59
12.66	12.22	12.32	9.67
10.35	10.37	7.56	7.93
10.96	9.91	8.68	9.33
8.25	8.83	7.84	10.55
12.14	11.53	10.69	11.25
7.52	7.26	6.53	13.2
8.92	16.93	10.74	12.91

Exercise 8.11. An experiment was conducted, and you received the following data. You are told that the underlying distribution is normal, but the standard deviation is unknown. Construct an approximate 99% CI for μ .

2.8	3.05	2.97	3.35
3.05	3.13	2.77	3.29
2.67	2.86	3.04	3.04
2.92	3.3	3.05	2.99
2.84	3.2	2.63	3.92
2.82	3.31	3.1	3.31
2.67	2.82	2.71	3.09
2.54	3.1	2.96	3.7
2.94	2.96	2.72	2.91
3.07	3.17	3.43	3.39

Exercise 8.12. An experiment was conducted, and you received the following data. You are told that the underlying distribution is normal, but the standard deviation is unknown. Construct an approximate 89% CI for μ .

16.99	20.54	17.37	19.45
20.79	20.58	24.58	30.54
19.53	18.79	21.57	19.17
25.32	24.44	24.64	19.33
20.7	20.74	15.11	15.86
21.91	19.82	17.37	18.67
16.51	17.67	15.68	21.1
24.29	23.07	21.39	22.5
15.05	14.52	13.05	26.39
17.84	33.86	21.48	25.82

Exercise 8.13. An experiment was conducted, and you received the following data. The underlying distribution is unknown. Construct an approximate 90% CI for the mean.

3.55	3.96	3.15	4.73
2.25	6.15	3.11	5.42
5.64	3.43	4.59	2.53
4.42	3.7	2.73	6.54
4.73	6.22	3.16	2.05
3.44	4.75	7.2	3.32
4.91	2.77	4.05	6.4
3.73	5.44	4.05	3.74
4.38	4.64	3.93	3.85
4.39	5.45	7.35	1.95

Exercise 8.14. An experiment was conducted, and you received the following data. There are 1000 observations (contained in file `1000obvs_2.csv` on ELMS). Use R to calculate the sample mean and an approximate 95% CI for the mean of the data.

Exercise 8.15. A warehouse manager received a shipment of batteries and wants to estimate the average time one of the batteries will last. They randomly select 1000 batteries, put them into

a battery stress-tester, and record the time it takes for them to fail (in minutes). The resulting data was recorded in `battery_life.csv` (on ELMS). Use R to calculate the mean time and a 90% approximate CI for the mean.

Exercise 8.16. A coffee shop wants to make sure its wait times are acceptable for its customers. Using a new software system, they measure the time it takes for a customer to receive their order. They are looking to have their average times be within 4-6 minutes or less. As their consultant, you collected data from 1000 orders (in `coffee_times.csv` on ELMS). Use R to calculate the mean time and a 95% approximate CI for the mean. Does the coffee shop seem to be hitting their goals?

Exercise 8.17. A team of scientists is conducting an experiment to measure blood sugar levels in a certain demographic group. Over the course of a year, they take blood sugar measurements from 1000 individuals. The data is collected in `blood_sugar.csv` on ELMS. Use R to calculate the mean blood sugar level and an 85% approximate CI.

Exercise 8.18. An advertising agency wants to compare the effectiveness of two ad campaigns, A and B. They randomly select users on a video streaming platform to watch either A or B, and measured how many seconds of the ad they watched before skipping (more is better). The data is collected in `ad_campaign.csv` on ELMS.

1. Calculate the mean time and an approximate 95% CI for both ad campaigns.
2. Which ad seemed to perform better based on the CIs/mean? Is there anything that complicates the interpretation?

Appendices

A | Sets

In the study of probability, we often run across situations where we need to talk about collections of outcomes as opposed to a single outcome. Such situations naturally lead us to the concept of *events*, i.e. collections of outcomes that we would like to compute probabilities for. The language of sets allows us a convenient language for talking about such collections of events.

Example A.0.1 1. In roulette, you have the option of betting on a collection of numbers that are “close together” as opposed to making a betting on a single outcome. A person making bets in such a situation would need to be able to compute the probabilities for these more complicated bets as well.

2. A researcher studying the effects of a certain drug may be interested in the time it takes for the drug to break down in the blood stream. In such a situation, the researcher would end up with a range of time that it takes for the drug to break down (what is more realistic? that it takes five hours for the drug to breakdown in every single person who takes it or that it breaks down between 4.5 - 5.5 hours?)

Sets are essentially (well-defined) collections of objects. They are usually denoted by upper case letters. The objects that belong to a set are called its *elements* or *members*. For example $\{1, 2, 3, 4, 5, 6\}$ is a set (with the curly braces indicating that we are looking at a set) and $1, 2, 3, 4, 5, 6$ are all elements of $\{1, 2, 3, 4, 5, 6\}$. We use $\in$ to denote membership. So we can write $1 \in \{1, 2, 3, 4, 5, 6\}$ to indicate that 1 is a member of (or alternatively is an element of) $\{1, 2, 3, 4, 5, 6\}$.

A sub-collection of objects from a set forms a *subset* of that set: we say A is a subset of B if every element of A is also an element of B . This fact is denoted by $A \subseteq B$. If A is not a subset of B (i.e. if there is some element of A that is not in B) we write $A \not\subseteq B$

1. $\{1, 2\} \subseteq \{1, 2, 5\}$ since $1, 2$ are both elements of $\{1, 2, 5\}$
2. $\{1, 2\} \not\subseteq \{1, 3, 4\}$ since 2 is not an element of $\{1, 3, 4\}$. Here $\not\subseteq$ should be read as “not a subset of”. We can denote 2 is not an element of $\{1, 3, 4\}$ by writing $2 \notin \{1, 3, 5\}$.
3. For any set C , the empty set $\emptyset \subseteq C$ (where the empty set is the set that contains no elements.)
4. For any set C , $C \subseteq C$

Set operations

As there are ways to combine numbers together to obtain a new number, there are ways to combine sets together to obtain different sets. We will discuss the following set operations:

- (a) Intersection
- (b) Union
- (c) Relative Complements

A.1.1 Intersection

Given a collection of sets it is reasonable to ask what these sets have in common (this goes hand in hand with the word “and”). The *intersection* of a collection of sets is the set that contains the elements that the collection has in common.

Example A.1.1

Suppose that $A = \{1, 2, 3\}$, $B = \{2, 3, 4\}$, $C = \{3, 4, 5\}$ and $D = \{4, 5, 6\}$. Then the intersection of A and B , denoted by $A \cap B$ is $\{2, 3\}$. Similarly $B \cap C = \{3, 4\}$ and $A \cap C = \{3\}$ etc. We can compute the intersection of three sets (or more) by using the same principles: $B \cap C \cap D = \{4\}$ as the only elements in common to all B, C and D is 4.

It is possible to extend intersections to an infinite sequence of sets as well (it is possible to extend it even further but that is not going to be useful to us in the context of the class). Given a sequence $A_1, A_2, \dots, A_n, \dots$ of sets, the intersection of the sets will be denoted by $\bigcap_{i=1}^{\infty} A_i$ and is the set that contains the elements common to all of the A_i .

A collection of sets is said to be *disjoint* if their intersection is empty. A collection of sets is said to be *pairwise disjoint* if any two sets in the collection is disjoint. Any pairwise disjoint collection of sets is disjoint but the converse may not be true.

Example A.1.2

Let $A_i = \{i\}$ for each positive integers i . So $A_1 = \{1\}, A_2 = \{2\}, A_3 = \{3\}, \dots$. The A_i are pairwise disjoint as $A_i \cap A_j = \emptyset$ whenever $i \neq j$.

The sets $A = \{1, 2\}$, $B = \{1, 3\}$ and $C = \{2, 3\}$ are disjoint. They are not pairwise disjoint (in this example, the intersection of any two sets from A, B, C is non-empty but $A \cap B \cap C = \emptyset$).

Exercise: Suppose that $A_n = \{n, n+1, n+2, \dots\}$ for all positive integers n . Explain why their intersection is empty, i.e. explain why $\bigcap_{n=1}^{\infty} A_n = \emptyset$

A.1.2 Union

Given a collection of sets it is also reasonable to ask what you can get by combining all the elements in these sets together (this goes hand in hand with the word “or”). The *union* of a collection of sets is the set that contains the elements that appear in *at least one* of the sets in the collection.

Example A.1.3

Suppose that $A = \{1, 2, 3\}$, $B = \{2, 3, 4\}$, $C = \{3, 4, 5\}$ and $D = \{4, 5, 6\}$. Then the union of A and B , denoted by $A \cup B$ is $\{1, 2, 3, 4\}$. Similarly $B \cup C = \{2, 3, 4, 5\}$ and $A \cup C = \{1, 2, 3, 4, 5\}$ etc. We can compute the unions of three sets (or more) by using the same principles: $B \cup C \cup D = \{2, 3, 4, 5, 6\}$.

Similar to intersections, it is possible to extend unions to an infinite sequence of sets as well. Given a sequence $A_1, A_2, \dots, A_n, \dots$ of sets, the union of the sets will be denoted by $\bigcup_{i=1}^{\infty} A_i$ and is the set that contains the elements appear in at least one A_i .

Exercise: Suppose that $A_n = [\frac{1}{n+1}, 1]$ for all positive integers n . Why is their union the interval $(0, 1]$?

A.1.3 Relative Complements

Given sets A, B another reasonable question to ask is what elements are unique to each set. Relative complements answer this question.

The set of elements in A but not in B is denoted by $A - B$ and is read A take away B .

Example A.1.4

Suppose that $A = \{1, 2, 3\}$ and $B = \{2, 3, 4\}$. Then $A - B = \{1\}$ and $B - A = \{4\}$.

Unlike union and intersection, there is no natural extension of the notion of a relative complement to a sequence of sets.

The notion of complement of a set A often denoted by A^c or A' will appear throughout our study of probability. A^c stands for the set of elements not in A . This however causes issues: say $A = \{1, 2\}$, then objects not in A contains, among other things, the math building, your laptop etc. This is unlikely to be useful (in fact it is not even a set if we were to do a more technical study of sets). To address this issue we introduce the concept of a *universal set*. The universal set is meant to contain “everything of interest”. Complements are actually relative complements taken with the universal set. For example if the universal set $U = \{1, 2, 3, 4, 5, 6\}$ and $A = \{1, 2\}$ then $A^c = U - A = \{3, 4, 5, 6\}$.

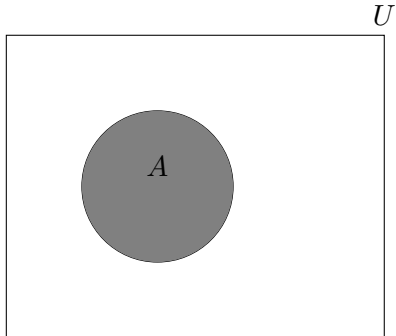
In the study of probability we take the universal set to be the sample space as every outcome of interest is in the sample space.

Venn diagrams and visualizing sets

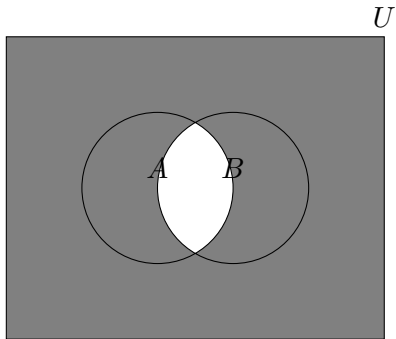
Venn diagrams allow us a nice way to visualize sets. The idea is to use a rectangle to denote the universal set and to use a different shape (most of the time circles) to denote various subsets of interest. We think of the interior of the circles we have drawn as containing the elements from that set.

Here are a couple of examples.

- (a) This Venn diagram has a subset A highlighted in gray.



- (b) This Venn diagram highlights $(A \cap B)^c$, i.e. the shaded area is the complement of the intersection of A and B .



A.2.1 Some useful identities

There are certain identities that are useful when working with sets. For us, these identities establish connections between the three set operations that we have mentioned in this note: union, intersection and taking (relative) complements.

Here is a list of useful identities:

- (a) $(A^c)^c = A$
- (b) $(A \cap B)^c = A^c \cup B^c$
- (c) $(A \cup B)^c = A^c \cap B^c$
- (d) $(\bigcap_{i=1}^n A_i)^c = \bigcup_{i=1}^n A_i^c$
- (e) $(\bigcup_{i=1}^n A_i)^c = \bigcap_{i=1}^n A_i^c$
- (f) $(\bigcap_{i=1}^\infty A_i)^c = \bigcup_{i=1}^\infty A_i^c$
- (g) $(\bigcup_{i=1}^\infty A_i)^c = \bigcap_{i=1}^\infty A_i^c$

Venn diagrams can be useful in establishing some of these identities: Draw a Venn diagram for the set on the left hand side of the equation, do the same for the set on the right hand side of the equation and if the shaded regions are the same then the equation holds!

The role of sets in Stat 400

As we discussed above, the idea for us that sets provide a *useful language for discussing events and combinations of events*. The idea is that for a given experiment that we are interested in, the set of all possible outcomes will act as the universal set, i.e. the role of the universal set will always be played by the sample space. Events can now be described as *subsets* of the

sample space.

Consider rolling a die. The possible outcomes are now 1, 2, 3, 4, 5, 6. The set that contains these outcomes is now $\Omega = \{1, 2, 3, 4, 5, 6\}$. Examples of events include

- (a) $\{1, 2\}$: you see either a 1 or a 2 when the die is rolled.
- (b) $\{4, 5, 6\}$: you see either a 4, a 5, or a 6 when the die is rolled.
- (c) $\emptyset$: the empty or null event indicates the (im)possibility that nothing happens when you conduct the experiment.
- (d) $\{1, 2, 3, 4, 5, 6\}$ you see either a 1 or a 2 or 3 or a 4 or a 5 or a 6 when the die is rolled

In general given an event A lists a collection of possible outcomes of the experiment of interest and $P(A)$ denotes the probability that an outcome listed in A takes place.

Cardinality

The last concept we will need from set theory is that of cardinality. Cardinality aims to measure how many elements are in a set. In order to define what it means for two sets to have the same cardinality, we need the notion of a bijection first.

Recall that a function $f : A \rightarrow B$ is a *bijection* if:

- (a) it is *surjective* (or *onto*), i.e for any $b \in B$ there is some $a \in A$ such that $f(a) = b$.
- (b) it is *injective* (or *one-to-one*), i.e for any a_1, a_2 in A with $f(a_1) = f(a_2)$, we have $a_1 = a_2$.

Definition A.1

Two sets A and B have the same cardinality (denoted by $|A| = |B|$) if and only if there is a *bijection* between A and B .

So $|A| = |B|$ implies that each element in A can be paired with a unique element of B . *Thus the number of elements must be the same as that of B .*

Example A.4.1 (a) Let $2\mathbb{N} = \{2, 4, 6, 8, 10, \dots\}$. Then

$$|2\mathbb{N}| = |\mathbb{N}|$$

The function $f : \mathbb{N} \rightarrow \mathbb{N}$

$$f(x) = 2x$$

is a bijection.

- (b) The interval $(-\frac{\pi}{2}, \frac{\pi}{2})$ has the same cardinality as $\mathbb{R}$. The function $f : (-\frac{\pi}{2}, \frac{\pi}{2}) \rightarrow \mathbb{R}$ where

$$f(x) = \tan(x)$$

is a bijection.

Note that in the example above, $2\mathbb{N} \subseteq \mathbb{N}$ and $\mathbb{N} \neq 2\mathbb{N}$ (sometimes denoted as $2\mathbb{N} \subsetneq \mathbb{N}$), and $(-\frac{\pi}{2}, \frac{\pi}{2}) \subsetneq \mathbb{R}$. It turns out that this happens when we have infinite sets. This is not the case with finite sets.

In our context, the notions of *countable* and *uncountable* sets are important: problems involving countable sets are usually handled using sums (and matrices), while those involving uncountable sets use integrals.

Definition A.2

A set A is said to be *countable* if it is empty or there is a *surjection*, or surjective function $f : \mathbb{N} \rightarrow A$. A set is *uncountable* if it is not countable.

Note that *finite sets* are countable too. For example, if

$$A = \{0, 1, 2, 3, 4, 5\}$$

$$f : \mathbb{N} \rightarrow A \quad f(x) = x \mod 6$$

(where $x \mod 6$ is the remainder when you divide x by 6).

We differentiate between countable sets that are finite and countable sets that are infinite by calling sets that are countable and infinite *countably infinite*.

Recognizing a set as finite or infinite is usually fairly straightforward. However recognizing an infinite set as countably infinite or uncountable can be difficult. Here, I will list out the most common countably infinite and uncountable sets.

Countable sets

- $\mathbb{N}$
- $\mathbb{Q}$
- If $A_1, A_2, \dots, A_n$ are countable, then $A = A_1 \times \dots \times A_n$ (also written $\prod_{i=1}^n A_i$, the set of all ordered n -tuples) is also countable.

Uncountable sets

- $\mathbb{R}$
- Any nontrivial interval, e.g. (a, b) where $a < b$
- The irrationals
- If $A_1, \dots, A_n$ are uncountable, then so is $A = A_1 \times \dots \times A_n$.

If you are up for a challenge, see if you can use the following theorem to show that $\mathbb{N}$ and $\mathbb{N} \times \mathbb{N}$ have the same cardinality.

Theorem A.3

Schröder-Bernstein theorem.

Suppose that there are one-to-one functions $f : A \rightarrow B$ and $g : B \rightarrow A$. Then $|A| = |B|$.

Exercises for Chapter A

Exercise A.1. Use Venn diagrams to determine whether the following relations always hold.

- (a) $(A \cap B)^c = A^c \cup B^c$
- (b) $(A - B) \cap C = (A \cup B) \cap (A \cup C)$
- (c) $(A - B) \cap (B - A) = \emptyset$

Exercise A.2. Let $A = \{3n : n \in \mathbb{N}\}$. Find a bijection from A to $\mathbb{N}$.

Exercise A.3. Let $\Omega = \mathbb{N}$, $A = 2\mathbb{N}$, $B = 3\mathbb{N}$. Describe the following sets.

- (a) $A \cap B$
- (b) $A \cup B$
- (c) $A - B$
- (d) $B - A$

Exercise A.4. Let $\Omega = \{1, 2, 3, 4, 5, 6\}$, $A = \{2, 3, 5\}$, $B = \{2, 4, 6\}$. Compute:

- (a) $A \cup B$
- (b) $A \cap B$
- (c) $A - B$
- (d) $B - A$

B | Tools from calculus

In this appendix we look at some topics that are useful in our study of statistics.

B.0.1 Techniques of integration

In this section we review some techniques that are useful for computations. This section is intended mainly for review. We assume all functions involved are differentiable (as many times as necessary) on $\mathbb{R}$ to avoid lengthy technical statements or techniques. For further details, you should refer to a calculus text.

Integration by substitution

We can evaluate

$$I = \int g(f(x))f'(x) \, dx$$

by setting $u = f(x)$ and obtaining

$$\begin{aligned} I &= \int g(u) \, du = G(u) + C \\ &= G(f(x)) + C \end{aligned}$$

Example B.0.1

Evaluate

$$\int 3x^2 (x^3 + 9)^{2024} \, dx$$

Take $u = x^3 + 9$. Now we have $du = 3x^2 \, dx$. Thus

$$\begin{aligned} \int 3x^2 (x^3 + 9)^{2024} \, dx &= \int u^{2024} \, du \\ &= \frac{u^{2025}}{2025} + C \\ &= \frac{(x^3 + 9)^{2025}}{2025} + C \end{aligned}$$

Integration by substitution yields the following useful rules.

- (a) The so called generalized power rule (valid for $r \neq -1$):

$$\int f'(x)(f(x))^r \, dx = \frac{f(x)^{r+1}}{r+1} + C$$

- (b)

$$\int \frac{f'(x)}{f(x)} \, dx = \ln |f(x)| + C$$

- (c)

$$\int f'(x)e^{f(x)} \, dx = e^{f(x)} + C$$

A list of useful substitutions

To evaluate the following integrals, we can use the substitutions listed along with them:

(a) $\int \sqrt{a^2 - x^2} \, dx$: $x = a \sin(\theta)$.

(b) $\int \sqrt{a^2 + x^2} \, dx$: $x = a \tan(\theta)$.

(c) $\int \sqrt{x^2 - a^2} \, dx$: $x = a \sec(\theta)$.

Integration by parts

$$\int f(x)g'(x) \, dx = f(x)g(x) - \int f'(x)g(x) \, dx$$

Alternately, by setting $u = f(x)$ and $v = g(x)$, it can be rewritten as

$$\int u \, dv = uv - \int v \, du$$

Example B.0.2

$$\int xe^x \, dx$$

$$\begin{aligned} u &= x & dv &= e^x \, dx \\ du &= dx & v &= e^x \end{aligned}$$

$$\begin{aligned} \int xe^x \, dx &= xe^x - \int e^x \, dx \\ &= xe^x - e^x + C \end{aligned}$$

Example B.0.3

$$\int \sin(x)e^x \, dx$$

$$\begin{aligned} u &= \sin(x) & dv &= e^x \, dx \\ du &= \cos(x)dx & v &= e^x \end{aligned}$$

$$\int \sin(x)e^x \, dx = \sin(x)e^x - \int e^x \cos(x) \, dx$$

Applying integration by parts to $\int e^x \cos(x) \, dx$:

$$\begin{aligned} u &= \cos(x) & dv &= e^x \, dx \\ du &= -\sin(x)dx & v &= e^x \end{aligned}$$

$$\begin{aligned} \int e^x \sin(x) \, dx &= e^x \sin(x) - \left(\cos(x)e^x - \int e^x (-\sin(x)) \, dx \right) \\ &= e^x \sin(x) - \cos(x)e^x - \int e^x \sin(x) \, dx \end{aligned}$$

Take $I = \int e^x \sin(x) \, dx$. Then the above can be rewritten as

$$\begin{aligned} I &= e^x \sin(x) - e^x \cos(x) - I \\ 2I &= e^x \sin(x) - e^x \cos(x) \end{aligned}$$

Hence, the final answer is

$$I = \frac{e^x \sin(x) - e^x \cos(x)}{2} + C$$

B.0.2 Power series

A power series is of the form

$$\sum_{n=0}^{\infty} a_n x^n$$

Recall that a power series may exhibit one of the following behaviors:

- (a) Converge only at $x = 0$.
- (b) Converge for all x .
- (c) Converge for all $|x| < R$ and diverge for all $|x| > R$, for some $R > 0$.

In case (a), we can consider $R = 0$, and in the case of (b) we can consider $R = \infty$. R is called the radius of convergence.

Suppose that $R > 0$. Within the radius of convergence, a power series "behaves like a polynomial" and so the following identities hold:

$$\begin{aligned}\frac{d}{dx} \sum_{n=0}^{\infty} a_n x^n &= \sum_{n=0}^{\infty} \frac{d}{dx} (a_n x^n) \\ \int_0^t \left(\sum_{n=0}^{\infty} a_n x^n \right) dx &= \sum_{n=0}^{\infty} \int_0^t a_n x^n dx\end{aligned}$$

Furthermore, the resulting power series will have the same radius of convergence.

We also know that if

$$f(x) = \sum_{n=0}^{\infty} a_n x^n \text{ for all } -R < x < R$$

then we also know that

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n$$

Example B.0.4

Compute $E(X)$ for $X \sim \text{Geom}(p)$ where $0 < p < 1$.

Note that

$$\begin{aligned}E(X) &= \sum_{n=1}^{\infty} np(1-p)^{n-1} \\ &= p \sum_{n=1}^{\infty} n(1-p)^{n-1}\end{aligned}$$

Take $q = 1 - p$. Now we note that $0 < q < 1$ and

$$\begin{aligned}E(X) &= p \sum_{n=1}^{\infty} nq^{n-1} \\ &= p \sum_{n=1}^{\infty} \frac{d}{dq} q^n\end{aligned}$$

Using the above results on power series,

$$E(X) = p \frac{d}{dq} \sum_{n=1}^{\infty} q^n$$

Written in this form, we can regard $q^1, q^2, q^3, \dots$ as a geometric series with a first term of q and a common ratio of q .

Using the fact that (for $-1 < r < 1$)

$$\frac{a}{1-r} = \sum_{n=0}^{\infty} ar^n$$

we complete the computation of the expectation:

$$\begin{aligned}E(X) &= p \frac{d}{dq} \left[\frac{q}{1-q} \right] \\ &= p \left[\frac{(1-q) + q}{(1-q)^2} \right] \\ &= p \left[\frac{1}{1-q} \right] \\ &= \frac{1}{p}\end{aligned}$$

Exercise: Use similar manipulations to the above, along with $V(X) = E(X^2) - (E(X))^2$ to compute $V(X)$ for $X \sim \text{Geom}(p)$.

C | Iterated integrals

When we first see integrals in calculus, it is with relation to computing the area under the curve for a non-negative continuous function $f : [a, b] \rightarrow \mathbb{R}$.

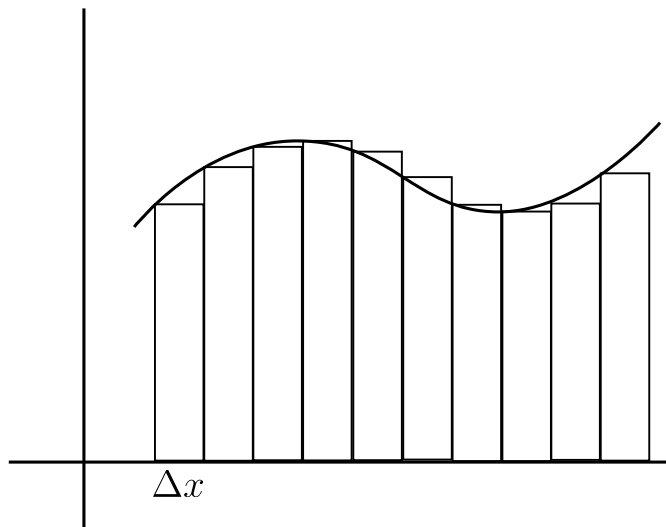


Figure C.1: Riemann sums for area under a curve

We define $\int_a^b f(x) \, dx$ as a limit of Riemann sums where the Riemann sums approximate the area under the curve using rectangles. The limit captures the idea that more rectangles (with the rectangle with the largest base going to 0) provide better approximations. These ideas can be extended to functions with different domains with the use of various tools (e.g. $\int_0^\infty f(x) \, dx = \lim_{b \rightarrow \infty} \int_0^b f(x) \, dx$.)

Given a continuous, non-negative function $f(x, y)$ defined on $R \subset \mathbb{R}^2$, we can ask a similar question: what is the volume under the curve (or surface)?

We can try to take a Riemann sum approach that utilizes cuboids. However, as in the case of Riemann sums for functions of a single variable, Riemann sums are cumbersome to use. Instead we use *iterated integrals*.

The idea behind iterated integrals is to look at *slices*. If we fix an x_0 (or a y_0), and take a slice of width Δx (or Δy) we can approximate the volume of this slice as follows:

$$\text{volume of slice} \approx \left(\int_{a(x_0)}^{b(x_0)} f(x_0, y) \, dy \right) \Delta x$$

Better approximations can be obtained by reducing the thickness of the slice. A sum of the following form provides an approximation to the total volume:

$$\sum_{i=1}^n \int_{a(x_i)}^{b(x_i)} f(x_i, y) \, dy \Delta x_i$$

By considering the limiting behavior, we obtain an expression for the volume as

$$\int_c^d \int_{a(x)}^{b(x)} f(x, y) \, dy \, dx$$

Note: There is quite a bit of technical details that go into establishing the equivalence of the Riemann sum approach to the iterated integral approach. The equivalence of these two approaches fails in certain situations. However, as these situations do not arise in this class, we do not discuss the related technical issues here.

Now we show an example of computing an iterated integral.

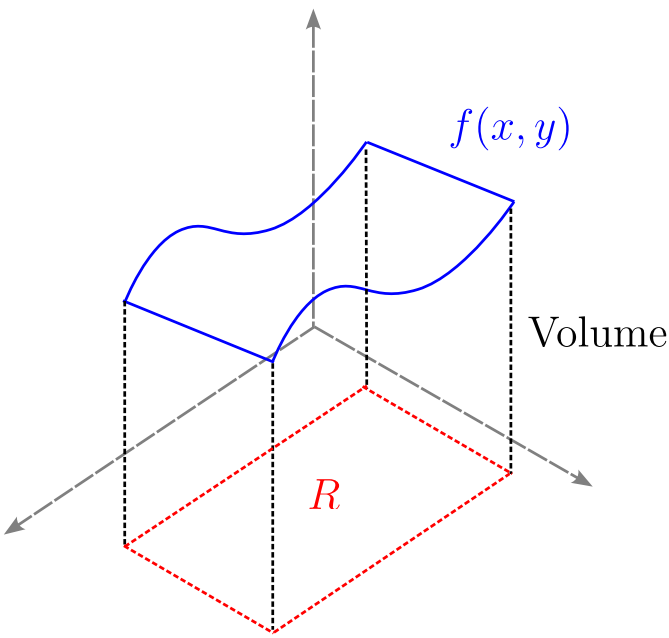


Figure C.2: Volume under the curve

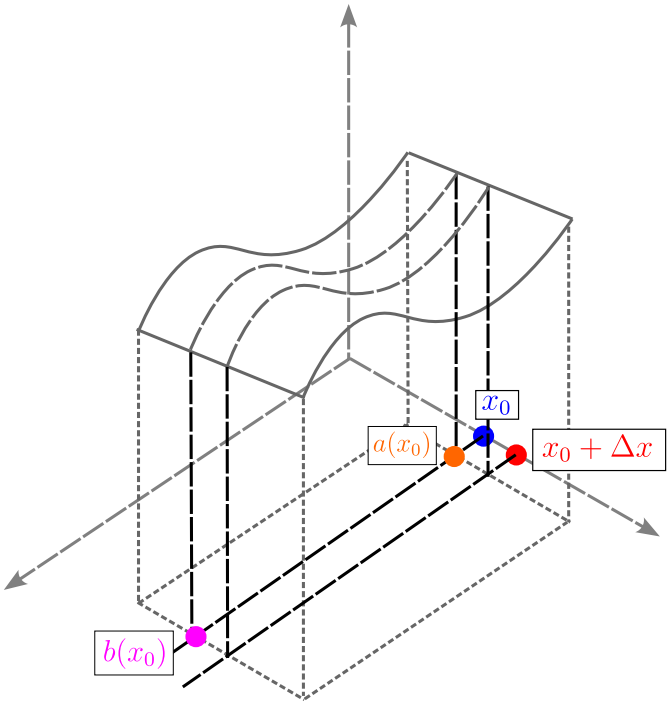


Figure C.3: Volume of a slice

Example C.0.1

Suppose that $f(x, y)$ is defined as

$$f(x, y) = \begin{cases} xy & (x, y) \in R \\ 0 & \text{otherwise} \end{cases}$$

where R is the triangular region with vertices $(0, 0)$, $(1, 1)$, $(1, 0)$. Compute the iterated integral $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) \, dx \, dy$.

Now we evaluate the integral (denoted by I). We start by choosing an order to compute the iterated integral in: we start by considering slices along the x -axis. Because $f(x, y)$ is zero outside of R , this yields

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) \, dy \, dx = \int_0^1 \int_0^x xy \, dy \, dx$$

The bounds of 0 and 1 on the outer integral (along with the dx at the end of the integral) indicate that we slice along the x -axis. The bounds of 0 and x in the inner integral are due to the fact that the different slices that we are considering, for different values of x have different lengths. We will evaluate this integral *starting with the innermost integral and working outwards, treating all variables with which we are not integrating with respect to (in this case x) as constant*. This gives

$$I = \int_0^1 x \times \frac{y^2}{2} \Big|_0^x \, dx = \int_0^1 \frac{x^3}{2} \, dx$$

which evaluates to $\frac{1}{8}$. As an exercise, you should setup the iterated integral taking slices along the y -axis.

Example C.0.2

Suppose that $f(x, y)$ is defined as

$$f(x, y) = \begin{cases} xe^{xy} & (x, y) \in R \\ 0 & \text{otherwise} \end{cases}$$

where R is the square region with vertices $(1, 1)$, $(1, 2)$, $(2, 1)$, $(2, 2)$. Compute the iterated integral $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) \, dx \, dy$.

Taking slices along the x -axis, we obtain that

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) \, dy \, dx = \int_1^2 \int_1^2 xe^{xy} \, dy \, dx$$

Evaluating the innermost integral as we had done previously, we obtain that this is equal to:

$$\int_1^2 e^{xy} \Big|_1^2 \, dx \text{ which in turn yields } \int_1^2 e^{2x} - e^x \, dx \text{ and hence we obtain } \frac{e^{2x}}{2} - e^x \Big|_1^2 \text{ which yields } \frac{e^4 - e^2}{2} - (e^2 - e^1) = \frac{e^4}{2} - \frac{3e^2}{2} + e.$$

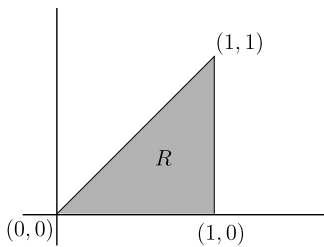


Figure C.4: Region where $f(x,y)$ is non-zero

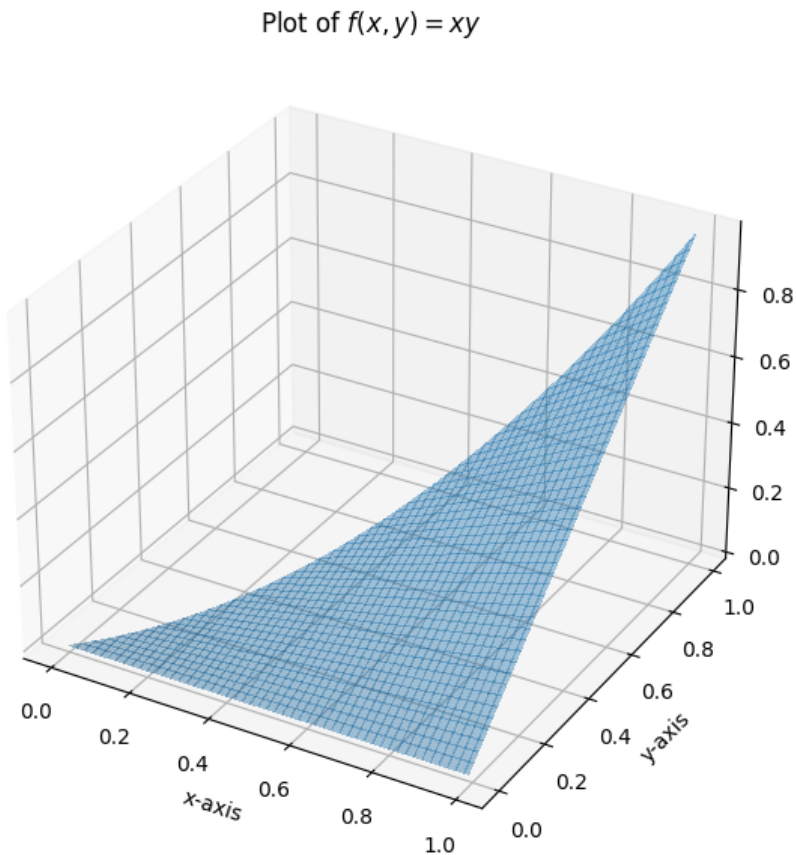


Figure C.5: Graph of $f(x,y)$

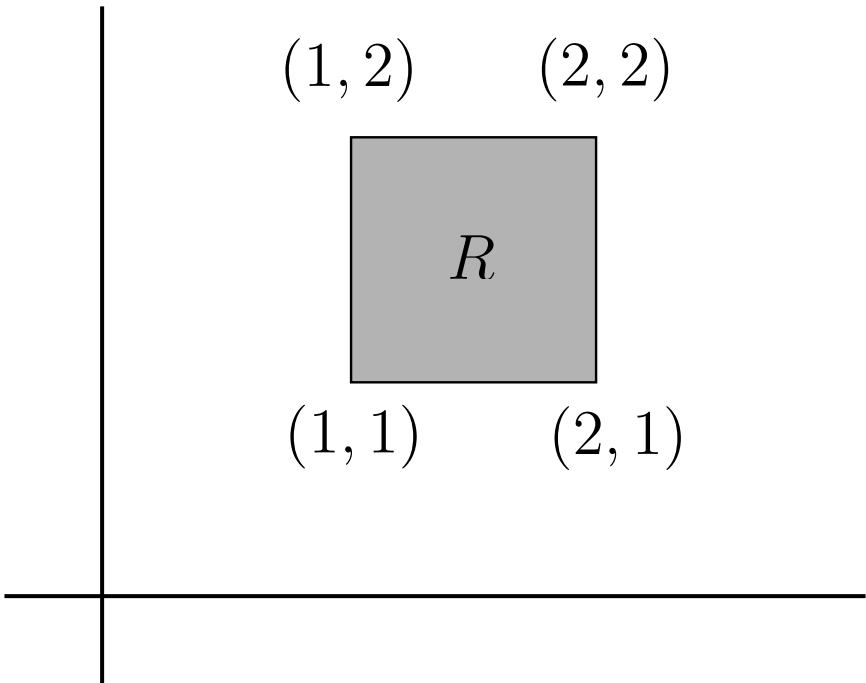


Figure C.6: Region where $f(x,y)$ is non-zero

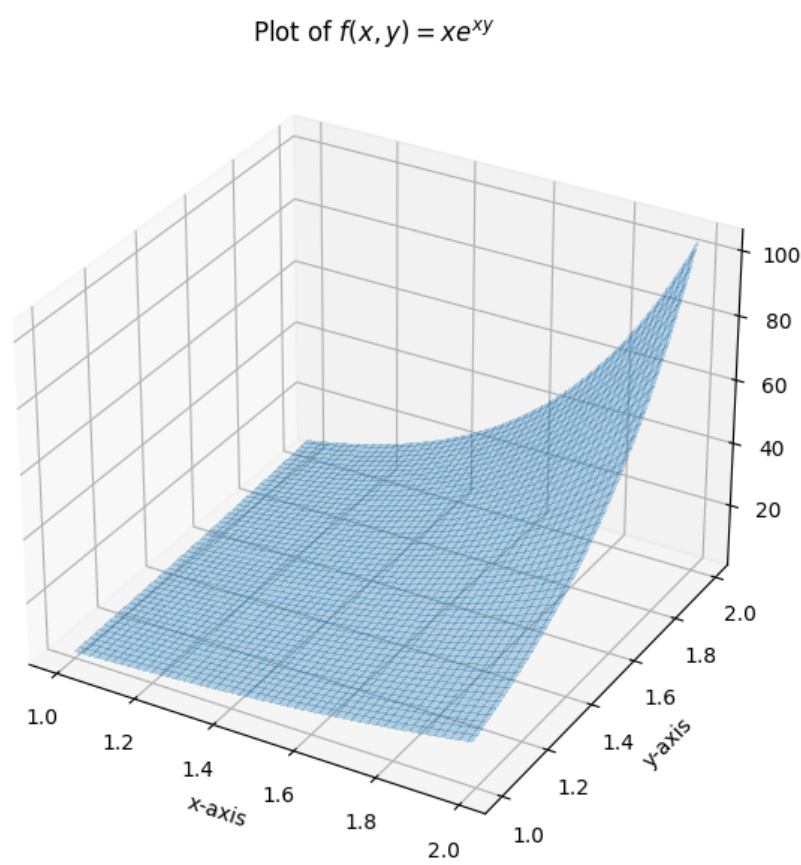


Figure C.7: Graph of $f(x, y)$

D | Descriptive statistics

Statistics involves learning from data. This process often includes multiple steps.

- (a) Gathering data
- (b) Summarizing and attempting to gain an initial understanding of the nature of the data that has been gathered.
- (c) Drawing inferences from the data.

Each step involves its own challenges and techniques. For example,

- (a) Determining the best way to gather the data: online, in person, by phone, etc.
- (b) Determining if the data has been correctly entered.
- (c) Determining how likely the next observed value could be.

Here we discuss some techniques that can be used to summarize, visualize, and (hopefully thereby) gain a better understanding of the data.

Measures of central tendency

Measures of central tendency are values that we use to represent the collection of values in some form. They are considered to be a "typical" value of the collection of values. We consider three such values here.

Average

The average of a collection of numbers $x_1, \dots, x_n$ is given by

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$$

Example D.1.1

The average of the numbers 10, 11, 14, 9, 12 can be found as

$$\bar{x} = \frac{10 + 11 + 14 + 9 + 12}{5} = 11.2$$

We often need to calculate the *sample variance* along with the mean. The sample variance of a collection of numbers is given by

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

and measures the variation in the values from $\bar{x}$. s is called the standard deviation.

To calculate the sample variance, we can first subtract 11.2 (the value of $\bar{x}$) from each sample. This is shown in the table below.

x_i	$x_i - \bar{x}$	$(x_i - \bar{x})^2$
10	-1.2	1.44
11	-0.2	0.04
14	2.8	7.84
9	-2.2	4.84
12	0.8	0.64

$$s^2 = \frac{1}{5-1} (1.44 + 0.04 + 7.84 + 4.84 + 0.64) = 3.7$$

$$s = \sqrt{3.7} = 1.92$$

Median

The median of a collection of numbers is obtained by first sorting the observation and then taking:

- the $\left(\frac{n+1}{2}\right)^{\text{st}}$ value if n is odd,
- the average of the $\left(\frac{n}{2}\right)^{\text{st}}$ and $\left(\frac{n}{2} + 1\right)^{\text{st}}$ values if n is even.

Example D.1.2

The median of the values

$$10, 11, 14, 9, 12$$

is obtained by first sorting the values to obtain

$$9, 10, 11, 12, 14$$

and then choosing the $\left(\frac{5+1}{2}\right)^{\text{st}} = 3^{\text{rd}}$ value.

Mode

The mode of a collection of values is the value that appears the most number of times.

Example D.1.3

The mode of the values 1, 3, 3, 5, 7 is 3.

In the case there are multiple values that appear most often, all such values qualify as modes and such a collection is called multimodal.

Example D.1.4

The collection of values 1, 1, 3, 3, 5 is multimodal - both 1 and 3 are modes.

Note:

- (a) s^2 can be computed using the formula

$$\frac{1}{n-1} \left(\left(\sum x_i^2 \right) - \bar{x}^2 \right)$$

- (b) Measures of variability other than s^2 also exist. For example, the range of a collection is obtained using

$$\text{range} = \text{maximum value} - \text{minimum value}$$

Visualizing data

There are multiple ways of visualizing data. You have probably run across a few at this point - such as pie charts, stem and leaf plots, and histograms.

In this section, we focus on histograms, particularly as it relates to frequencies/relative frequencies as follows.

Discrete values from counting

Given a collection of values that has arisen as a result of some counting process, we can construct a (relative) frequency histogram for the data as follows:

- (a) Determine frequencies/relative frequencies for observed values
- (b) Mark observed values along horizontal axis
- (c) For each marked value x , draw a rectangle with base $[x - 1/2, x + 1/2]$ (along the horizontal axis) and height equal to the (relative) frequency.

Example D.2.1

The observed data is 1, 3, 3, 4, 4, 4, 5, 8. We first summarize the values and their (relative) frequencies:

Note that in the case of a relative frequency histogram, the sum of the areas of the rectangles is equal to 1. The data is visualized in Figure D.1.

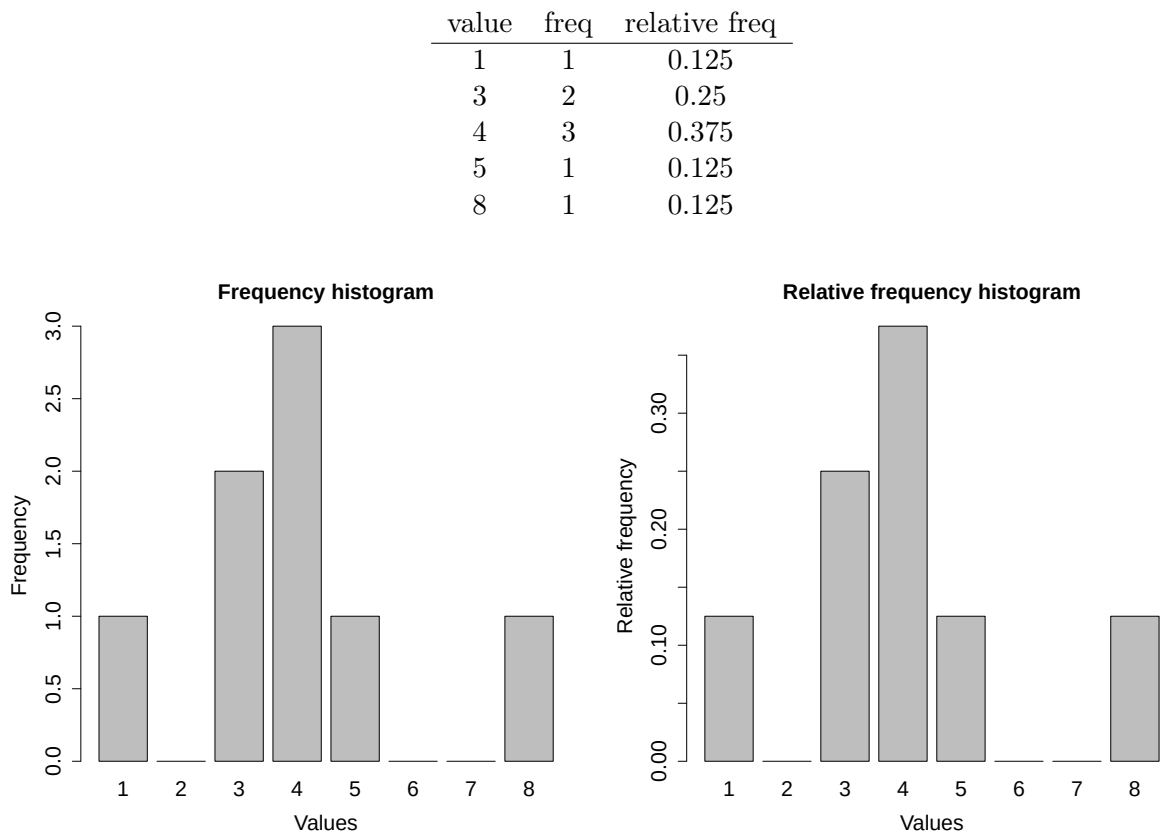


Figure D.1: Frequency and relative frequency histograms for Example D.2.1

Continuous data

In case we have continuous data we create histograms as follows:

- (a) Select ranges for the values (equal width is preferred but may not be possible)
- (b) Determine frequency/relative frequency.
- (c) Along the horizontal axis, mark the classes from step 1, and draw rectangles with height equal to:
 - *the frequency*, if drawing a frequency histogram
 - $(\text{relative frequency})/(\text{class width})$, if drawing a relative frequency histogram

We show examples of drawing histograms for continuous data with equal and unequal class widths.

Example D.2.2

We draw a histogram for the data

0.1, 0.5, 1.1, 1.5, 1.7, 2.3, 2.5, 2.7, 2.8, 2.9, 3.3, 3.4

We use classes of width 1 (denoted using the interval notation $[0, 1)$, $[1, 2)$, etc.) and count the number of values falling into the appropriate classes in the table below. The plots are shown in Figure D.2.

Class	Freq.	Rel. freq.
$[0, 1)$	2	0.1667
$[1, 2)$	3	0.25
$[2, 3)$	5	0.4167
$[3, 4)$	2	0.1667

Example D.2.3

We draw a histogram for the data

0, 0.1, 0.2, 0.5, 4.1, 4.1, 4.3, 4.5, 5, 5.3, 5.5, 5.7, 5.7, 5.8, 6.5, 10

We use classes of unequal width to illustrate the spread of the data. The classes, counts, relative frequencies, and heights of the resulting histogram bars are summarized in the table below. The plots are shown in Figure D.3.

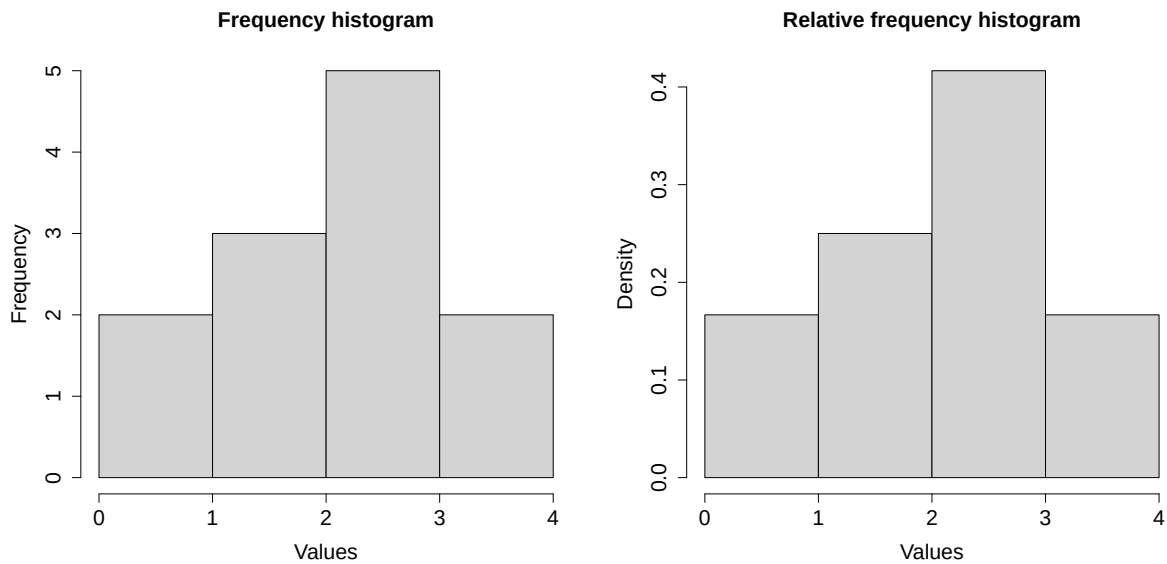


Figure D.2: Frequency and relative frequency histograms for Example D.2.2.

Class	Freq.	Rel. Freq.	Height
[0, 1)	4	0.25	0.25
[1, 4)	0	0	0
[4, 6)	10	0.625	0.3125
[6, 10)	2	0.125	0.03125

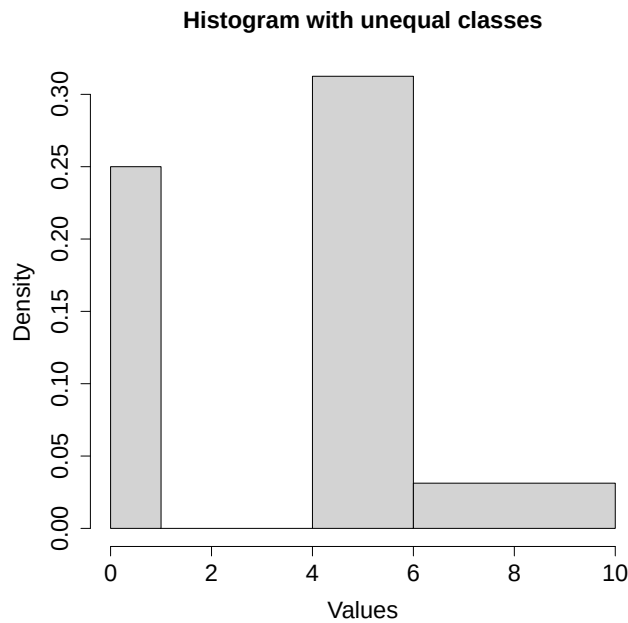


Figure D.3: Histogram with unequal class widths for Example D.2.3.

Note:

- The sum of the areas of the rectangles when using relative frequencies here is 1 as well.
- The values obtained by using the equation

$$\text{height} = \frac{\text{relative freq}}{\text{class width}}$$

are called *densities*.

- We interpret probabilities as long run averages. Under this interpretation, the relative frequencies for a large sample will closely resemble the associated probabilities. Thus there are links between pdfs and histograms!
- Much of the work involving summarizing and visualizing data can be tedious and error-prone when done by hand. It is possible to use statistical software such as R to avoid these issues. We present the basics of R in the appendix.